TRAINER SLIDES · IBF-STS ACCREDITED · FULL CODE EDITION

# Machine Learning 101 for Financial Trading

FSE-DAT-3019-1.1 · Data Analytics and Computational Modelling · Proficiency Level 3

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

**Version 2.3 · 22 October 2026**

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# Attendance and IBF-STS Funding

1

## Sign in every session

Mark attendance at the start of each session and after lunch.

%

## 75% attendance

IBF-STS funding requires at least 75% attendance.

✓

## Pass the assessment

Funding also requires passing both assessment components.

!

## Assessment attendance

A separate record is signed before the assessment begins.


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# About the Trainer




Complete this panel with your own details before delivery.

**NAME**

**TITLE / DESIGNATION**

**QUALIFICATIONS**

**AREAS OF EXPERTISE**

**TRAINING & INDUSTRY EXPERIENCE**

**CONTACT**

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YOUR TRAINER

# About the Trainer




## Dr Alfred Ang

Course Developer · Trainer · Assessor
for this IBF-STS programme.

**ROLE**
Course Developer, Trainer and Assessor

**CERTIFICATION**
ACTA · DACE · PhD

**FINANCIAL SERVICES**
IBF-STS programmes in AI, algorithmic trading and fintech

**DELIVERS**
AI, machine learning and data programmes for financial institutions

**FOUNDER**
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ICE-BREAKER

# Let's Know Each Other

- Your name, your institution and your role.
- Do you trade, invest, analyse or build systems for markets today?
- Have you used TradingView, a broker API or Python before?
- One trading rule you believe works — we will test it this week.


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HOUSEKEEPING

# Ground Rules

### Be on time
Sessions restart promptly after each break.

### Phones on silent
Take calls outside the room.

### Ask as we go
Questions are welcome at any point.

### Hands on the keyboard
Twenty labs — follow along on your own laptop.

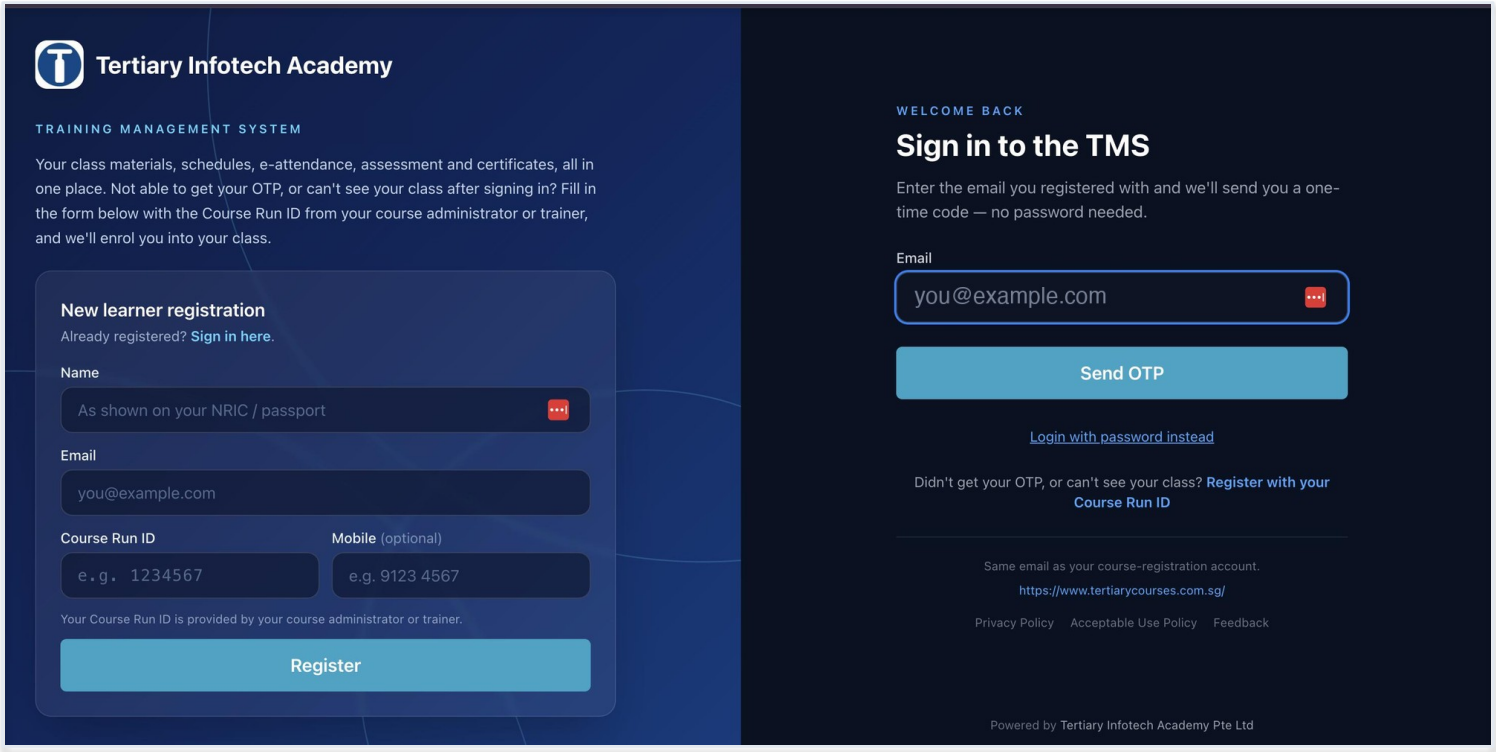
### Paper trading only
No real money is used anywhere in this course.

### Not investment advice
Lion City Capital and all scenarios are fictitious.


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COURSE PORTAL

# Login to TMS and Download Course Material




Log in at lms-tms.tertiaryinfotech.com with your registered email + the OTP from your inbox, then download the slides, Learner Guide and labs.


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COURSE PORTAL · STEPS

# Download Course Material

1. Go to lms-tms.tertiaryinfotech.com
2. Enter your registered email and click Send OTP
3. Key in the one-time password from your inbox
4. Open this course: download the Learner Slides + Learner Guide PDFs
5. Activities / Lab: download the labs folder (lab manuals + code)

**Keep the Learner Guide open all course — the assessment is open book.**


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COURSE PORTAL · GITHUB

# Download the Labs and Simulator (GitHub)

## Course repository

github.com/tertiarycourses/TGS-2023018794-Machine-Learning-101-for-Financial-Trading

Learner Guide, Learner Slides, Lesson Plan and all 20 lab folders (instruction PDFs, .py and .pine files) - Full Code edition.

Code > Download ZIP, or git clone.

## Trading simulator software

github.com/alfredang/algorithmic-trader-bot

The quant / algo trading simulator the labs run in: backtest, forward test, paper trading, analytics.

Install steps: next section and the Learner Guide.


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LEARNING OUTCOMES

# What You'll Achieve

1

**LO1 · A1, A6**

Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.

2

**LO2 · A2, A3**

Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.

3

**LO3 · A4, A5**

Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.

4

**LO4 · A7, A8, A9, A10**

Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.


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COURSE JOURNEY

# The Course Journey

1

## Topic 1 — Quant and Algo Trading Foundations

Day 1 · Labs 1–4 · A1, A6, K1, K9 · Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents

2

## Topic 2 — Indicators, Strategies and Metrics

Day 1 · Labs 5–9 · A2, A3, K3, K5, K7 · Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca

3

## Topic 3 — ML Models and A/B Testing

Day 2 · Labs 10–14 · A4, A5, K2, K4, K6 · Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals

4

## Topic 4 — End-to-End Trading Pipeline

Day 2 · Labs 15–20 · A7, A8, A9, A10, K7, K8 · Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis


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# AI writes the code. You own the algorithm.

This Full Code edition is for learners who want full customisation in Python. Every lab builds one stage of a single Python pipeline: the complete scripts are provided in each lab folder, with the prompts that generate them, and an AI coding assistant writes and changes the code. What matters is still the big concepts: the hypothesis, how each algorithm makes its decision, how it is tested without bias and with costs, how risk is controlled, and how the results are judged.


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THE PARADIGM SHIFT

# Your Role versus the AI's Role

1

### Your job: the hypothesis

What edge do you expect, why, and how will you know it is real?

2

### Your job: the algorithm's logic

The rule or model, its inputs, and when it buys or sells.

3

### Your job: testing and judgement

Out-of-sample, next-bar, net of costs; read the metrics.

4

### Your job: risk and accountability

Limits, kill switch, approval - you answer for every trade.

AI

### AI's job: the code

AI assistants write, run and fix the Python and Pine; you review it.

AI

### AI's job: the drudgery

Data plumbing, charts, reports - you review what it produces.


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# TSC Alignment — Abilities

1 **A1**
Develop hypotheses for implementing data analytics and computational modelling activities

2 **A2**
Identify appropriate statistical algorithms and data models to test hypotheses and/or theories

3 **A3**
Use appropriate analytics platforms, analytical tools, statistical methods and analytical approaches given specific analytics and reporting requirements

4 **A4**
Conduct statistical modelling of data to derive patterns and solutions

5 **A5**
Perform A/B testing based on different hypotheses


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# TSC Alignment — Abilities (continued)

1. **A6**
   Identify possible big data applications

2. **A7**
   Perform coding and configuration of software agents or programs based on selected models or algorithms

3. **A8**
   Conduct tests on the actions taken and outcomes to assess effectiveness of the models

4. **A9**
   Diagnose unintended outcomes produced by statistical models

5. **A10**
   Draw relevant trends and insights from data analysis to support decisions and report accordingly


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# TSC Alignment — Knowledge

1. **K1**
   Data analytics and modelling business use cases
2. **K2**
   Types of statistical analyses, data models, algorithms and advanced computational methods
3. **K3**
   Range and application of various statistical methods and algorithms
4. **K4**
   Range and application of various types of data models
5. **K5**
   Usage of analytics platforms and tools


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# TSC Alignment — Knowledge (continued)

1. **K6**
   Statistical modelling techniques

2. **K7**
   Coding languages for programming of algorithms and signals

3. **K8**
   Potential reasons for unintended outcomes

4. **K9**
   Forms of data within financial services


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COURSE SCHEDULE

# Lesson Plan — Day 1 (1 of 2)

09:00 - 18:30 · 8 instructional hours · excluding 1-hour lunch and two tea breaks

| | | |
|---|---|---|
| **09:00 - 09:30** | Course Overview, Trainer Introduction and Ground Rules | |
| **09:30 - 10:45** | T1.1-1.4 Quant Success Stories, the Evolution of Quant, the ML-to-LLM Shift, and Agentic AI versus AI Agents | K1, K9 |
| **10:45 - 11:00** | **Tea Break** | |
| **11:00 - 11:30** | T1.5-1.6 Forms of Financial Data, Big Data Applications and Trading Hypotheses | A1, A6, K1, K9 |
| **11:30 - 11:50** | **Lab 1: Pipeline Skeleton, Configuration and the Alpaca Paper Account** | K5 |
| **11:50 - 12:10** | **Lab 2: Data-Validation Stage: Cross-Check Pipeline Bars against TradingView** | A3, K5, K9 |
| **12:10 - 12:30** | **Lab 3: Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality** | A6, K9 |
| **12:30 - 13:30** | **Lunch Break** | |
| **13:30 - 13:45** | **Lab 4: Hypothesis Stage: from a Written Card to an Automated Statistical Test** | A1, A6, K1 |
| **13:45 - 14:30** | T2.1-2.2 Technical Indicators and Standard Algorithmic Trading Strategies | A2, K3 |


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COURSE SCHEDULE

# Lesson Plan — Day 1 (2 of 2)

09:00 - 18:30 · 8 instructional hours · excluding 1-hour lunch and two tea breaks

| | | |
|---|---|---|
| 14:30 - 14:55 | **Lab 5: Indicator Stage: Pine Script v6 Indicators and their Python Twins** | A3, K3, K7 |
| 14:55 - 15:25 | **Lab 6: Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation** | A2, A3, K3, K7 |
| 15:25 - 15:40 | **Tea Break** | |
| 15:40 - 16:05 | **Lab 7: Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals** | A2, K3, K7 |
| 16:05 - 16:50 | T2.3-2.4 Analytics Platforms, Coding Languages and Order Placement | A3, K5, K7 |
| 16:50 - 17:20 | **Lab 8: Execution Stage: an Order Manager that Trades the Paper Account in Any Session** | A3, K5 |
| 17:20 - 17:50 | T2.5 Performance and Risk Metric Definitions | A3, K3 |
| 17:50 - 18:15 | **Lab 9: Evaluation Stage: Performance and Risk Metrics with a Benchmark Table** | A3, K3 |
| 18:15 - 18:30 | Day 1 Recap and Q&A | |


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COURSE SCHEDULE

# Lesson Plan — Day 2 (1 of 2)

09:00 - 18:30 · 8 instructional hours · excluding 1-hour lunch and two tea breaks

| | | |
|---|---|---|
| **09:00 - 09:15** | Day 1 Recap and Day 2 Overview | |
| **09:15 - 09:40** | T3.1-3.2 Statistical Analyses, Algorithms and Supervised Learning Models | A4, K2, K4 |
| **09:40 - 10:00** | **Lab 10: Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay** | A2, A4, K2 |
| **10:00 - 10:20** | **Lab 11: ML-Signal Stage: a Walk-Forward Classifier from Probability to Position** | A4, K4, K6 |
| **10:20 - 10:35** | **Tea Break** | |
| **10:35 - 10:55** | T3.3-3.4 Unsupervised Learning and Statistical Modelling Techniques | A4, K4, K6 |
| **10:55 - 11:10** | **Lab 12: Regime-Filter Stage: PCA and K-Means as a Gate on the Signals** | A4, K4, K6 |
| **11:10 - 11:25** | T3.5-3.6 A/B Testing, and Pairs Trading as a Hedging Algorithm | A4, A5, K2, K6 |
| **11:25 - 11:40** | **Lab 13: A/B-Test Stage: a Statistical Decision before Capital Moves** | A5, K2 |
| **11:40 - 12:00** | **Lab 14: Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve** | A4, K6 |
| **12:00 - 12:30** | T4.1-4.2 Pipeline Architecture and Backtesting with Trading Costs | A7, A8, K7 |
| **12:30 - 13:30** | **Lunch Break** | |


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COURSE SCHEDULE

# Lesson Plan — Day 2 (2 of 2)

09:00 - 18:30 · 8 instructional hours · excluding 1-hour lunch and two tea breaks

| | | |
|---|---|---|
| **13:30 - 13:55** | **Lab 15: Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even** | A7, A8, K7 |
| **13:55 - 14:05** | T4.3 Forward Testing with Automated Paper Trading | A7, A8 |
| **14:05 - 14:30** | **Lab 16: Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading** | A7, A8, K7 |
| **14:30 - 14:50** | **Lab 17: Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution** | A7, K5, K7 |
| **14:50 - 15:00** | T4.4 Potential Reasons for Unintended Outcomes | A9, K8 |
| **15:00 - 15:20** | **Lab 18: Diagnostics Stage: Automated Bias Checks before Go-Live** | A9, K8 |
| **15:20 - 15:35** | **Tea Break** | |
| **15:35 - 15:45** | T4.5 Performance Analysis, Insights and Reporting | A10 |
| **15:45 - 16:10** | **Lab 19: Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report** | A8, A10 |
| **16:10 - 16:30** | **Lab 20: Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval** | A7, A10, K1 |
| **16:30 - 17:15** | **Written Assessment (Short Answer Questions)** | K1 - K9 |
| **17:15 - 18:30** | **Practical Performance Assessment** | A1 - A10 |


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BEFORE THE ASSESSMENT

# Briefing for Assessment

### Open book
You may use your Learner Guide, lab code and your own notes.

### Two components
Written Assessment (45 min) and Practical Performance (75 min).

### Both must be passed
Competent in BOTH is required to be awarded the module.

### All your own work
Collaboration results in a Not Yet Competent outcome.

### Ask if unclear
The assessor can explain a question, but not how to answer it.

### Not Yet Competent?
You will be counselled on the gaps and offered a re-assessment.


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# Final Assessment

## Written Assessment

Short Answer Questions · 45 minutes · open book.

9 open-ended questions — one for each knowledge statement K1 - K9.

No multiple-choice questions.

## Practical Performance

Python + Alpaca paper trading · 75 minutes · open book.

4 practical tasks covering ability statements A1 - A10.

A new trading scenario, using the same pipeline tool-kit as the labs.


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ON ASSESSMENT DAY

# Assessment Flow

1. Sign the assessment attendance record
2. Assessor briefing (not assessed)
3. Sit the WA (45 min) then the PP (75 min) — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

**Both components are open book. You must be Competent in BOTH to be awarded the module.**


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GETTING STARTED

01

# Set Up Alpaca and TradingView

**LEARNING OUTCOME**

Create the free accounts and the Python environment every lab uses. Everything runs on paper — no real money at any point.


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GETTING STARTED

# Download, Install and Start the Trading Simulator Software

Full step-by-step guide: Learner Guide, 'Platform Guides'.

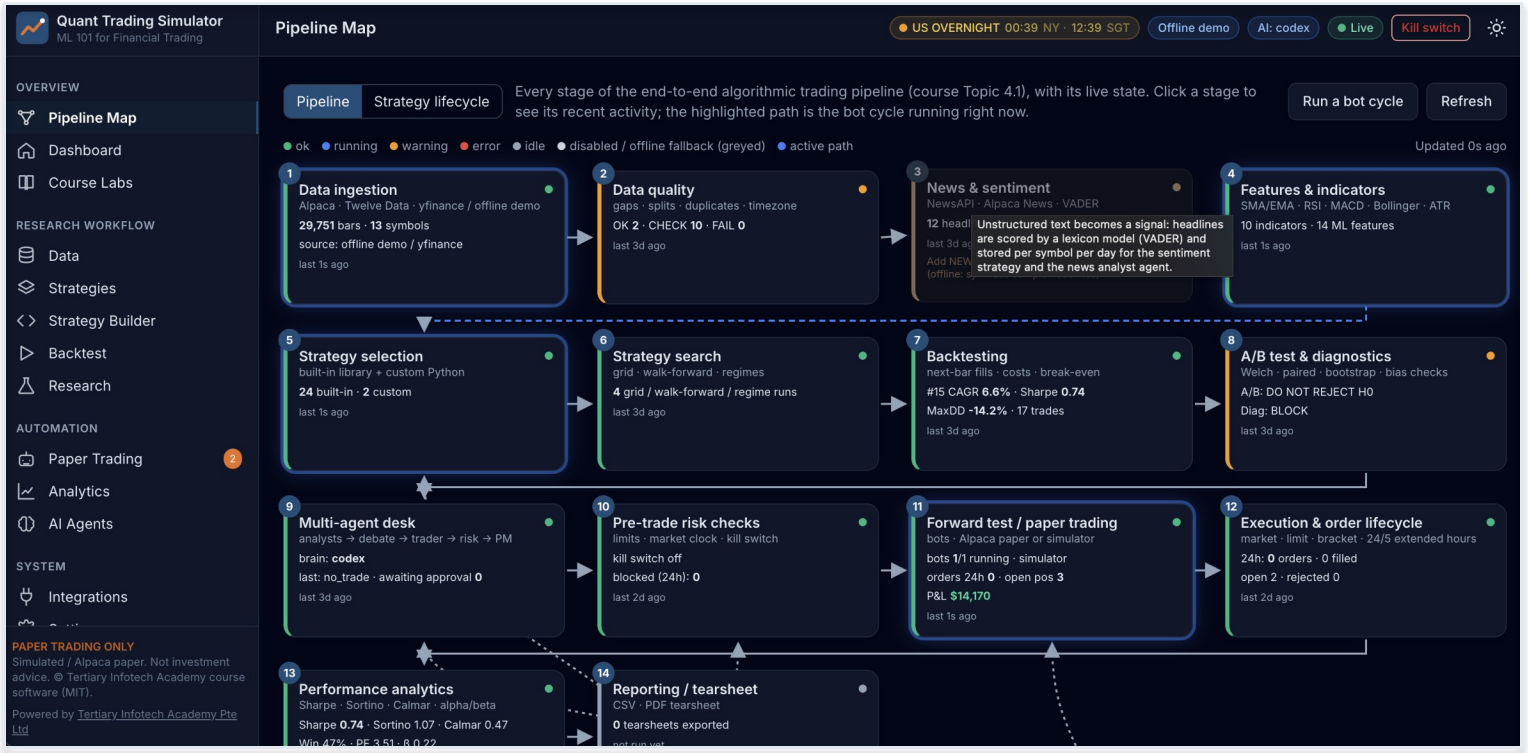
1. Install Git, Python 3.11+ and uv (or Docker)
2. git clone the simulator - or Download ZIP
3. Copy .env.example to .env
4. uv sync > uv run trader seed > uv run trader serve --with-worker
5. Or: docker compose up -d --build > docker compose exec backend trader seed
6. Open http://localhost:8000 > Course Labs

**Simulator: github.com/alfredang/algorithmic-trader-bot - the README has the full start-up steps and troubleshooting.**


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GETTING STARTED

# The Trading Simulator — Pipeline Map




Once the simulator is running, Pipeline Map shows every stage of the automated pipeline and its live state. Click a stage to see its most recent activity.


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GETTING STARTED

# Sign Up for Alpaca and Open the Paper-Trading Account

Full step-by-step guide: Learner Guide, 'Platform Guides'.

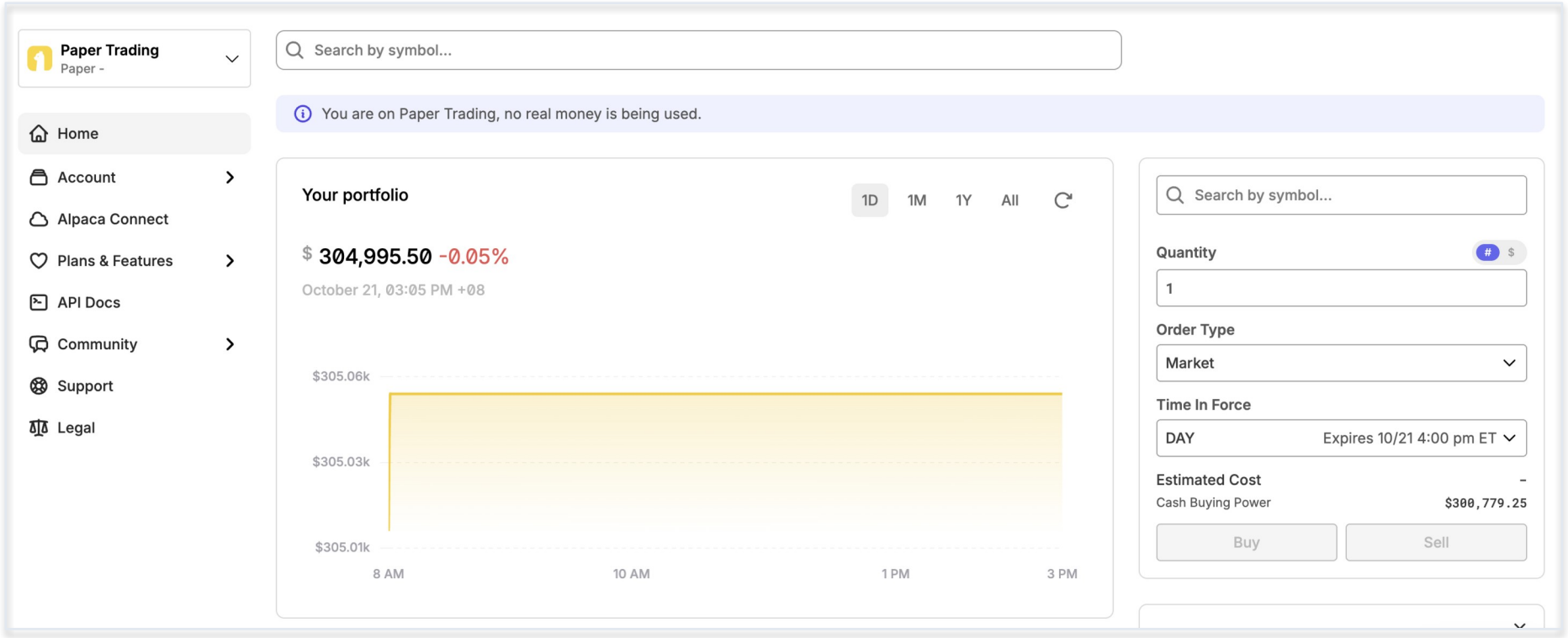
1. Sign up at alpaca.markets and verify your email
2. Log in; choose 'Paper Trading' (upper left)
3. API section: generate key - copy the secret once
4. Paste keys into labs/.env
5. Run python -m pipeline check - must say ONLINE

**No funding needed. Paper keys only work on paper-api.alpaca.markets. No account? Every lab also runs offline.**


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GETTING STARTED

# Your Alpaca Paper-Trading Dashboard

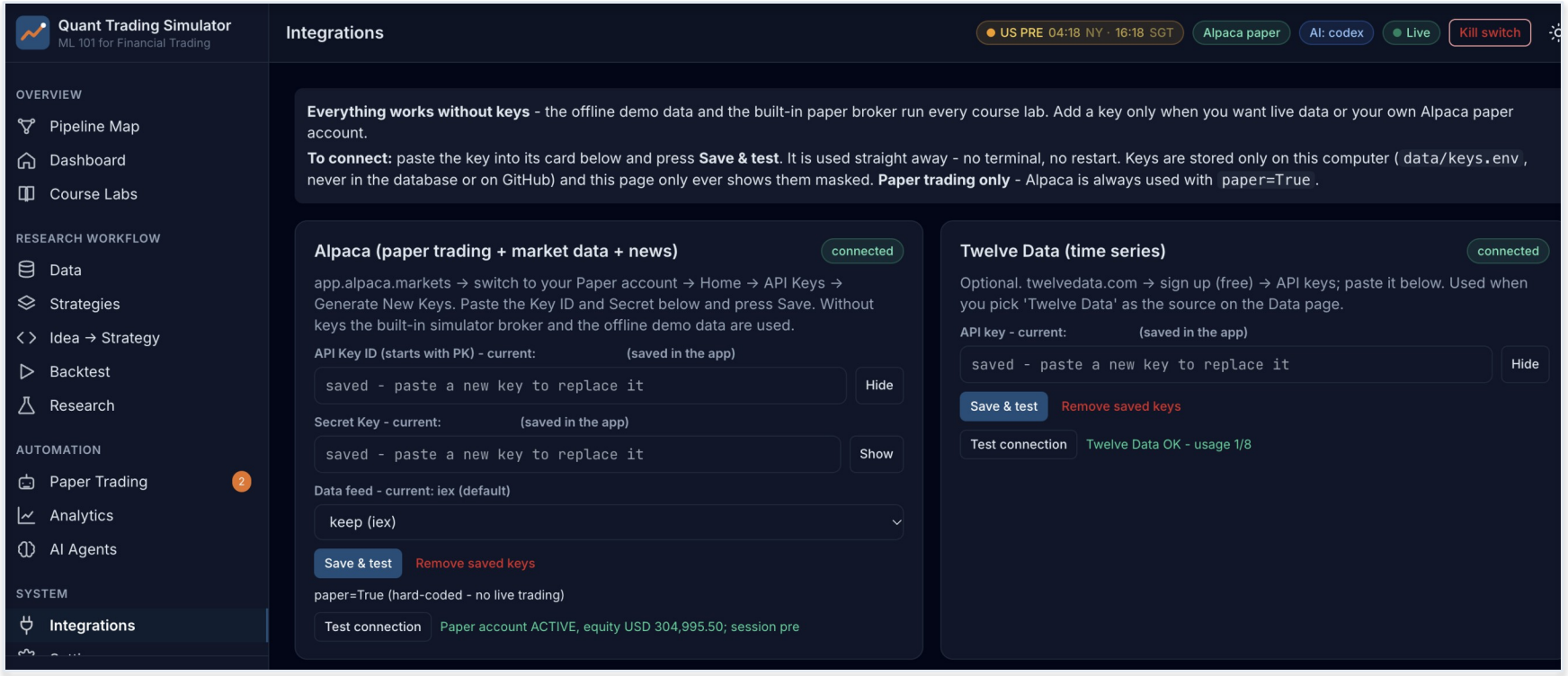



Check that the account selector shows 'Paper'. The banner confirms that no real money is being used. The API keys for the labs are generated from this account.


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GETTING STARTED

# Connect Alpaca to the Simulator — Integrations




Paste the paper Key ID and Secret into the Alpaca card, then press Save & test. Keys stay on your computer and are only shown masked. Without keys, every lab still runs offline.


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GETTING STARTED

# Sign Up for TradingView and Find the Tools You Will Use

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Sign up at tradingview.com (email, Google or Apple)
2. Keep the free plan
3. Products > Supercharts; search NASDAQ:AAPL or SGX:D05
4. Set time zone to Singapore
5. Find Indicators, Pine Editor, Strategy Tester, Trade

**The free plan covers Labs 2, 5 and 6. Webhook alerts in Lab 17 need a paid plan - the lab includes a local simulator.**


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GETTING STARTED

# TradingView — Chart, Watchlist and Symbol Panel




Search for a symbol, pick the interval, and add Indicators. Pine Script and the Strategy Tester open from the panel below the chart, and Trade opens Paper Trading.


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# How the Full Code Labs Work

## Set up once

Open a terminal in labs_fullcode: python3 -m venv .venv, activate it, then pip install -r requirements.txt. Alpaca paper keys go in .env (or leave blank: offline).

## Every lab folder

LabNN_Instructions.pdf (the manual), LabNN_Prompts.pdf (step -> prompt), README.md (how to run), the _stage.py runner and solution/ - the complete code.

## One command per lab

python3 LabNN-.../<stage>_stage.py runs the stage. Prompt_Library.html has every prompt with a Copy button for your AI coding assistant.


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PLATFORM · THE IDEA THAT MATTERS MOST

# Paper first. Always.

Every strategy in this course is tested on history, then forward-tested on a paper account with automatic orders, before anyone would consider real capital. That discipline is the course.


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TOPIC 1

01

# Quant and Algo Trading Foundations

**LEARNING OUTCOME**

Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.

Abilities A1, A6 · Knowledge K1, K9 · Day 1 · Labs 1–4 · Assessed by PP Task 1 and WA Q1, Q9


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# Quant Success Stories

1

### Ed Thorp - Princeton Newport · 1969

First market-neutral quant fund: about 20% a year after fees for two decades by hedging warrants and convertibles

2

### Renaissance - Medallion · 1988

Purely model-driven; about 66% a year before fees, 39% after (1988-2018); closed to outsiders, employee-owned

3

### D.E. Shaw · 1988

Founded by a computer-science professor from Morgan Stanley's quant group; models and software; about US$60-65bn

4

### Man AHL · 1987

Systematic trend following from hand-typed price histories to momentum models across 800+ markets

5

### Two Sigma · 2001

AI, machine learning and distributed computing applied to trading; about US$70bn under management

6

### Jane Street · 2024

Quant market maker: about US$20.5bn net trading revenue; nearly a quarter of primary US ETF activity


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# Spotlight: Renaissance Technologies and the Medallion Fund

1

## Founded by a mathematician

Jim Simons founds Monemetrics in 1978; renamed Renaissance Technologies in 1982

2

## Medallion fund, 1988

The flagship fund trades purely on mathematical models of market data

3

## About 66% a year before fees

Average gross return 1988-2018, about 39% after its high fees; closed to outsiders, employee-owned

4

## The lesson for this course

The edge came from data, rigorous testing, costs and risk control - not from one clever indicator


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# What the Success Stories Have in Common

1

**Hypothesis first**

Each started from a testable idea about prices, not from a tool

2

**Data and testing**

Clean data and out-of-sample tests before capital

3

**Many small edges**

Diversified, market-neutral bets rather than one big call

4

**Costs and risk**

Execution cost, leverage and risk limits decide survival - see LTCM 1998

5

**Citadel Securities**

Quant market maker estimated to handle about a quarter of all US equities trading


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TOPIC 1 · DEFINITION

# Algorithmic trading

Using computer programs that follow predefined rules on price, timing, volume or models to generate and execute orders automatically, with limited or no human intervention.


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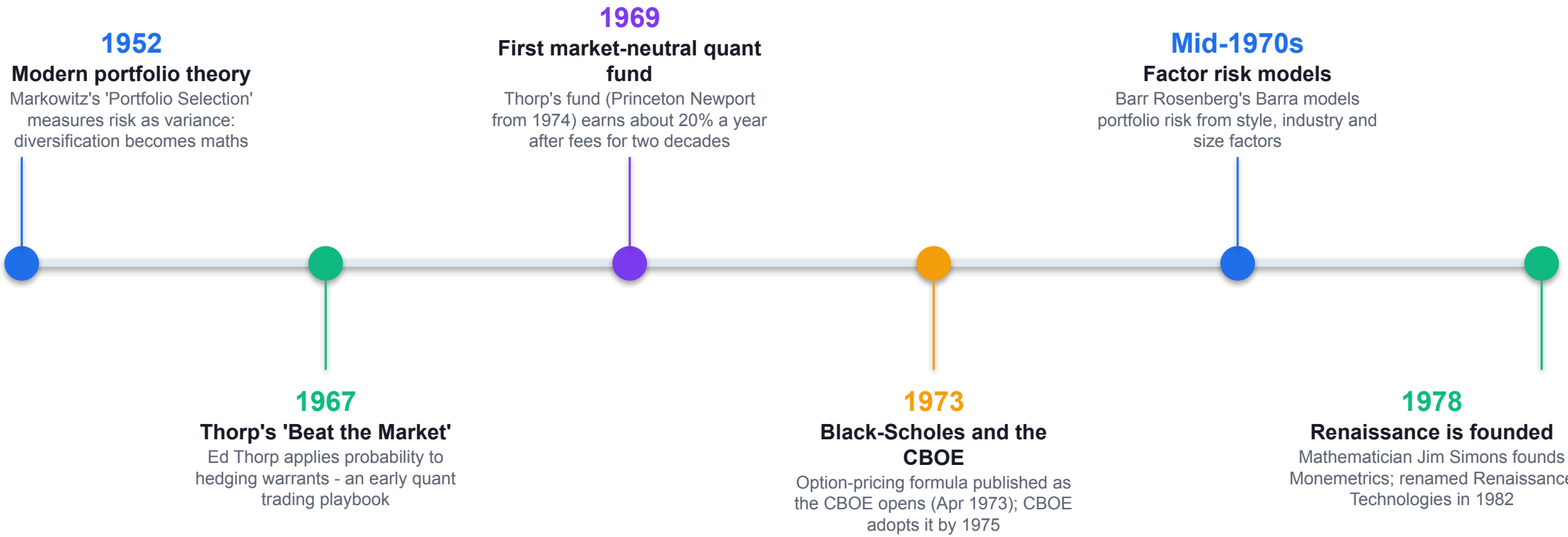
# The Evolution of Quant: From Thorp to HFT to AI

1. 1950s-70s: portfolio theory, arbitrage and option pricing
2. 1970s-80s: factor risk models and program trading
3. 1980s-90s: statistical arbitrage and pairs trading
4. 1988 on: fully systematic signal mining (Renaissance, D.E. Shaw)
5. 2000s: electronic market making and HFT
6. 2010s: speed race - co-location, microwave, microseconds
7. 2020s: ML on alternative data, then LLM and agentic quant

**Each era kept the last era's discipline - hypothesis, data, testing, costs, risk - and added speed or new data.**


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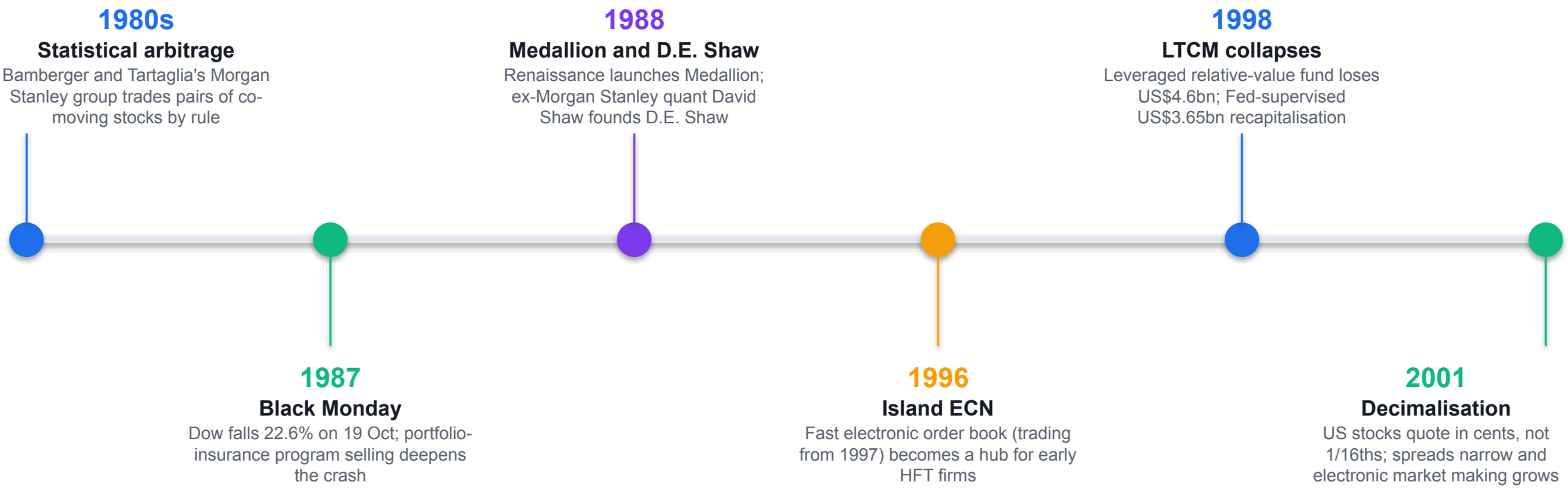
# The Evolution of Quant and Algorithmic Trading (1 of 4)





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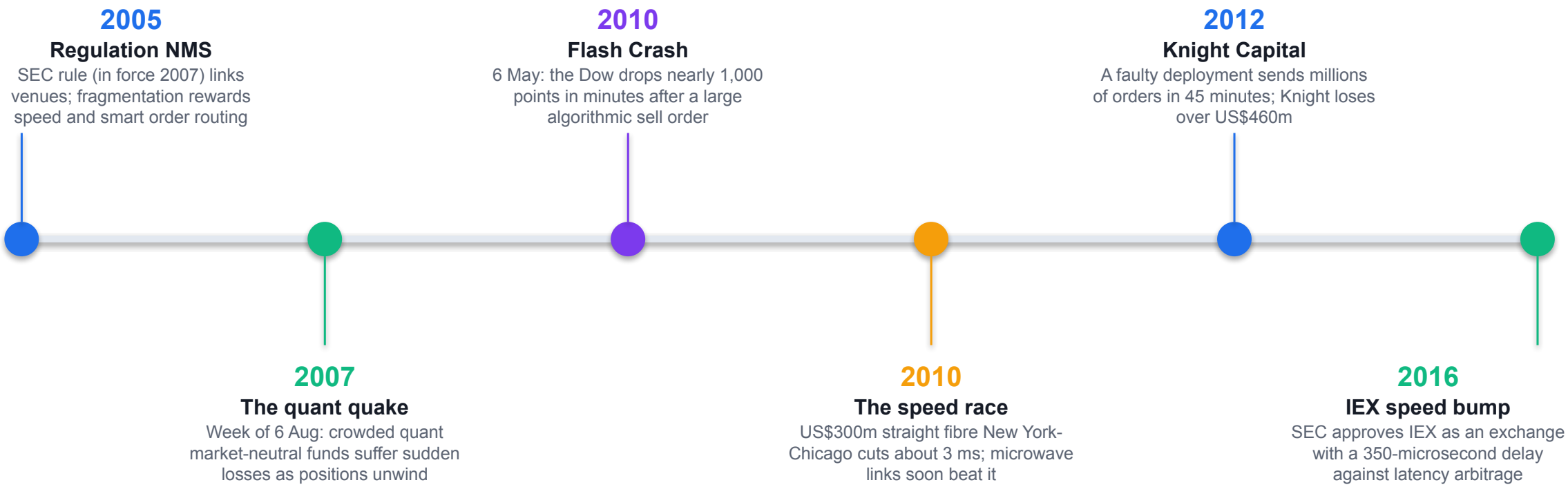
# The Evolution of Quant and Algorithmic Trading (2 of 4)





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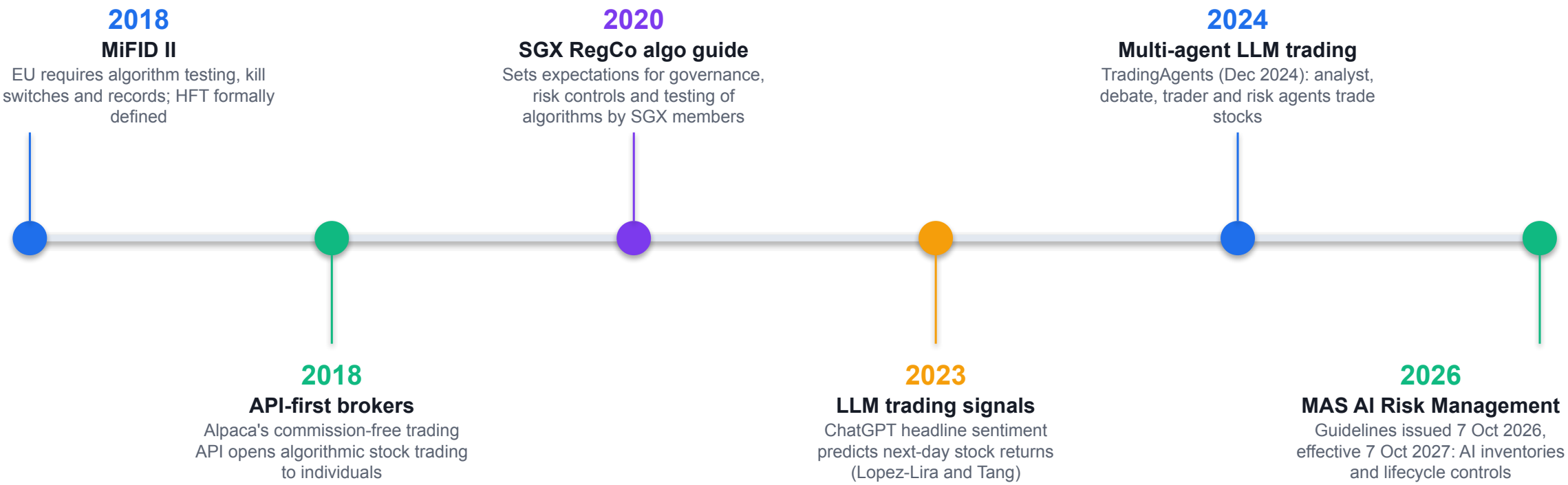
# The Evolution of Quant and Algorithmic Trading (3 of 4)





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# The Evolution of Quant and Algorithmic Trading (4 of 4)





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# Six Paradigm Shifts in Quant and Algorithmic Trading

1

## From intuition to probability · 1950s-1970s

Markowitz, Thorp and Black-Scholes make risk, hedging and option value mathematical

2

## From stocks to factors · 1970s-1980s

Barra factor models and program trading treat portfolios as baskets of risk exposures

3

## From factors to statistical arbitrage · 1980s-2000s

Morgan Stanley's pairs group, D.E. Shaw and Renaissance harvest many small market-neutral edges

4

## From floors to electronic speed · 1996-2010s

ECNs, decimalisation and Reg NMS enable HFT: co-location, microwave links, microseconds

5

## From statistics to machine learning · 2000s-2010s

Models learn non-linear patterns from prices, order books and alternative data

6

## From writing code to directing AI · 2023-present

AI writes and runs the code; quants own the big concepts - hypothesis, algorithm logic, testing, risk


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# Why Algorithms - and What Can Go Wrong

## Benefits

- Speed: reacts and executes in milliseconds
- Discipline: rules remove fear, greed and fatigue
- Backtesting: test ideas on historical data first
- Lower costs: split large orders to reduce market impact
- Scale: monitors many markets and assets around the clock

## Risks

- Technical failure: bugs and outages (Knight Capital, 2012)
- Overfitting: great backtests that fail on live data
- Black swans and regime change the model has never seen
- Feedback loops between algorithms (2010 Flash Crash)
- Set-up, data and compliance costs; no guaranteed profit


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# The Eras of AI in Trading

1

**Symbolic AI · 1956-1980s**

Hand-written rules: expert systems and rule-based program trading; brittle when markets change

2

**Machine learning · 1990s-2000s**

Models learn from data: regression, tree and boosting signals; statistical arbitrage at scale

3

**Deep learning · 2012-2017**

Many-layer networks on GPUs find non-linear patterns in prices, order books and alternative data

4

**Generative AI · 2017-2023**

Transformers and LLMs turn news, filings and earnings calls into trading signals

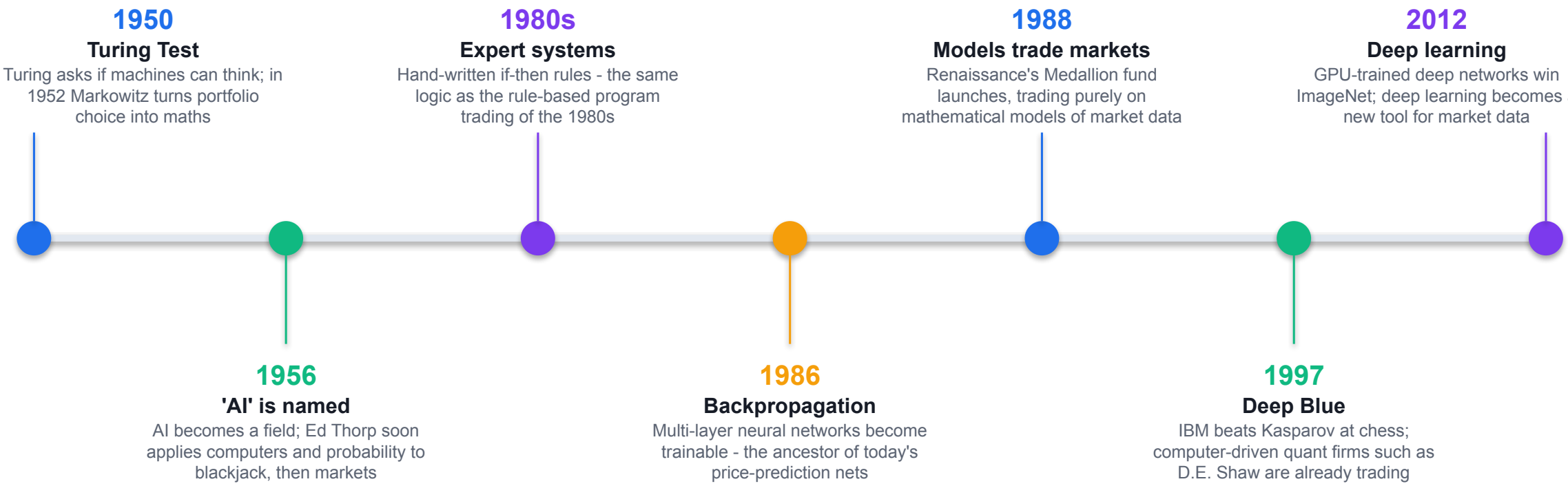
5

**Agentic AI · 2024-present**

LLM agents research, code, backtest and paper-trade strategies under hard risk limits


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# AI Milestones and Their Trading Counterparts (1 of 2)





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# AI Milestones and Their Trading Counterparts (2 of 2)





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# Paradigm Shift: Machine-Learning Quant versus LLM-Era Quant

| | Machine-learning quant | LLM-era quant |
|---|---|---|
| **Main data** | Structured: prices, volumes, fundamentals | Adds unstructured text: news, filings, earnings calls, research |
| **Features** | Hand-engineered by the quant (lags, RSI, factors) | Text read directly; LLM scores sentiment and events |
| **Model building** | Fit regressions, trees, boosting on labelled data | Prompt or fine-tune pretrained LLMs; agents propose alphas |
| **Who writes the code** | The quant, line by line | AI writes it; the quant directs and reviews |
| **Research workflow** | The quant writes every line of code | Coding agents draft code and backtests; the quant reviews |
| **Alpha decay** | Signals fade as more funds trade them | LLM signals fade too as LLM adoption rises |
| **New risks** | Overfitting; look-ahead in features | LLM training-data look-ahead and 'distraction'; homogeneous factors |
| **Human role** | Model builder | Supervisor: hypothesis, testing, risk sign-off |


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# Impact of LLMs on Quant

1

## Text becomes a signal

ChatGPT headline scores predict next-day returns, strongest for small stocks and bad news (Lopez-Lira and Tang)

2

## Finance-specific LLMs

BloombergGPT (2023): 50bn parameters trained with 363bn tokens of Bloomberg financial data

3

## Automated alpha mining

Alpha-GPT turns ideas into formulaic alphas; AlphaAgent targets alpha decay and crowding

4

## Backtests need new controls

LLMs may 'know' in-sample history: test after the knowledge cutoff and anonymise firms (Glasserman and Lin)

5

## Faster crowding

Strategy returns decline as LLM adoption rises - prices become more efficient (Lopez-Lira and Tang)

6

## Agents still need limits

Agents propose trades; hard-coded risk checks and humans approve them


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# Agentic AI versus AI Agents in Trading

## Agentic AI - the capability

The capability of AI to plan and act: it calls tools, uses skills and memory, and follows instructions to complete multi-step work when it is asked.

Trading example: On request, an LLM fetches a stock's bars and news through tools, checks its memory of past trades and proposes an order within its instructions

## AI Agent - always on, 24/7

A special class of agentic AI that stands by 24/7 and works all the time: always on, woken by schedules, events and messages, with its own long-term memory.

Examples: OpenClaw, Hermes Agent, OpenAI Dots.

Trading example: A round-the-clock trading agent that watches prices and news overnight and proposes paper trades within risk limits


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# AI Agents vs Agentic AI - Side by Side

| | AI Agents | Agentic AI |
|---|---|---|
| **What it is** | A special class of agentic AI that is always on | Capability to plan and act with tools, skills, memory |
| **Runtime** | Persistent: stands by 24/7 on a server or PC | Runs when invoked: a chat, workflow or pipeline |
| **Trigger** | Schedules, events, messages - unprompted | A user prompt or a pipeline step |
| **Memory** | Long-term memory across days and tasks | Session or task memory; can use long-term memory |
| **Building blocks** | Agentic AI + always-on runtime and scheduler | LLM + tools + skills + memory + instructions |
| **Examples** | OpenClaw, Hermes Agent, OpenAI Dots | Claude Code, ChatGPT with tools, TradingAgents |
| **Trading example** | 24/7 market and news monitor proposing trades | On-demand multi-agent analysis before a trade |
| **Main risks** | Acting unsupervised overnight; runaway loops | Hallucination; prompt injection; tool misuse |


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# From Chatbot to Always-On AI Agent

1

1 · Chatbot: Answers one prompt at a time; no tools, no actions

2

2 · Agentic AI: Plans and calls tools, skills, memory and instructions when asked

3

3 · Multi-agent agentic system: Specialist agents coordinate: analysts, debate, trader, risk manager

4

4 · AI agent (always on): Stands by 24/7, wakes on schedules and events, works continuously within limits

**An AI agent is agentic AI that never switches off - so its limits, kill switch and monitoring matter most.**


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# Roles in a Multi-Agent Trading System

1

## Fundamental Analyst

Reads filings, earnings and balance sheets to judge intrinsic value

2

## Technical Analyst

Reads price, volume and indicators such as MACD and RSI

3

## Sentiment Analyst

Gauges mood from social media, analyst ratings and options flow

4

## News Analyst

Tracks news wires, press releases and macro events

5

## Bull / Bear Researchers

Argue for and against the trade so weak ideas are challenged

6

## Trader

Combines the reports into a buy, sell or hold call with timing and size

7

## Risk Manager

Checks volatility, liquidity, exposure and drawdown limits

8

## Portfolio Manager

Approves or rejects the proposal and fits it into the whole portfolio


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# Stock-Trading Agent Frameworks (1 of 2)

1

**TradingAgents (Xiao et al., 2024)**

Simulated stock-trading firm: fundamental, sentiment, news and technical analysts, bull/bear debate, trader, risk team

2

**FinMem (Yu et al., 2023)**

LLM trading agent with profiling and layered memory making Buy / Sell / Hold decisions on individual stocks

3

**FinAgent (Zhang et al., 2024)**

Multimodal agent reading prices, news and chart images, with dual-level reflection; tested on stocks and crypto

4

**StockAgent (2024)**

LLM investor agents trade a simulated stock market under macro, policy and event shocks; avoids test-set leakage


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# Stock-Trading Agent Frameworks (2 of 2)

1

**QuantAgent (2025)**

Price-driven multi-agent system for high-frequency trading: Indicator, Pattern, Trend and Risk agents

2

**AI Hedge Fund (open source)**

Investor-persona agents (e.g. Warren Buffett), a risk manager and a portfolio manager decide trades for chosen tickers

3

**Agentic Trading Lab (Open-Finance-Lab)**

Agent-run stock portfolios with backtests and paper trading through the Alpaca API


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# Risks and Controls for AI Agents in Trading

| | What goes wrong | Control |
|---|---|---|
| **Hallucination** | An LLM states a wrong earnings figure or invents a signal with confidence | Ground answers in tool or RAG data, log the reasoning, review unusual calls |
| **Prompt injection** | Malicious text in news, web pages or files hijacks the agent's instructions | Input and output guardrails, least-privilege API keys, no secrets in prompts |
| **Overtrading and runaway loops** | Agents trade too often or amplify each other, compounding losses | Order and position caps, daily-loss circuit breaker, kill switch |
| **Model and backtest risk** | Overfitting, look-ahead bias and regime change: strong backtests, weak live results | Walk-forward tests, paper trade first, monitor drift and costs |
| **Accountability** | Unclear who answers for an autonomous agent's trade or a vendor's AI | The firm stays accountable: human sign-off, audit trails, clear owners |
| **Regulation (MAS)** | FEAT principles (2018) and MAS AI Risk Management Guidelines (issued Oct 2026) | Board oversight, AI inventory, materiality assessment and lifecycle controls |


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# Forms of Data within Financial Services

1

## Market data

OHLCV bars, quotes, trades, order-book depth — Alpaca serves bars, quotes, trades, news.

2

## Reference & corporate actions

Symbols, calendars, splits, dividends — unadjusted splits fake a crash.

3

## Fundamental & macro

Statements, ratios, earnings; rates, inflation, central-bank decisions.

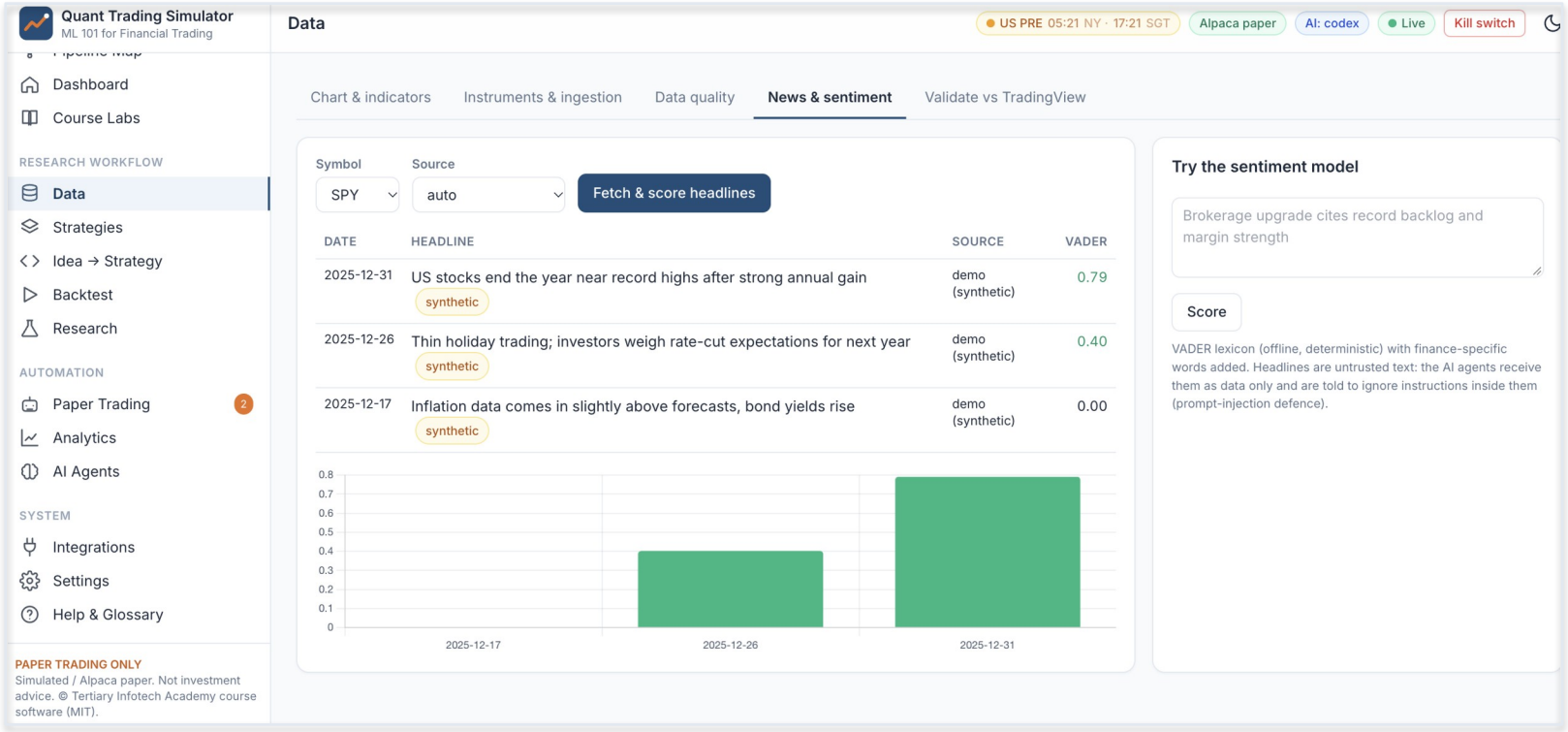
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## Alternative data

News and social sentiment, web traffic, satellite, card spend, ESG.


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# Unstructured Data in Practice — Headlines and Sentiment




Data > News & sentiment: headlines (unstructured text) are scored by a lexicon model and stored as one sentiment number per symbol per day - structured data a strategy can use.


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# Big Data in Trading — the Four Vs

1

## Volume

Tick data runs to billions of rows per year for one exchange.

2

## Velocity

Streaming quotes and news arrive in milliseconds.

3

## Variety

Prices, text, images, transcripts — structured and unstructured.

4

## Veracity

Bad ticks, survivorship, restatements — verify before you trust.


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# Big Data Applications

1

## Execution analytics

Tick and quote data to measure and cut slippage.

2

## Sentiment and event trading

News and social text scored by NLP and LLMs.

3

## Regime detection

Cross-asset returns clustered into market states.

4

## Surveillance and risk

Order and trade data scanned for market abuse; intraday risk.


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K1

# Data Analytics and Modelling Use Cases

1

## Signal generation

Find repeatable return — judged by out-of-sample, net Sharpe.

2

## Execution

VWAP / TWAP / shortfall — judged by slippage in basis points.

3

## Risk forecasting & sizing

Volatility and VaR models — judged by limit breaches.

4

## Research at scale

LLMs read filings and news — judged by accuracy and hours saved.


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# What Makes a Hypothesis Testable

1

**Instrument and universe**

e.g. SPY, daily bars, 2018 – 2025.

2

**Signal and holding period**

e.g. three down closes in a row; hold 5 days.

3

**Expected effect and benchmark**

Forward return exceeds the unconditional 5-day return.

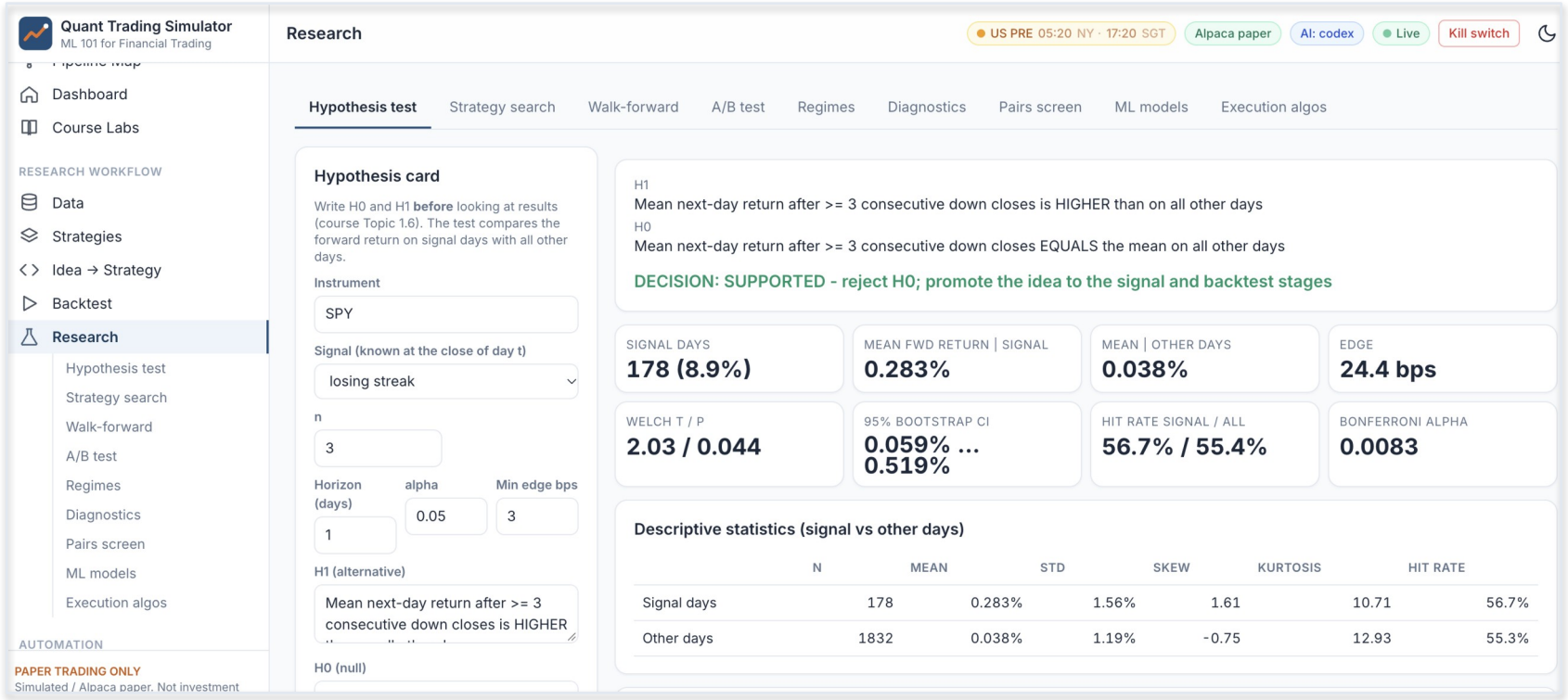
4

**Written BEFORE the test**

H0, H1 and the significance level chosen first.


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# A Testable Hypothesis in Practice




Research > Hypothesis test: H1 and H0 are written before the result, then tested with a Welch t-test, a bootstrap confidence interval and a minimum edge (Lab 4).


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# Machine Learning Implementation Steps

1. Frame the problem and hypothesis
2. Collect and clean the data
3. Define target and features
4. Split by time; train and validate
5. Test out of sample
6. Paper trade, then review

**A model that skips the out-of-sample test and the paper trade has not been tested.**


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# Pipeline Skeleton, Configuration and the Alpaca Paper Account

## LAB 01

Create the pipeline package and its config.yaml (universe, strategy parameters, risk limits, costs, secret names), build a validating configuration loader and a market-clock module, and run `python -m pipeline check` against the Alpaca paper account or the offline simulator.

### WORKFLOW

1. Set up Python and your AI coding assistant
2. Store Alpaca PAPER keys outside the code
3. Write the pipeline configuration
4. Build the configuration loader with safety validation
5. Add the market clock for US and Singapore time
6. +3 more — see the Learner Guide

### YOU'LL PRODUCE

**The pipeline skeleton: config.yaml, risk limits and a verified Alpaca paper account (or simulator).**

Duration
**20 min**


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A3 K5 K9

# Data-Validation Stage: Cross-Check Pipeline Bars against TradingView

**LAB 02**

Read daily bars for SGX:D05 and NASDAQ:AAPL from the TradingView Data Window, record them, and build a validation stage that compares TradingView with the pipeline's raw and dividend-adjusted bars and writes a validation report.

## WORKFLOW

1. Add the data-check panel to SGX:D05
2. Repeat for NASDAQ:AAPL and switch the adjustment
3. Record what TradingView shows
4. Build the validation stage
5. Run the validation stage
6. +3 more — see the Learner Guide

## YOU'LL PRODUCE

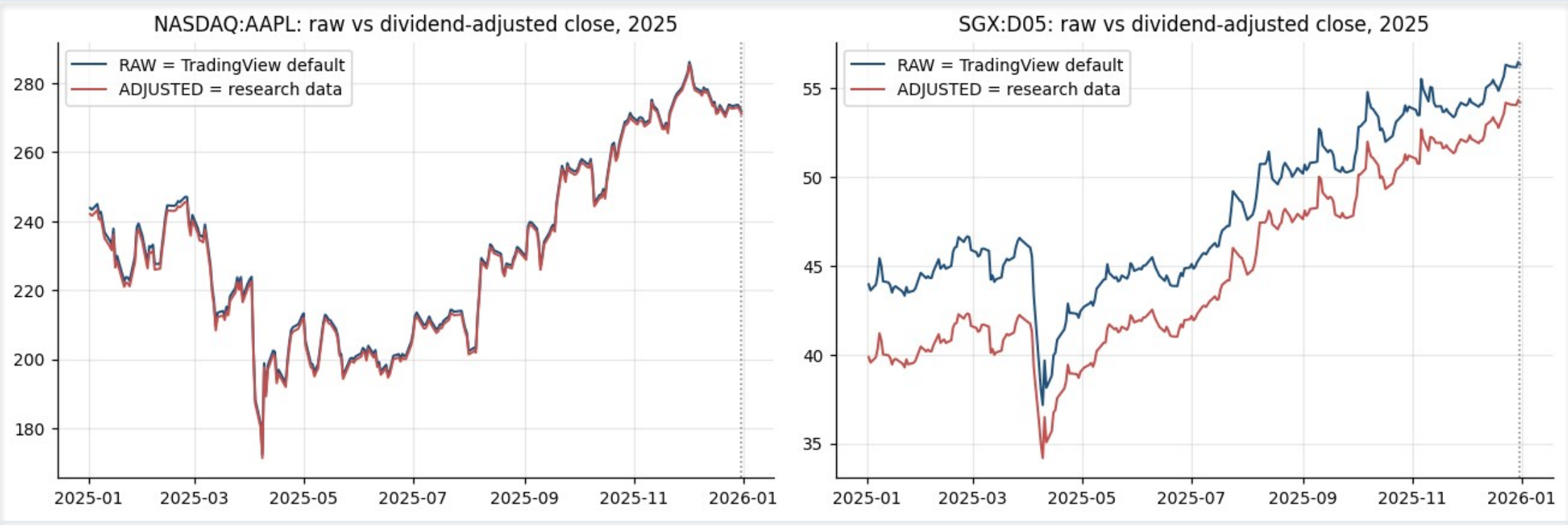
**A data-validation report reconciling pipeline bars with TradingView for SGX:D05 and AAPL.**

Duration
**20 min**


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LAB 02 · OUTPUT

# Lab 02 — What You Should See




Output of Lab02-tradingview-charts in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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# Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality

## LAB 03

Build `python -m pipeline ingest` to fetch and cache daily, weekly, monthly and hourly bars for the universe, run automated data-quality checks, ingest headlines as unstructured data, and describe the forms of data and the 3 Vs of big data.

### WORKFLOW

1. Understand how bars are requested
2. Build the cache and load_bars()
3. Add automated data-quality checks
4. Ingest an unstructured source: news headlines
5. Build the ingest run with the report
6. +3 more — see the Learner Guide

### YOU'LL PRODUCE

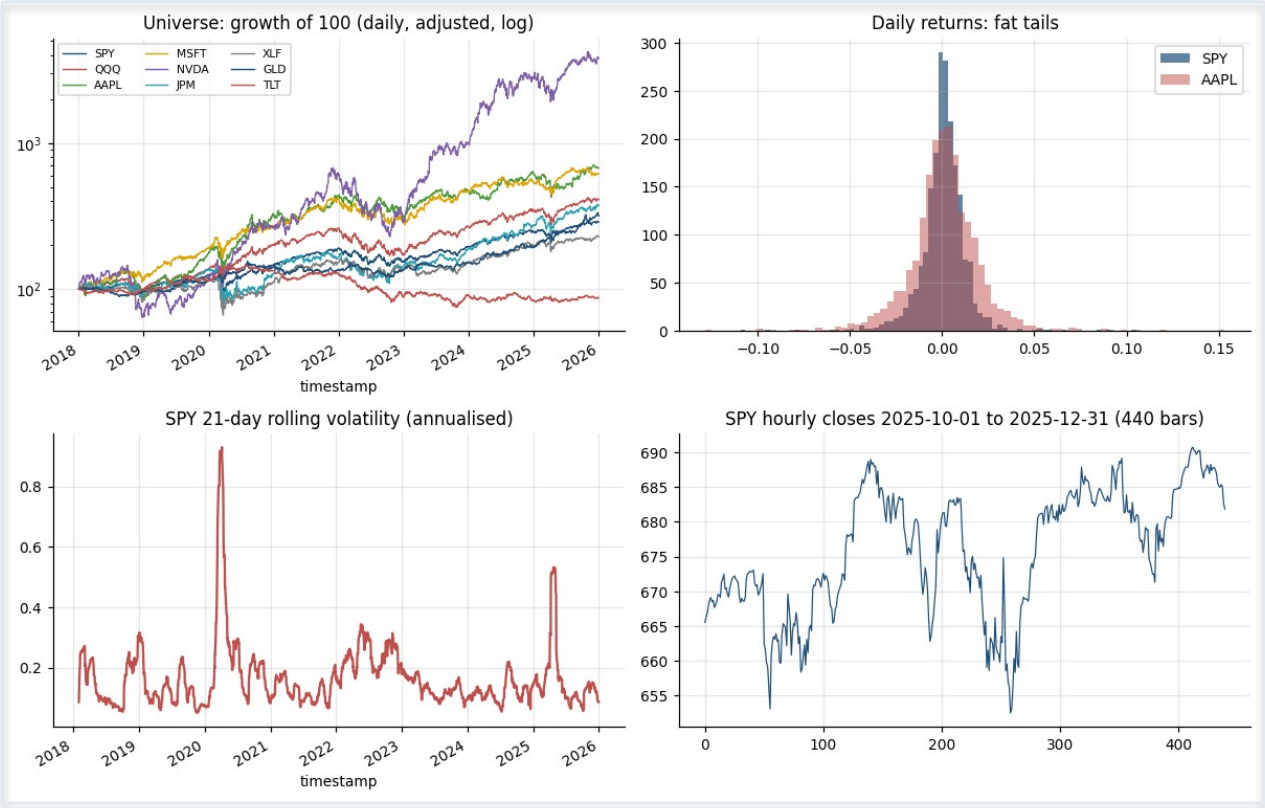
**An ingestion stage caching multi-timeframe bars and headlines with automated data-quality checks.**

Duration
**20 min**


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LAB 03 · OUTPUT

# Lab 03 — What You Should See




Output of Lab03-alpaca-market-data in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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# Lunch Break

1 hour · see you at 13:30


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A1 A6 K1

# Hypothesis Stage: from a Written Card to an Automated Statistical Test

## LAB 04

Write a falsifiable hypothesis card in YAML (H1, H0, signal known at the close, outcome, test, decision rule, big-data use case) and build `python -m pipeline hypothesis` to test it with a Welch t-test, a bootstrap confidence interval and a robustness grid.

### WORKFLOW

1. Read the hypothesis card
2. Write a card for your own idea
3. Build the test engine
4. Add the robustness grid, chart and result file
5. Run the test on the desk's card
6. +3 more — see the Learner Guide

### YOU'LL PRODUCE

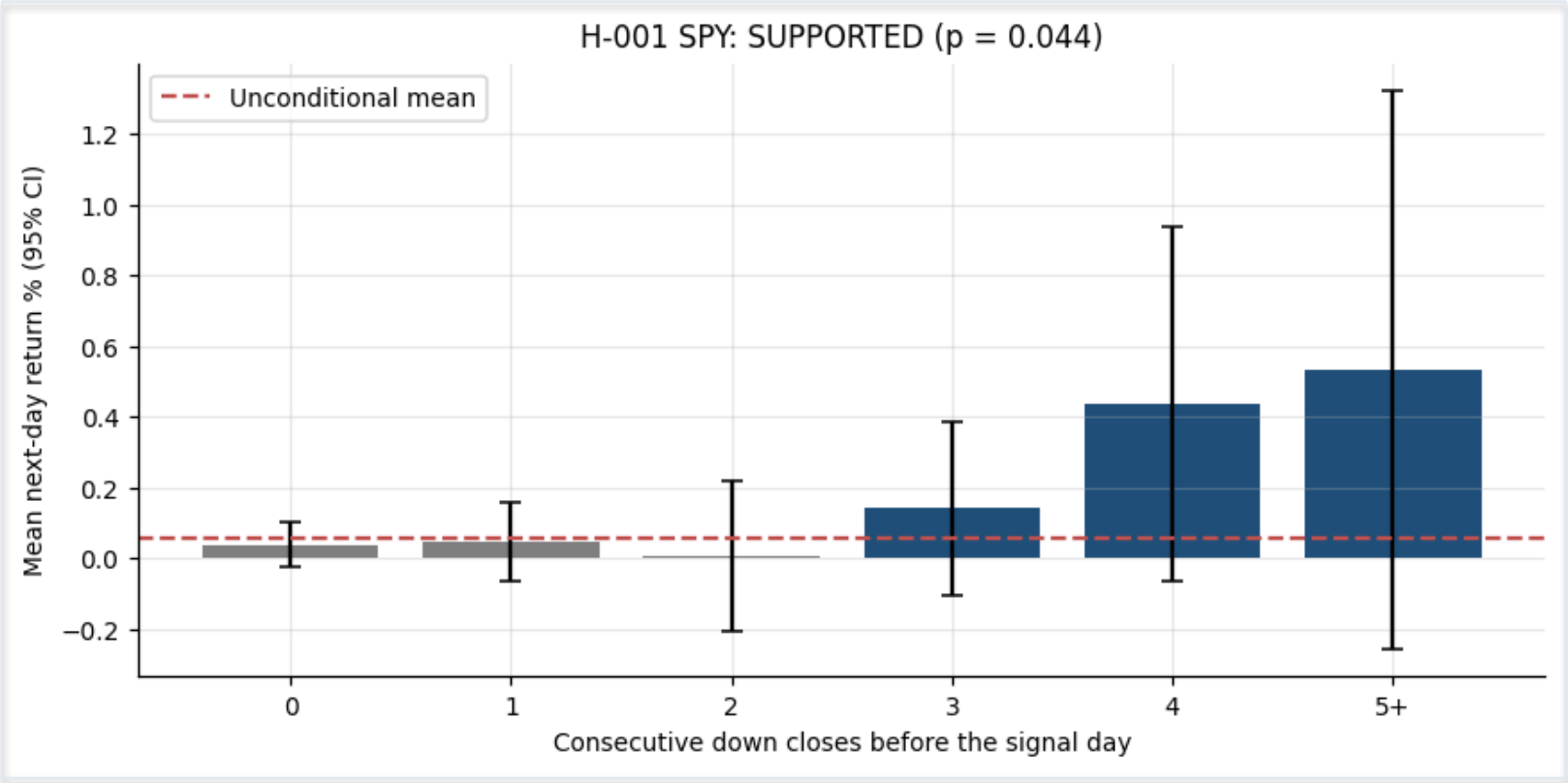
**A hypothesis card and an automated statistical test of the SPY dip-after-streak hypothesis.**

Duration
**15 min**


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LAB 04 · OUTPUT

# Lab 04 — What You Should See




Output of Lab04-trading-hypothesis in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A1 A6 K1 K9

# Topic 1 Recap

1

**A1**
Develop hypotheses for implementing data analytics and computational modelling activities

2

**A6**
Identify possible big data applications

3

**K1**
Data analytics and modelling business use cases

4

**K9**
Forms of data within financial services


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TOPIC 2

02

# Indicators, Strategies and Metrics

**LEARNING OUTCOME**

Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.

Abilities A2, A3 · Knowledge K3, K5, K7 · Day 1 · Labs 5–9 · Assessed by PP Task 2 and WA Q3, Q5, Q7


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K3

# Trend and Momentum Indicators

1

## SMA / EMA

Average close over n bars; EMA weights recent bars by 2/(n+1).

2

## MACD

EMA12 − EMA26 with a 9-EMA signal line; crosses flag momentum.

3

## RSI (14)

100 − 100/(1+RS); above 70 overbought, below 30 oversold.

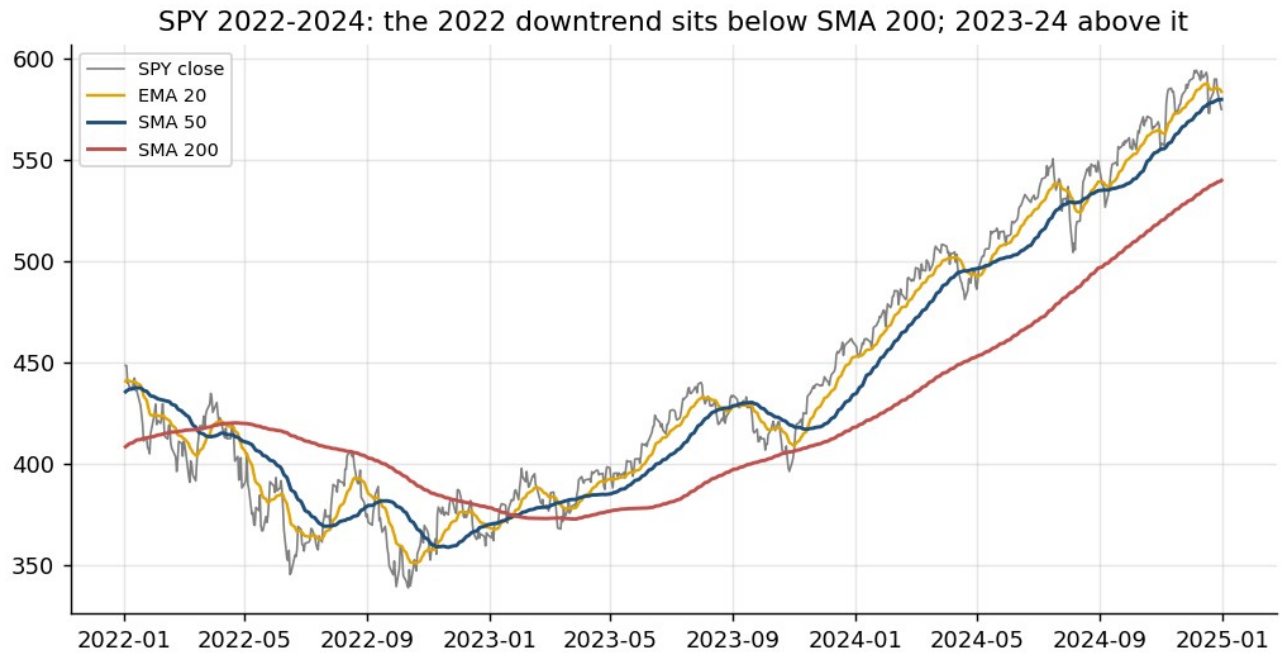
4

## Donchian channel

Highest high / lowest low of n bars — the breakout level.


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# Indicator: Moving Averages (SMA and EMA)




**DEFINITION**

The average closing price over the last n bars - a smoothed line that shows the trend. The EMA weights recent bars more, so it turns sooner.

**FORMULA**

SMA(n) = (C1 + C2 + ... + Cn) / n. EMA(t) = a x C(t) + (1 - a) x EMA(t-1), with a = 2 / (n + 1).

**HOW TO READ IT**

Price above a rising average = uptrend. The 50 / 200-day pair is the classic trend filter; EMA 20 tracks the short-term trend.

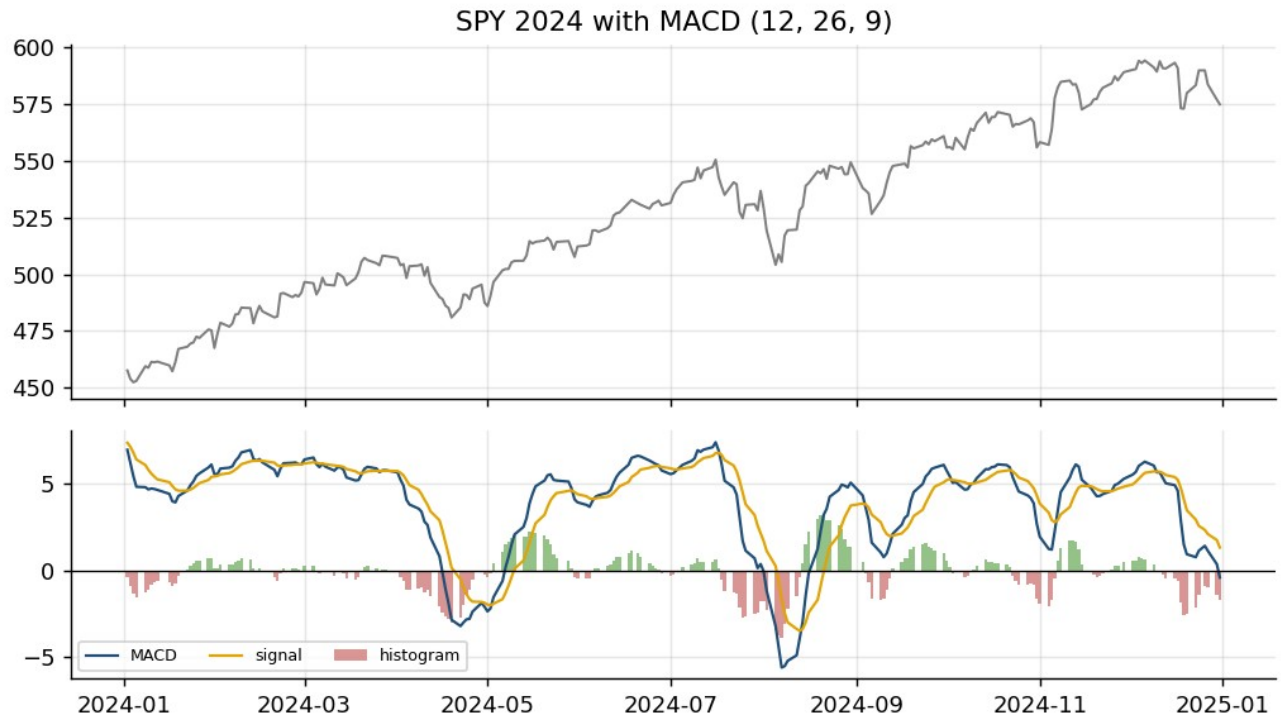
**INSIGHT**

Every average lags: the longer n, the smoother - and the later each signal. In sideways markets price whipsaws across it.


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K3

# Indicator: MACD (Moving Average Convergence Divergence)




**DEFINITION**

A momentum-of-trend gauge: the gap between a fast and a slow EMA, with a signal line and a histogram that show whether momentum is growing.

**FORMULA**

MACD = EMA12(C) - EMA26(C). Signal = EMA9(MACD). Histogram = MACD - Signal.

**HOW TO READ IT**

MACD crossing above its signal line = momentum turning up; MACD above zero = the 12-day trend is above the 26-day trend.

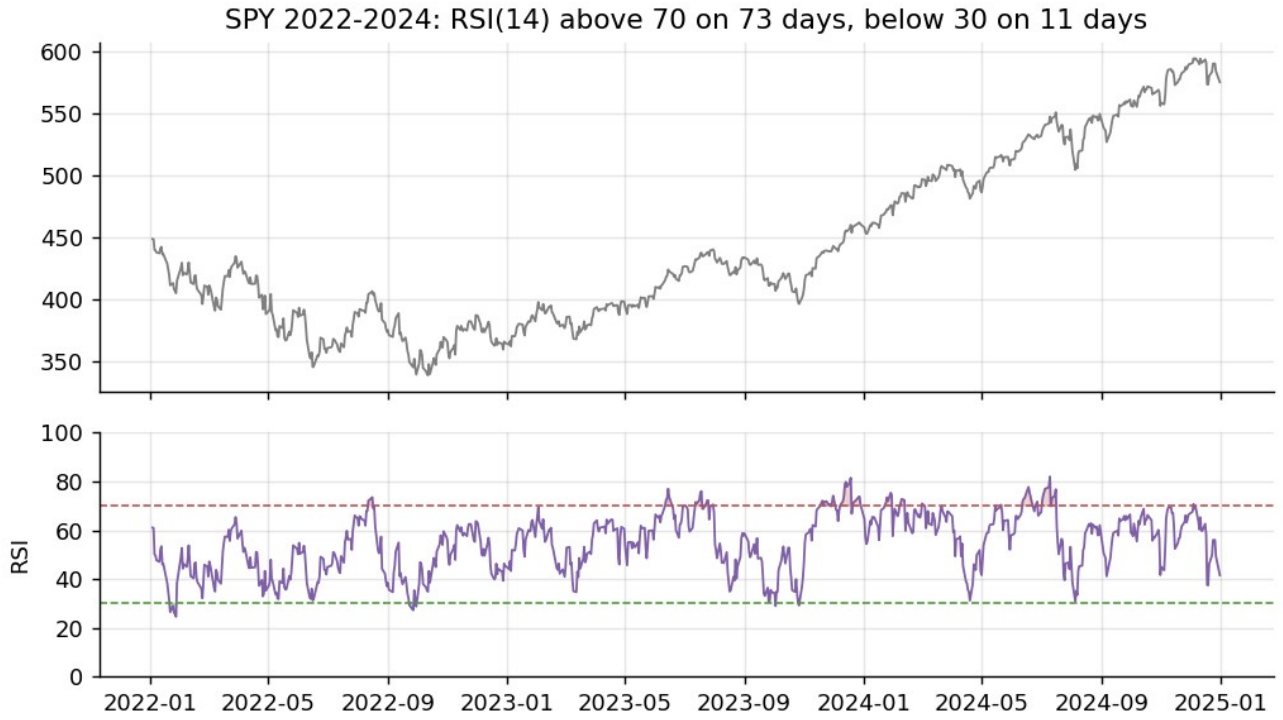
**INSIGHT**

A shrinking histogram warns that a move is losing strength before price turns - but it is built from EMAs, so it lags and whipsaws in ranges.


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K3

# Indicator: RSI (Relative Strength Index)




**DEFINITION**

An oscillator from 0 to 100 that compares the size of recent gains with recent losses - how stretched the latest move is.

**FORMULA**

RS = average gain / average loss over n bars (Wilder smoothing, n = 14). RSI = 100 - 100 / (1 + RS).

**HOW TO READ IT**

Above 70 = overbought, below 30 = oversold. Short look-backs such as RSI(2) are used for mean-reversion entries.

**INSIGHT**

In a strong trend RSI can stay above 70 for weeks: overbought is not a sell signal by itself. Test the thresholds - they are conventions.


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# RSI(N): the Number Is the Look-back Period

| | RSI(2) - short look-back | RSI(14) - Wilder's standard (1978) |
|---|---|---|
| **Look-back** | The last 2 bars (days on a daily chart) | The last 14 bars - about 3 trading weeks |
| **Reacts to** | 1-2 day moves: fast and jumpy | Sustained moves: smooth and slower |
| **How often extreme** | Often: SPY RSI(2) < 10 on 9.6% of days | Rarely: SPY RSI(14) < 30 on 1.7% of days |
| **Typical levels** | Buy below 10, exit above 70 (simulator default) | Buy below 30, sell above 70 |
| **SPY, 10 Oct 2025 (-2.7% after a -0.3% day)** | 5.8 - deeply oversold: the RSI(2) rule buys | 42.5 - neutral: no signal |
| **Next trading day, 13 Oct** | SPY +1.5% - the bounce is captured | No trade |
| **Simulator strategy** | RSI(2) mean reversion (Connors style) | RSI oversold / overbought |


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# The Number in Brackets Is Always the Look-back

1

## RSI(N) formula

RS = average gain / average loss over the last N bars; RSI = 100 - 100 / (1 + RS). Mostly up days: near 100. Mostly down: near 0. Equal: 50.

2

## SMA(50) and EMA(20)

SMA(50): average of the last 50 closes. EMA(20): exponential average over about 20 bars, weighting recent bars more.

3

## Bollinger(20, 2) and ATR(14)

Bollinger: 20-day average with bands 2 standard deviations away. ATR(14): average true range over 14 bars.

4

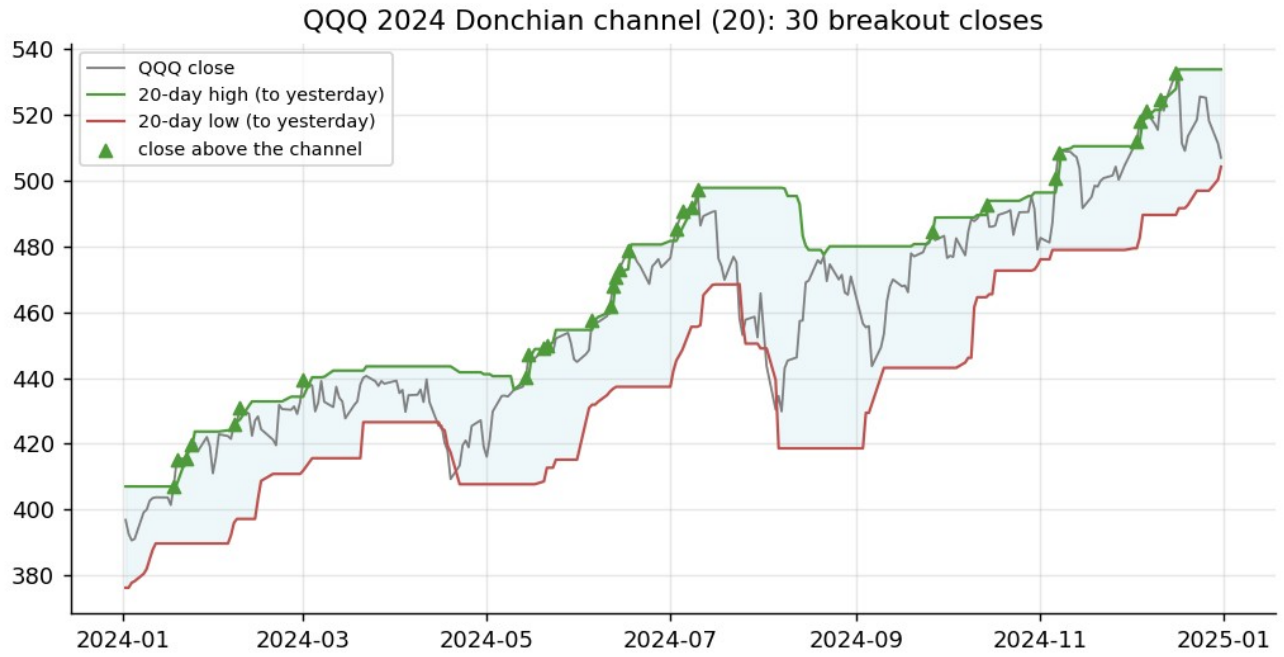
## Where to change N

TradingView: the indicator's Settings > Inputs > Length. Simulator: the strategy's parameter (e.g. RSI length n) before you backtest.


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K3

# Indicator: Donchian Channel




**DEFINITION**

A channel drawn through the highest high and the lowest low of the last n bars - the levels a breakout must clear.

**FORMULA**

Upper(n) = max(High, last n bars). Lower(n) = min(Low, last n bars). Classic n = 20 (the Turtle traders used 20 and 55).

**HOW TO READ IT**

A close above the prior upper band = upside breakout; exit on a break of a shorter (e.g. 10-day) low.

**INSIGHT**

Breakout systems win rarely but big. Compare today's close with the channel up to YESTERDAY - including today's high is look-ahead.


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# Volatility and Volume Indicators

1

## Bollinger Bands

SMA20 ± 2 standard deviations; width tracks volatility.

2

## ATR (14)

Average true range — position sizing and volatility stops.

3

## VWAP

Volume-weighted average price — the execution benchmark.

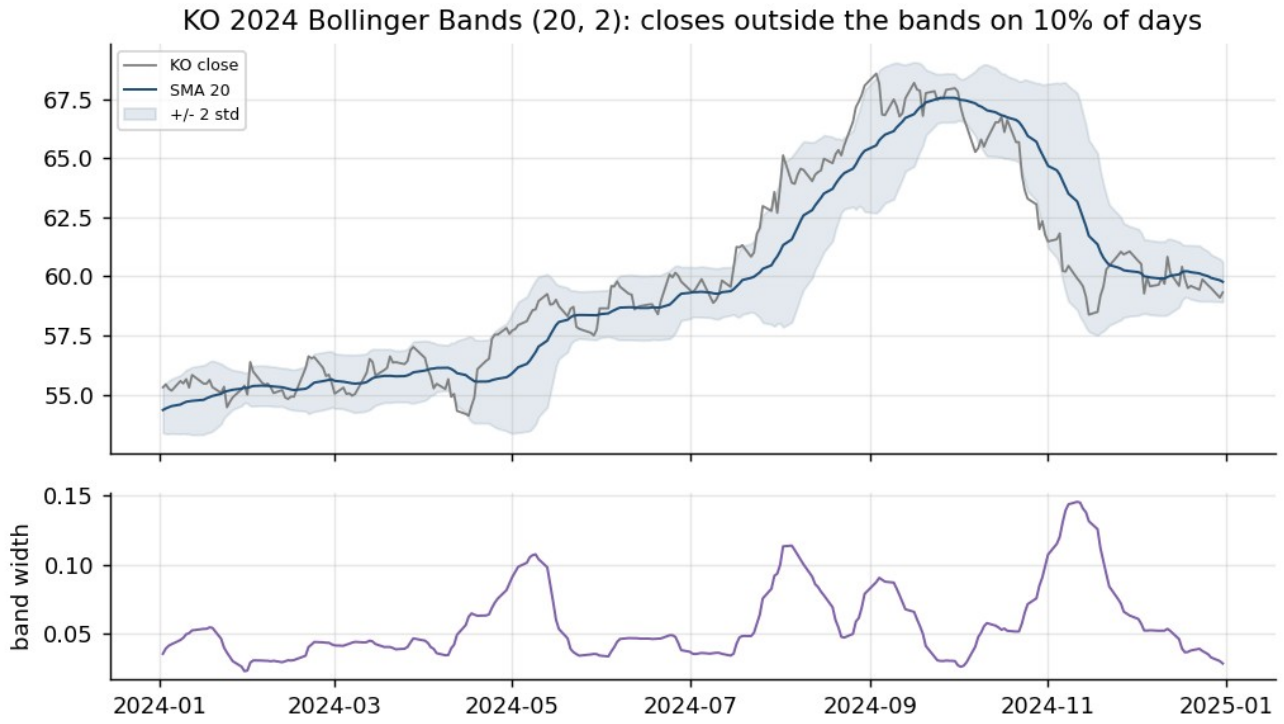
4

## OBV

Adds volume on up days, subtracts on down days.


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# Indicator: Bollinger Bands




**DEFINITION**

A moving average with bands set a number of standard deviations above and below it, so the bands widen and narrow with volatility.

**FORMULA**

Middle = SMA20(C). Upper / Lower = Middle +/- 2 x StDev20(C). %B = (C - Lower) / (Upper - Lower).

**HOW TO READ IT**

Mean reversion: buy a close below the lower band, exit at the middle. A 'squeeze' (narrow bands) often comes before a big move.

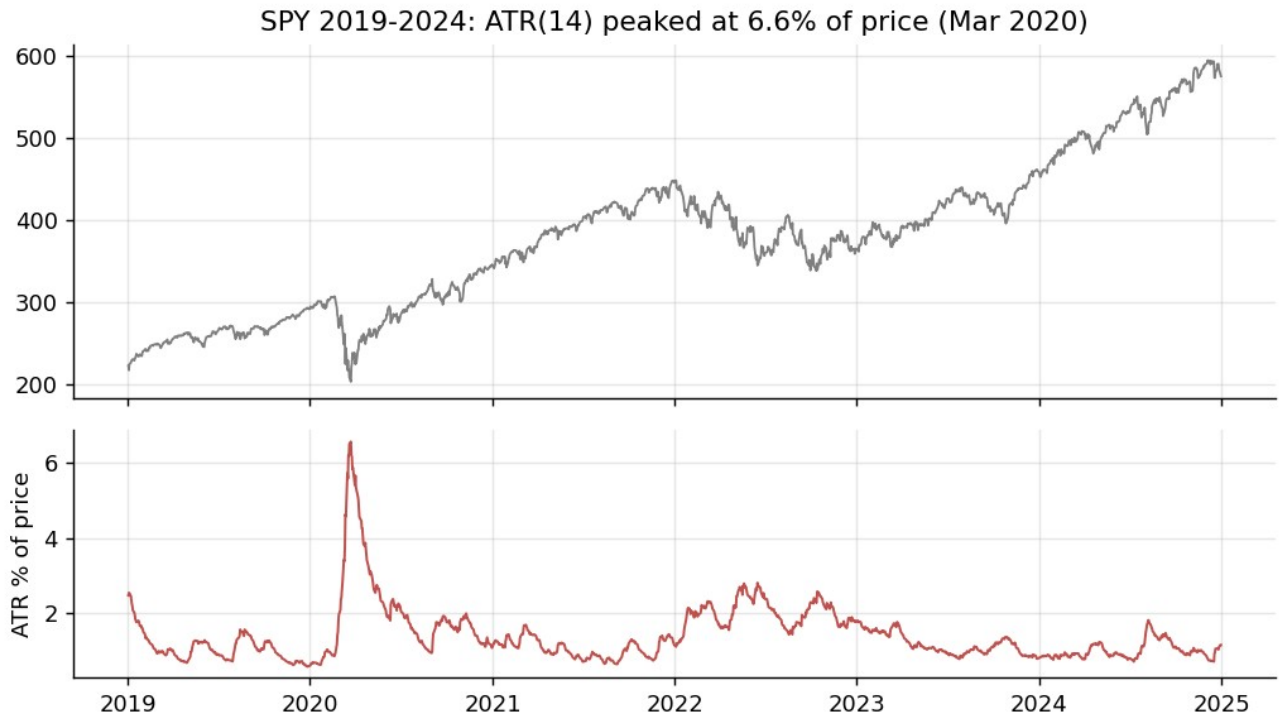
**INSIGHT**

Returns are not normal: closes fall outside the bands far more often than the 5% a normal curve implies. In a trend, 'oversold' keeps falling.


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K3

# Indicator: ATR (Average True Range)




**DEFINITION**

The average size of a day's price range, including overnight gaps - a volatility ruler in price units.

**FORMULA**

TR = max(H - L, |H - previous C|, |L - previous C|). ATR = average of TR over n bars (Wilder smoothing, n = 14).

**HOW TO READ IT**

Set stops at 2-3 x ATR and size positions as risk budget / (k x ATR), so every trade carries similar risk.

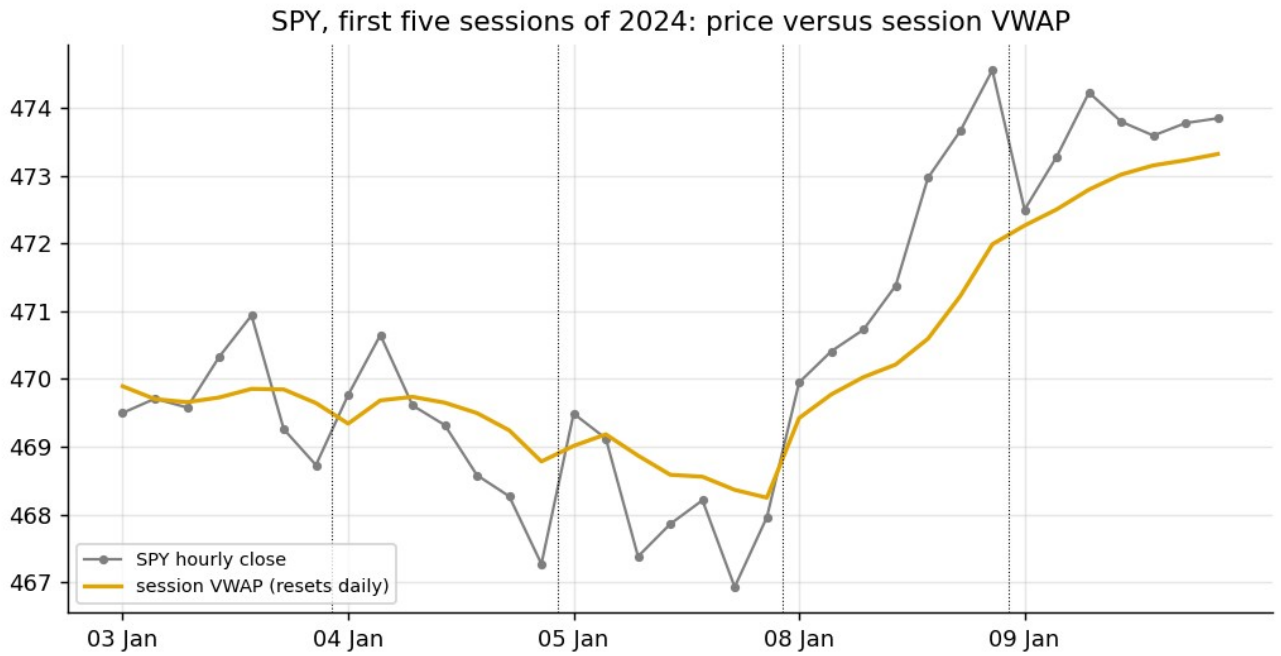
**INSIGHT**

ATR says how far, not which way. Compare instruments with ATR as a % of price, since ATR rises with the price level.


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# Indicator: VWAP (Volume-Weighted Average Price)




**DEFINITION**

The average price paid during the session, weighted by the volume traded at each price. It resets at every session open.

**FORMULA**

Typical price P = (H + L + C) / 3. VWAP = sum(P x Volume) / sum(Volume), summed from the session open.

**HOW TO READ IT**

The execution benchmark: buying below VWAP is a good fill. Intraday traders read price above VWAP as buyers in control.

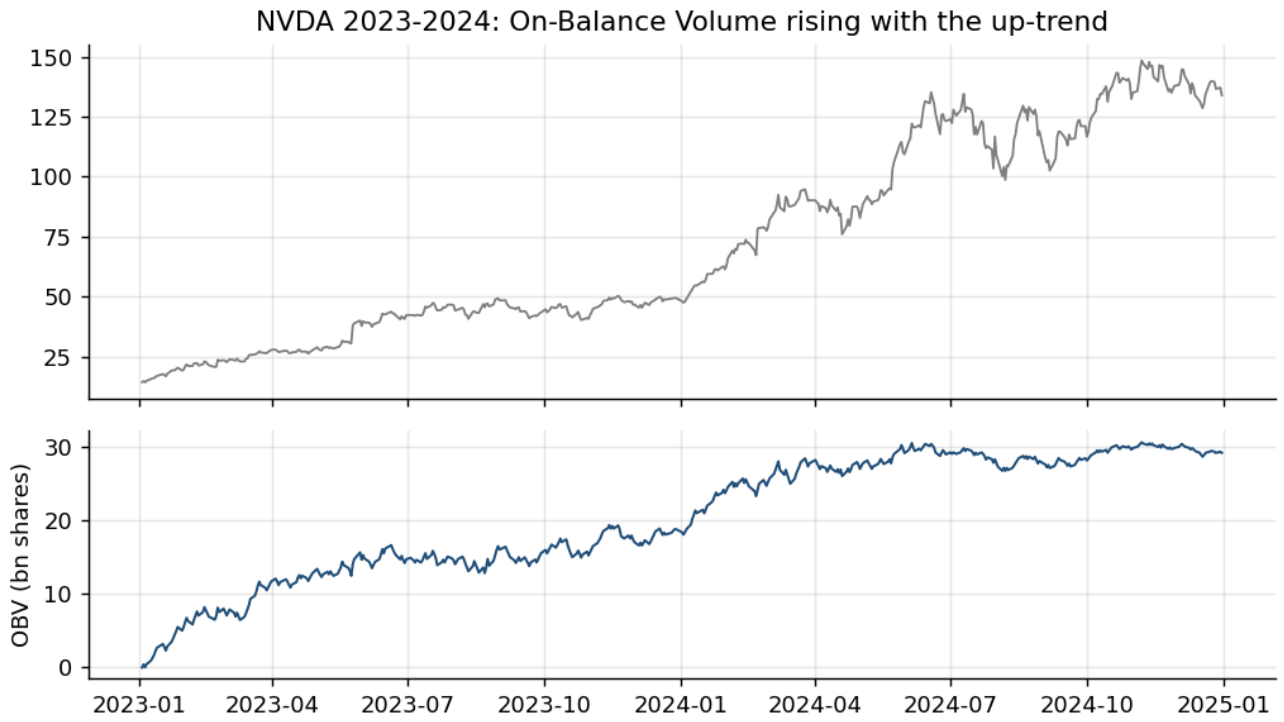
**INSIGHT**

Institutions judge execution algorithms against VWAP. It needs intraday data - on daily bars it is only an approximation.


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# Indicator: OBV (On-Balance Volume)




**DEFINITION**

A running total of volume that adds a day's volume when the close rises and subtracts it when the close falls.

**FORMULA**

OBV(t) = OBV(t-1) + Volume(t) if C(t) > C(t-1); - Volume(t) if C(t) < C(t-1); unchanged if equal.

**HOW TO READ IT**

A rising OBV confirms an up-trend; OBV making new highs before price is read as accumulation. Read the slope, not the level.

**INSIGHT**

Volume shows the conviction behind a move. An OBV that falls while price rises (divergence) is a warning, not a signal on its own.


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# Matching the Method to the Hypothesis

1

**'Does the signal beat the benchmark?'**

Backtest + t-test or bootstrap on returns.

2

**'Will tomorrow be up?'**

Classifier: logistic regression, random forest, XGBoost.

3

**'What state is the market in?'**

Clustering: K-means on return and volatility features.

4

**'Which parameters?'**

Grid search in-sample, confirm out-of-sample.


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# Coding Languages for Algorithms and Signals

1

**Pine Script v6**

//@version=6 · ta.ema · ta.crossover · strategy.entry.

2

**Python**

Signals as pandas Series of +1 / 0 / −1 aligned to bars.

3

**C++ / Java**

Low-latency execution engines and exchange gateways.

4

**R, SQL, MQL**

Statistics, data stores, MetaTrader expert advisors.


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# Order Types and Time in Force

1

**Market and limit**

Certainty of fill versus control of price.

2

**Stop, stop-limit, trailing stop**

Trigger on a price; protect or enter on breaks.

3

**Bracket / OCO / OTO**

Entry with attached take-profit and stop-loss.

4

**day · gtc · opg · cls · ioc · fok**

How long an order stays working.


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# Performance and Risk Metrics

1

## Return

Total return · CAGR = (end/start)^(1/years) − 1.

2

## Risk

Volatility = σ(daily) × √252 · maximum drawdown.

3

## Risk-adjusted

Sharpe · Sortino · Calmar = CAGR / |MaxDD|.

4

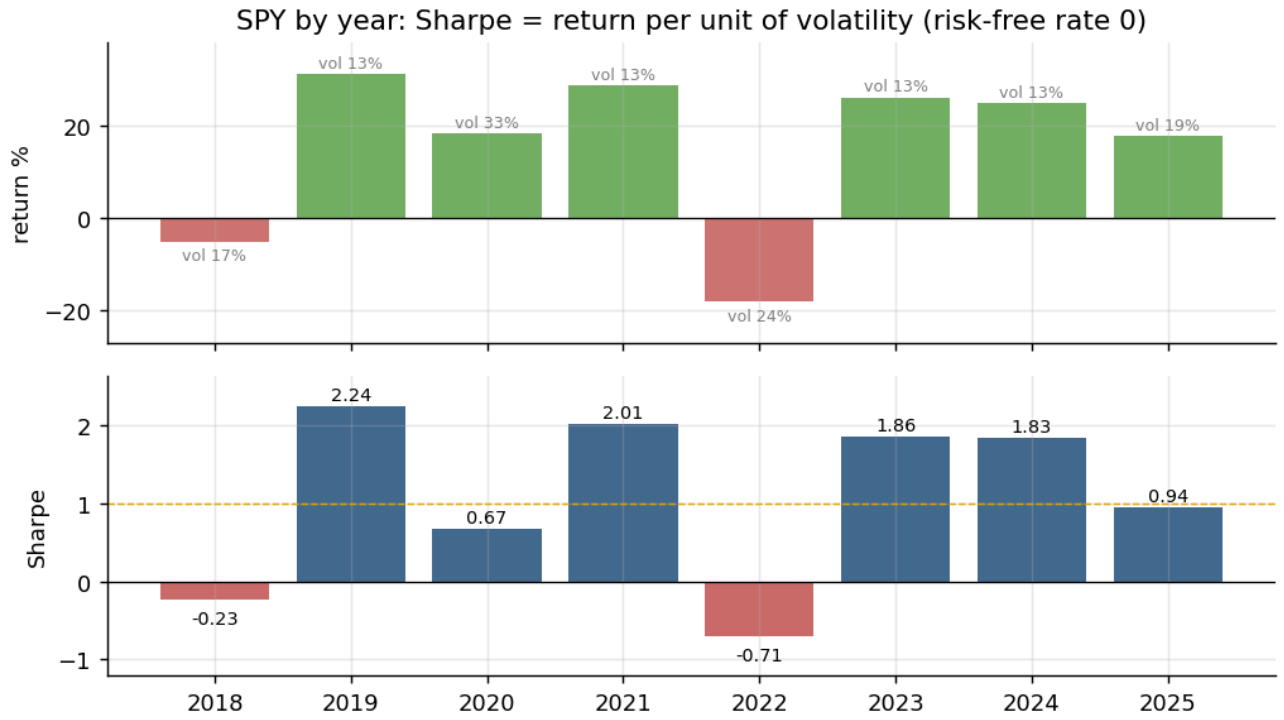
## Trade quality

Win rate · profit factor · expectancy · turnover · exposure.


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# Metric: Sharpe Ratio




**DEFINITION**

Return earned per unit of risk taken: the average excess return divided by its volatility, annualised.

**FORMULA**

Sharpe = (mean daily return - daily risk-free rate) / std of daily returns x sqrt(252).

**HOW TO READ IT**

Below 0.5 weak, about 1 good, above 2 rare for a daily strategy - check for errors. Always compare with buy-and-hold over the same period.

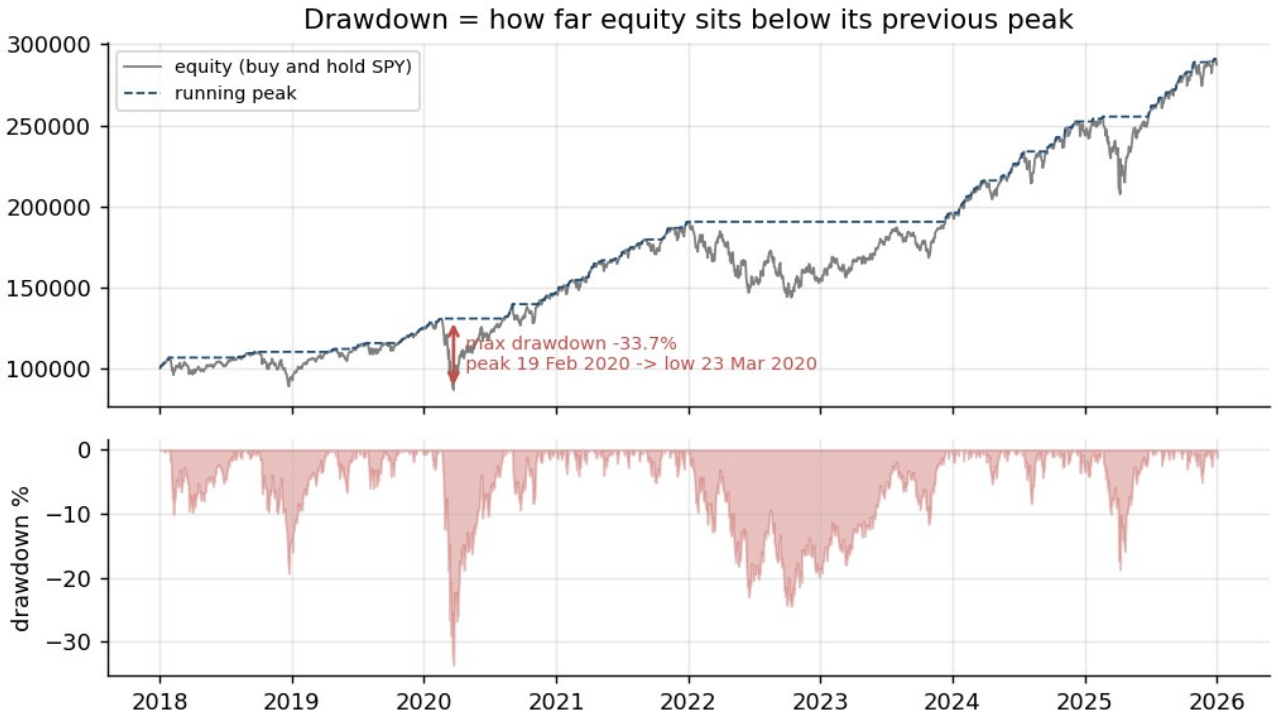
**INSIGHT**

Volatility counts as risk whether it is up or down, and a short or costless backtest inflates it - 2020 returned 18% but at 33% volatility, a Sharpe of only 0.67.


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# Metric: Maximum Drawdown




**DEFINITION**

The largest fall of the equity curve from a previous peak to a later low - the worst loss an investor would have lived through before a new high.

**FORMULA**

Drawdown(t) = equity(t) / running peak(t) - 1. Max drawdown = the most negative drawdown over the period.

**HOW TO READ IT**

A -20% drawdown needs +25% to recover; -50% needs +100%. Compare with buy-and-hold and with the loss the investor can tolerate.

**INSIGHT**

One path-dependent number: it does not say how long recovery took, and a short backtest may simply miss a bad period. SPY 2018-2025: -33.7% in five weeks of 2020.


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# Classification of Trading Algorithms by Purpose

| | Purpose | Examples |
|---|---|---|
| **Execution algorithms** | Work a large order at low cost and market impact | VWAP, TWAP, POV, implementation shortfall, iceberg, smart order routing |
| **Market making** | Quote bids and offers, earn the spread, manage inventory | Passive market making with resting limit orders |
| **Arbitrage** | Capture mispricing between related products or markets | ETF versus its basket, index futures versus stocks |
| **Directional / alpha** | Take a view on the direction of price | Trend following, momentum, breakout, mean reversion, news |
| **Statistical arbitrage and pairs** | Hedged: long one asset, short a related one; trade the spread | Pairs trading, cointegration, market-neutral baskets |
| **Factor and portfolio** | Build and rebalance portfolios from return drivers and risk | Value, momentum, size factors; mean-variance optimisation |
| **Hedging and risk** | Offset unwanted exposure and enforce limits | Delta hedging, beta hedging, pairs hedge, kill switch |


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# SEC Classification of High-Frequency Strategies (2010)

1

## Passive market making

Submits non-marketable resting bids and offers that provide liquidity at specified prices

2

## Arbitrage

Captures pricing inefficiencies between related products or markets, e.g. an ETF and its basket

3

## Structural

Exploits structural vulnerabilities in the market or its participants, e.g. stale quotes via faster data

4

## Directional

Takes a short-term position anticipating an intraday price move in a particular direction


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# Classification by Holding Period

| | Horizon | Typical algorithms |
|---|---|---|
| **High-frequency** | Microseconds to seconds | Market making, ETF and latency arbitrage |
| **Intraday** | Minutes to hours | VWAP / TWAP execution, intraday momentum |
| **Swing** | Days to weeks | Mean reversion, pairs trading |
| **Position** | Weeks to months | Trend following, factor portfolios |


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# Classification by Model Type

1. **Rule-based**
   Fixed if-then rules on indicators, e.g. a moving-average crossover

2. **Statistical**
   Econometric tests and models: cointegration, regression, t-tests

3. **Machine learning**
   Models learn from labelled data: logistic regression, random forest, XGBoost

4. **Reinforcement learning**
   Learns a policy from rewards, e.g. order-execution research

5. **LLM and agentic**
   Language models read text and agents plan, code and act under limits


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# Long and Short Positions

| | Long | Short |
|---|---|---|
| **What you hold** | Shares you own | Shares you borrowed and sold - you owe them back |
| **Opening trade** | Buy | Sell (the borrowed shares) |
| **Closing trade** | Sell | Buy back ('buy to cover') and return the shares |
| **You profit when** | The price rises | The price falls |
| **Example** | Buy SPY at $500, sell at $550: +$50 | Sell SPY at $500, buy back at $450: +$50 |
| **Maximum loss** | What you paid - a price cannot go below $0 | Unlimited - a price can keep rising |
| **Extra requirements** | Only the cash to pay for the shares | A margin account (Alpaca: $2,000+ equity); borrow fees on hard-to-borrow stocks (Alpaca: $0 on easy-to-borrow) |


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# Long and Short in Practice

1

### 'Long = buy, short = sell' is shorthand

Opening a long is a buy; opening a short is a sell. Selling shares you already own only closes the long - you go flat, not short.

2

### Pine Script (TradingView)

strategy.entry("Long", strategy.long) opens a buy; strategy.close("Long") sells back to flat; strategy.short opens a short.

3

### Trading Simulator

Most strategies are long-only: buy, then sell back to cash. 'RSI oversold / overbought' has an allow_short option that goes short above RSI 70.

4

### Pairs trading (Lab 14)

Long the cheap stock and short the rich one at the same time: the trade profits from the gap closing, whichever way the market moves.


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# Strategy Families at a Glance (1 of 2)

1

## Trend following

Rides sustained moves in the direction of the trend; cuts losers quickly - e.g. Buy when the 50-day moving average crosses above the 200-day

2

## Mean reversion

Bets that prices stretched far from their average will snap back - e.g. Buy when price closes below the lower Bollinger Band (20-day, 2 std dev)

3

## Momentum and breakout

Buys recent winners or prices breaking out of a range, expecting them to keep going - e.g. Buy on a close above the 20-day high (Donchian channel breakout)

4

## Statistical / pairs arbitrage

Trades the spread between two related assets when it drifts from its normal level - e.g. Short the rich leg, buy the cheap leg when the spread z-score exceeds 2


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# Strategy Families at a Glance (2 of 2)

1

## Execution algorithms

Split a large order to minimise cost and market impact: VWAP, TWAP, POV, implementation shortfall - e.g. POV: trade 10% of market volume until the order is filled

2

## Market making

Continuously quotes buy and sell prices, earning the spread while managing inventory - e.g. Quote bid and ask around mid-price; skew quotes as inventory builds

3

## Index rebalancing / event-driven

Trades predictable flows around index changes, earnings and corporate events - e.g. Position ahead of announced index inclusions on the rebalance date

4

## Sentiment / NLP

Turns news, filings and social media text into trading signals using NLP or LLMs - e.g. Go long when the average FinBERT headline sentiment is strongly positive


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# Strategy Family: Trend Following

## The idea

Prices that have started to move tend to keep moving. Buy when a fast average is above a slow one (or price is above its 200-day average); exit when the trend turns.

## When it works

Long, persistent moves - bull or bear markets, macro trends lasting months. A few big winners pay for many small losses.

## When it will not work

Sideways, choppy markets: price crosses the averages again and again (whipsaws), every crossing costs money, and signals arrive late at turns.

**Lab 7: EMA 20/50 on SPY**


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# Strategy Family: Mean Reversion

## The idea

Like a rubber band, short-term over-reactions snap back. Buy when price is stretched below its normal range (lower Bollinger Band, RSI(2) below 10) and sell when it returns to the average.

## When it works

Range-bound, liquid markets and stable large stocks, where extremes tend to snap back within days.

## When it will not work

A real trend or a news shock: 'cheap' keeps getting cheaper. A high win rate hides rare large losses - as in March 2020.

**Lab 7: Bollinger 20 / k = 2.0 and 1.5 on KO**


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# Mean Reversion Rules in the Simulator

| | Buy when 'too low' | Exit |
|---|---|---|
| **Bollinger Band mean reversion** | Close below the lower band (20-day mean - 2 standard deviations) | Price returns to the middle band (the 20-day mean) |
| **RSI oversold / overbought** | RSI(14) below 30 - oversold | RSI(14) above 70 - overbought |
| **RSI(2) mean reversion** | RSI(2) below 10 - a sharp 1-2 day drop | RSI(2) above 70 - the bounce |
| **Dip buy after a losing streak** | 3 or more down closes in a row (the Lab 4 hypothesis) | After the holding horizon, e.g. the next day |


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# Mean Reversion versus Trend Following

| | Mean reversion | Trend following / momentum |
|---|---|---|
| **Belief** | Extremes snap back to the average | What is moving keeps moving |
| **After a big drop** | Buy | Sell or stay out |
| **Works best in** | Range-bound, sideways markets | Strongly trending markets |
| **Typical horizon** | Short: days | Longer: weeks to months |
| **Win rate** | High - many small wins | Low - few but large wins |
| **Main danger** | A rare big loss when the dip becomes a crash | Many small losses in choppy markets |


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# Mean Reversion: the Main Risk and Three Safeguards

### The risk

The rule assumes the average holds. After a real change - a crash, bad news, a broken business - price does not come back: buying the dip becomes catching a falling knife.

### 1 Stop-loss or time limit

Cap the rare big loss: exit at a fixed loss, or after n days if the bounce has not come.

### 2 Trend filter

Only buy dips while price is above its 200-day SMA, so you are not buying into a bear market.

### 3 Stable markets

Liquid, defensive stocks such as KO, and pairs trading (Lab 14: the spread between two linked stocks) revert more reliably.

**Lab 7: Bollinger on KO - high win rate, but losses larger than wins: many small wins, rare big losses.**


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# Strategy Family: Momentum

## The idea

Winners keep winning for months. Buy what has risen over the last 6-12 months (time-series) or what has risen most among its peers (cross-sectional).

## When it works

Persistent market or sector themes, spread across many assets and rebalanced regularly.

## When it will not work

Sharp reversals - 'momentum crashes' such as the 2009 rebound, when last year's losers rally hardest - and choppy markets that churn the portfolio.

**Lab 7: 126-day time-series momentum on QQQ**


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# Strategy Family: Breakout

## The idea

A close above the recent range (for example the 20-day high, the Donchian channel) signals the start of a new move: buy the breakout, exit on a shorter low.

## When it works

Volatility expanding after a quiet period and a new trend starting - indices and commodities in strong regimes.

## When it will not work

False breakouts in a range: price pokes above the high and falls back. A low win rate, so exits must be disciplined and costs kept low.

**Lab 7: Donchian 20 / 10 on QQQ**


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TOPIC 2 · HOW THE ALGORITHMS WORK

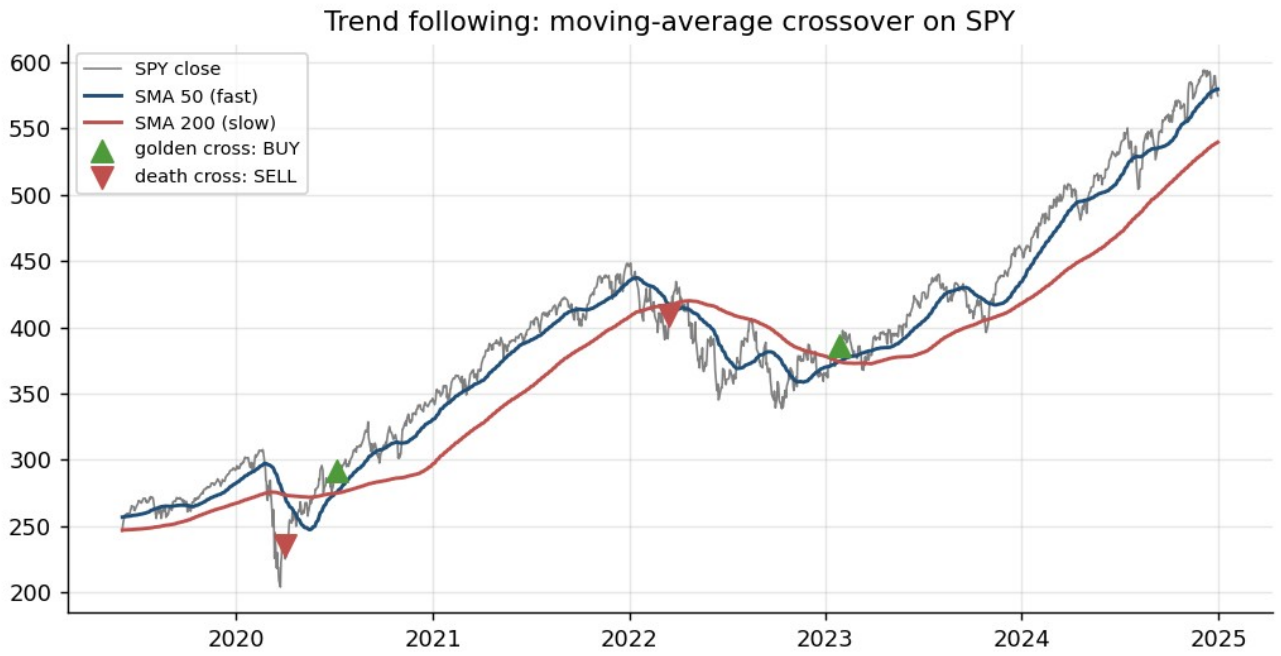
# Concepts over code

Every algorithm below is coded for you - by the simulator or an AI assistant. Learn what it does, when it works and what can go wrong; let AI handle the syntax.


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# Algorithm: Moving-Average Crossover (Trend Following)




**HOW IT WORKS**

Two moving averages of the close: a fast one reacts quickly, a slow one shows the trend.

**RULE**

Fast crosses ABOVE slow (golden cross): buy. Fast crosses BELOW (death cross): sell or go flat.

**WORKS BEST WHEN**

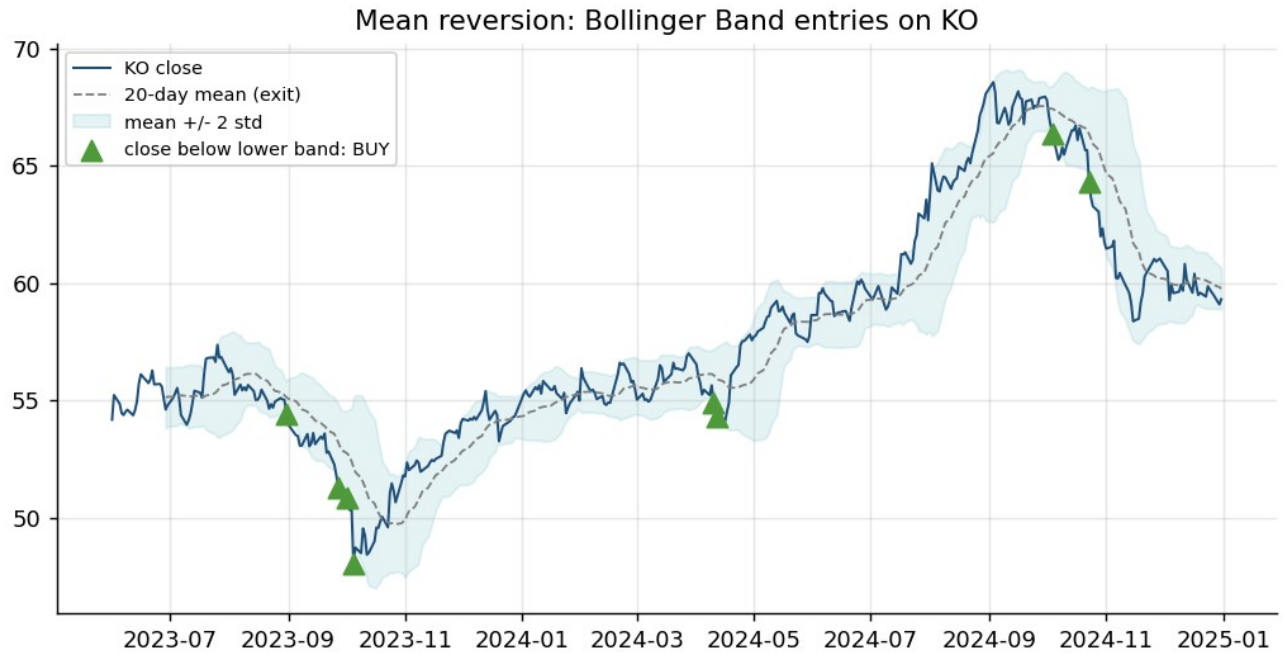
Markets trend for months - it rides the move and cuts losers when the trend turns.

**WATCH OUT FOR**

Whipsaws in sideways markets; signals lag; trade on the next bar, not the crossing bar.


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# Algorithm: Bollinger Band Mean Reversion




**HOW IT WORKS**

Bands = 20-day mean +/- 2 standard deviations. A close outside the band is 'stretched'.

**RULE**

Close below the lower band: buy. Exit when the price returns to the middle band (the mean).

**WORKS BEST WHEN**

Range-bound markets where over-reactions snap back, e.g. a defensive stock such as KO.

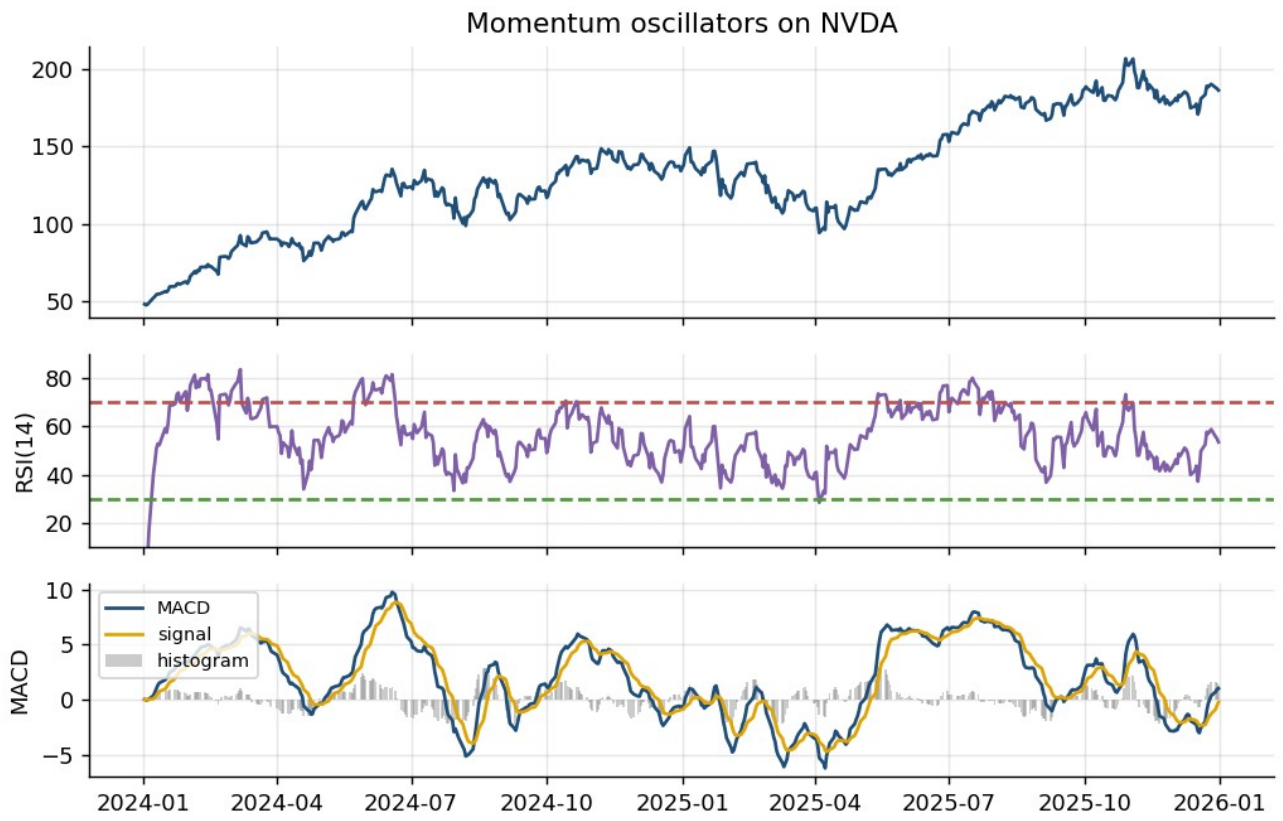
**WATCH OUT FOR**

Strong trends keep 'stretching' - add a stop-loss or a trend filter; many small wins, rare big losses.


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# Algorithm: RSI and MACD Momentum Oscillators




**HOW IT WORKS**

RSI = 100 - 100/(1+RS), RS = average gain / average loss over 14 bars. MACD = EMA12 - EMA26.

**RULE**

RSI < 30 oversold / > 70 overbought. MACD crossing above its 9-EMA signal line = momentum buy.

**WORKS BEST WHEN**

As filters or confirmation for a trend or mean-reversion rule, not as stand-alone predictors.

**WATCH OUT FOR**

Oscillators can stay overbought in strong trends; RSI and MACD are highly correlated with price.


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# Algorithm: Cross-Sectional Momentum




**HOW IT WORKS**

Rank a universe by past return (e.g. 12 months, skipping the last month) each month.

**RULE**

Hold the top-ranked assets, avoid or short the bottom-ranked; rebalance monthly.

**WORKS BEST WHEN**

Information spreads slowly and winners keep winning for several months.

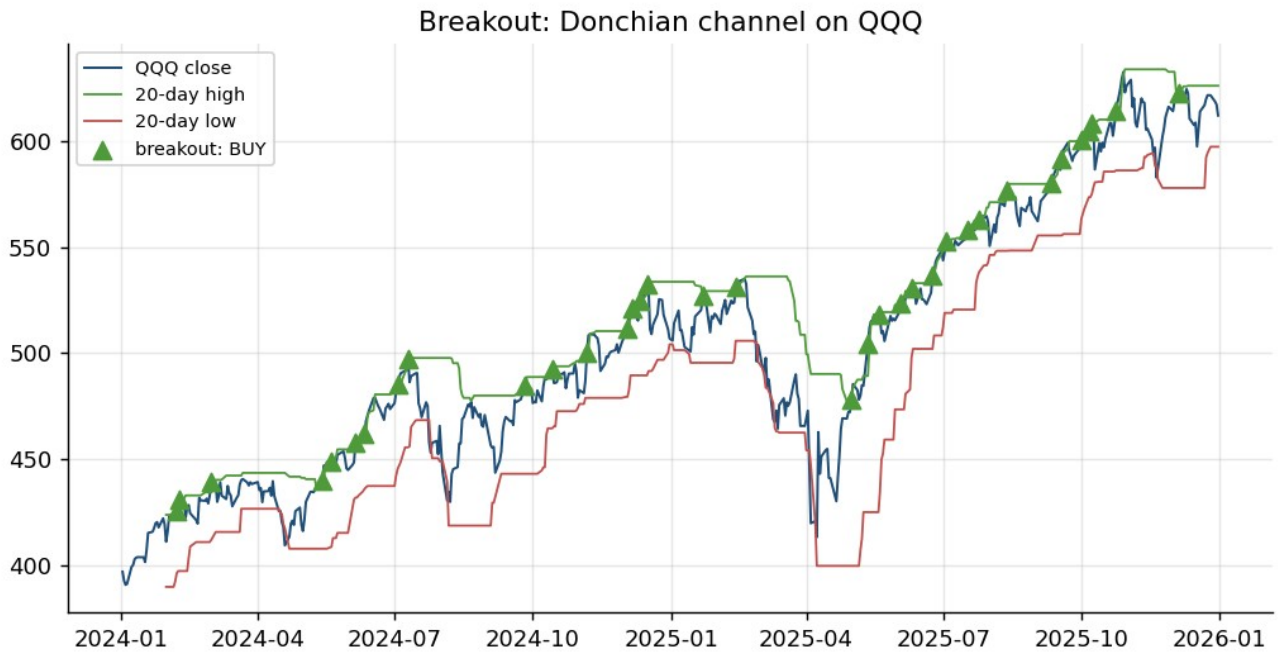
**WATCH OUT FOR**

Sharp 'momentum crashes' when markets reverse; turnover costs at every rebalance.


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# Algorithm: Donchian Channel Breakout




**HOW IT WORKS**

Track the highest high and lowest low of the last n bars (n = 20 here).

**RULE**

Close above the 20-day high: buy the breakout. Exit below the 20-day low (or a trailing stop).

**WORKS BEST WHEN**

New trends start with a break of a range - the rule used by the 1980s 'Turtle Traders'.

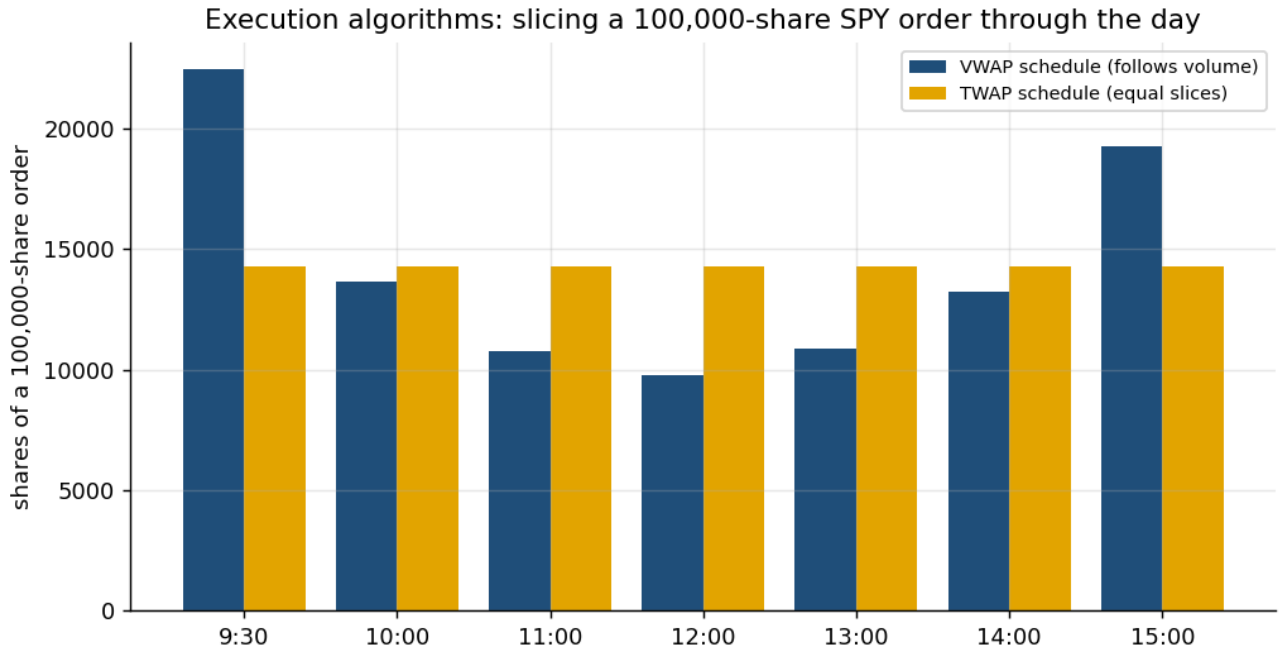
**WATCH OUT FOR**

False breakouts in choppy markets; the channel must use PREVIOUS bars only (no look-ahead).


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# Algorithm: VWAP and TWAP Execution




**HOW IT WORKS**

Split a large parent order into child orders through the day to reduce market impact.

**RULE**

TWAP: equal slices per time interval. VWAP: slices proportional to the typical volume curve.

**WORKS BEST WHEN**

Working client or fund orders too large to trade at once; benchmarked to the day's VWAP.

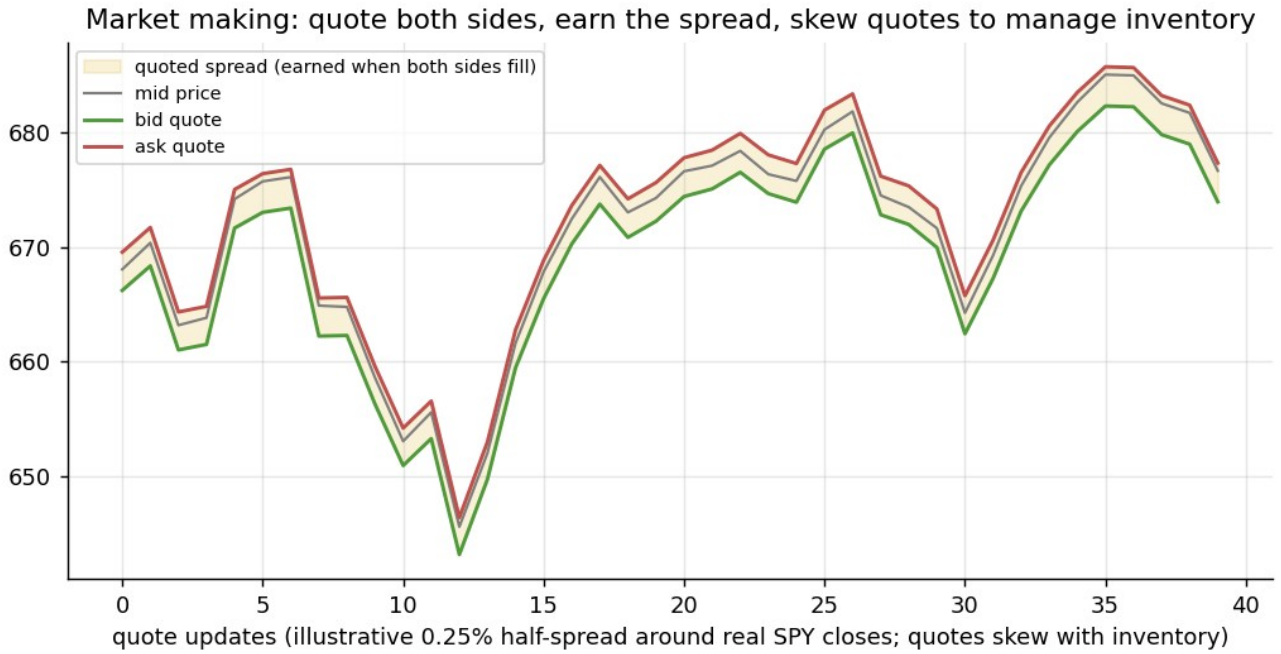
**WATCH OUT FOR**

VWAP relies on a volume forecast; both are predictable to other traders - add randomisation.


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# Algorithm: Market Making




**HOW IT WORKS**

Quote a bid below and an ask above the mid price at the same time.

**RULE**

Earn the spread when both sides fill; skew quotes down when long, up when short, to cut inventory.

**WORKS BEST WHEN**

Liquid markets with steady two-way flow - the business of quant market makers.

**WATCH OUT FOR**

Adverse selection by better-informed traders; inventory risk in fast moves; needs low latency.


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# TradingView versus Alpaca

| | TradingView | Alpaca |
|---|---|---|
| **What it is** | Charting and strategy-design platform | API-first US broker for automated trading |
| **Markets** | Global charts incl. SGX, US, FX, crypto | US stocks and ETFs, options, crypto |
| **Strategy design** | Pine Script v6 indicators and strategies | Write it yourself in Python (alpaca-py) |
| **Backtesting** | Strategy Tester (simplified costs and fills) | None built in - you build a cost-aware backtest |
| **Paper trading** | Built-in Paper Trading account from the chart | Paper account mirroring the live API |
| **Automation** | Alerts; webhooks on paid plans | Trading API: orders, positions, clock; 24/5 trading |
| **Data** | Charts and Data Window | Market Data API: bars, quotes, trades, news (IEX free) |
| **Machine learning** | Not supported | Any Python library (scikit-learn, XGBoost) |
| **Best used for** | Designing, visualising and first-testing a rule | Executing and forward-testing it automatically |


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# Paper Trading on TradingView, Step by Step

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Supercharts > Trade > Paper Trading > Connect
2. Set starting capital and commission
3. Order ticket: side, quantity, Market / Limit / Stop
4. Add stop loss and take profit; place order
5. Track Positions, Orders, Account History
6. Close the position; reset when done

**Paper fills use TradingView's prices but cannot show your own market impact - compare them with the backtest's assumed costs.**


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# Paper Trading on Alpaca, Step by Step - Manual and Automated

Full step-by-step guide: Learner Guide, 'Platform Guides'.

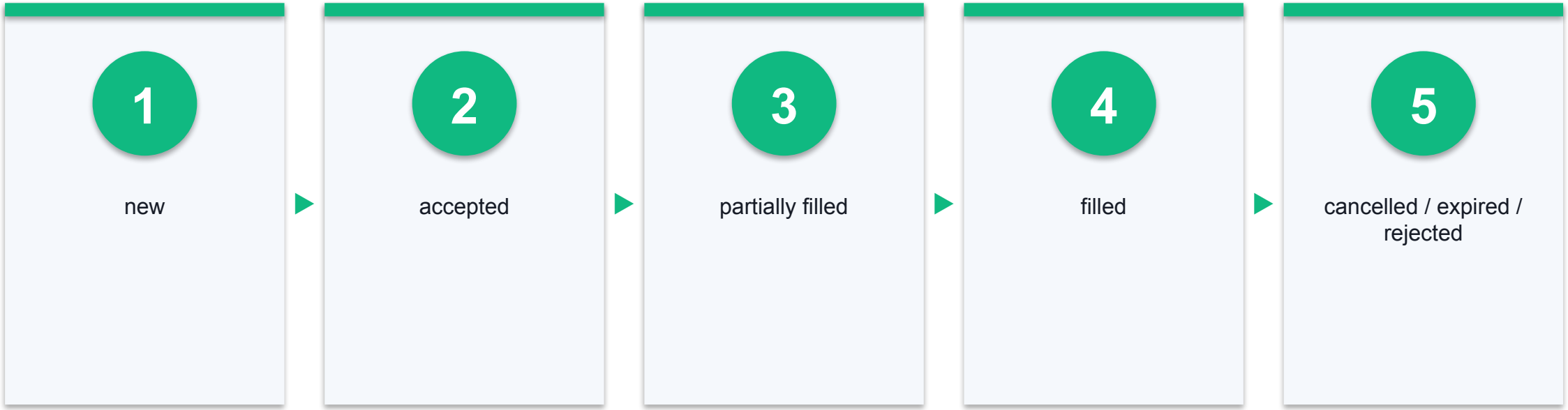
1. Dashboard: Paper account > search SPY > Buy
2. Watch Orders: new > accepted > filled
3. Outside US hours: extended-hours LIMIT orders
4. python -m pipeline execute (order manager)
5. python -m pipeline run --loop --live-paper
6. Check dashboard; kill switch to stop

**From Singapore the US regular session is 21:30-04:00 (22:30-05:00 in northern winter); class-time orders use Alpaca's 24/5 extended-hours limit orders.**


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# The Order Lifecycle

A program must read the status — never assume a fill.




**Bracket orders protect a position even if your program stops running.**


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# From Idea to Tested Rule on TradingView

1. Chart the instrument and timeframe
2. Add indicators (Pine)
3. Write the strategy() script
4. Set commission and slippage
5. Read the Strategy Tester
6. Compare with buy-and-hold

**The Strategy Tester is a first filter; the full test with costs runs in Python.**


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A3 K3 K7

# Indicator Stage: Pine Script v6 Indicators and their Python Twins

## LAB 05

Add the desk's Pine Script v6 indicators to a TradingView chart, build pipeline/indicators.py with twins of ta.ema, ta.sma, ta.rsi, ta.bb, ta.atr, ta.macd, ta.crossover and ta.crossunder, and cross-check the values against the TradingView Data Window.

### WORKFLOW

1. Add the trend and bands indicator
2. Add the RSI pane
3. Record the TradingView values
4. Build the Pine twins
5. Build the cross-check run
6. +3 more — see the Learner Guide

### YOU'LL PRODUCE

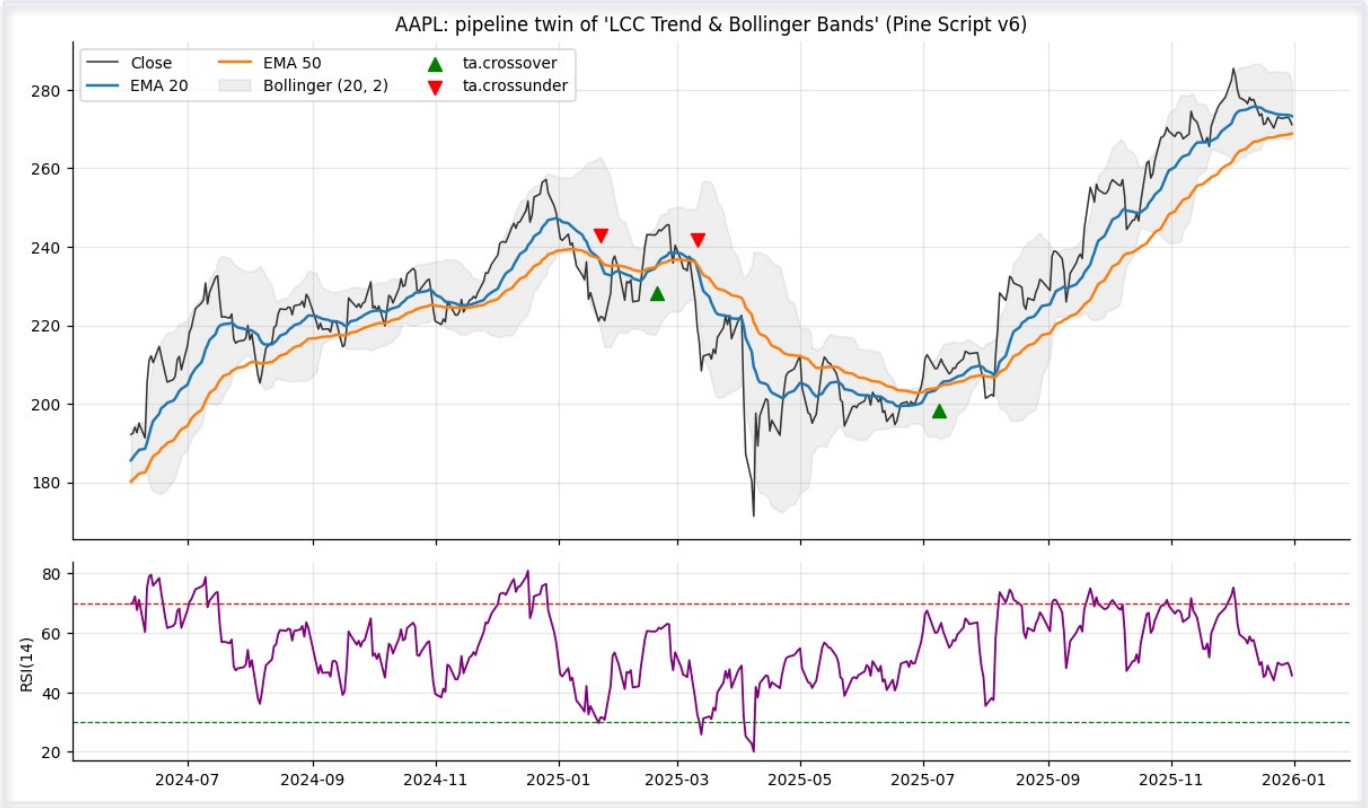
**Pine Script v6 indicators on TradingView and their Python twins, cross-checked value by value.**

Duration
**25 min**


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LAB 05 · OUTPUT

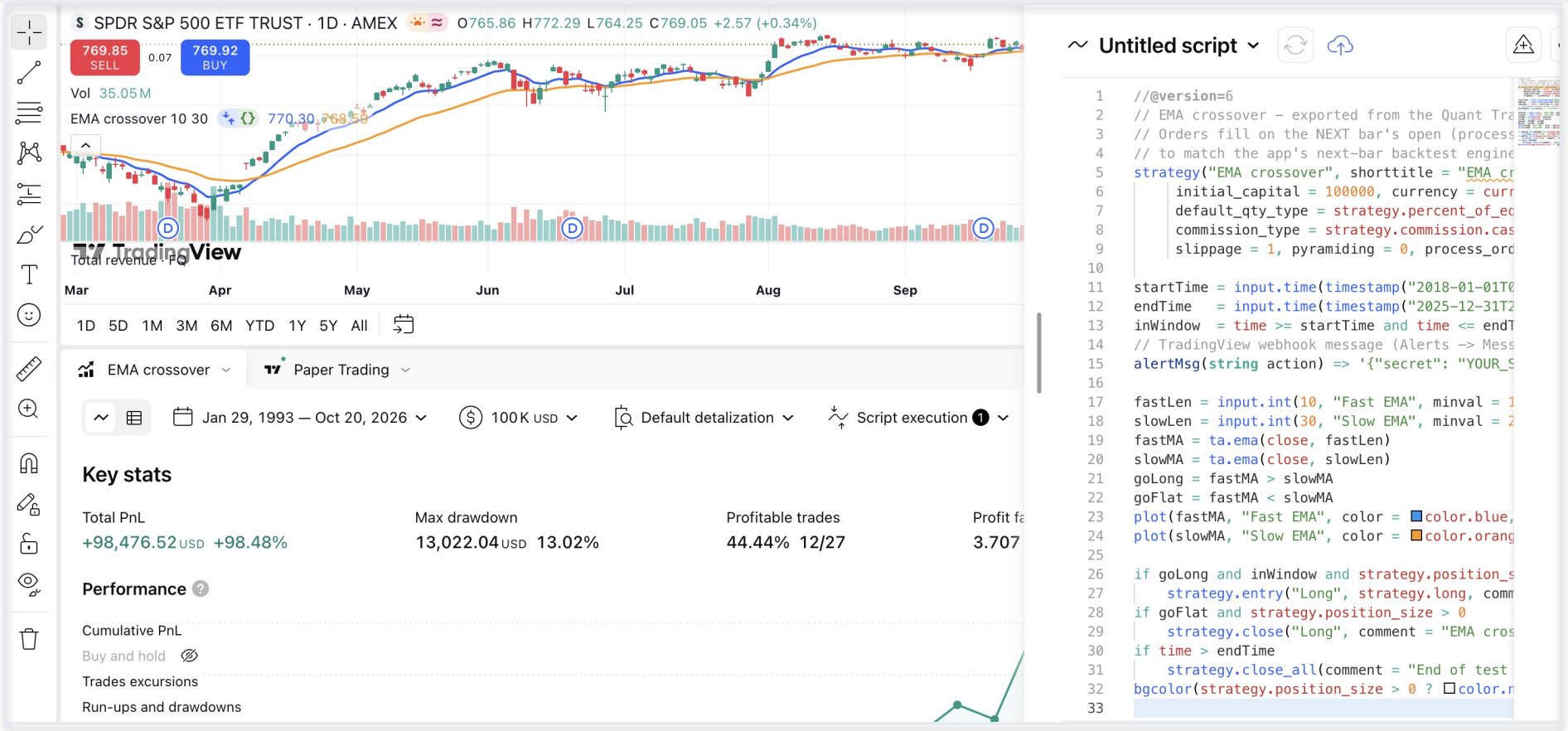
# Lab 05 — What You Should See




Output of Lab05-pine-indicators in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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# Lab 05 — The Exported Strategy in TradingView




The EMA crossover exported from the simulator as Pine Script v6, pasted into TradingView's Pine Editor and added to the SPY chart: the Strategy Tester reports PnL, drawdown and profitable trades to compare with the simulator's backtest.

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TOPIC 2 · HANDS-ON LAB

A2 A3 K3 K7

# Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation

**LAB 06**

Configure a Pine Script v6 strategy with costs, read the Strategy Tester, build `python -m pipeline reconcile` to check the Pine settings and reproduce the results, and place and manage paper trades from the chart with TradingView Paper Trading (and optionally an Alpaca paper account).

## WORKFLOW

1. Read the strategy code
2. Run it in the Strategy Tester and record the results
3. Build the reconciliation stage
4. Reconcile TradingView with the pipeline
5. Explain the differences
6. +5 more — see the Learner Guide

## YOU'LL PRODUCE

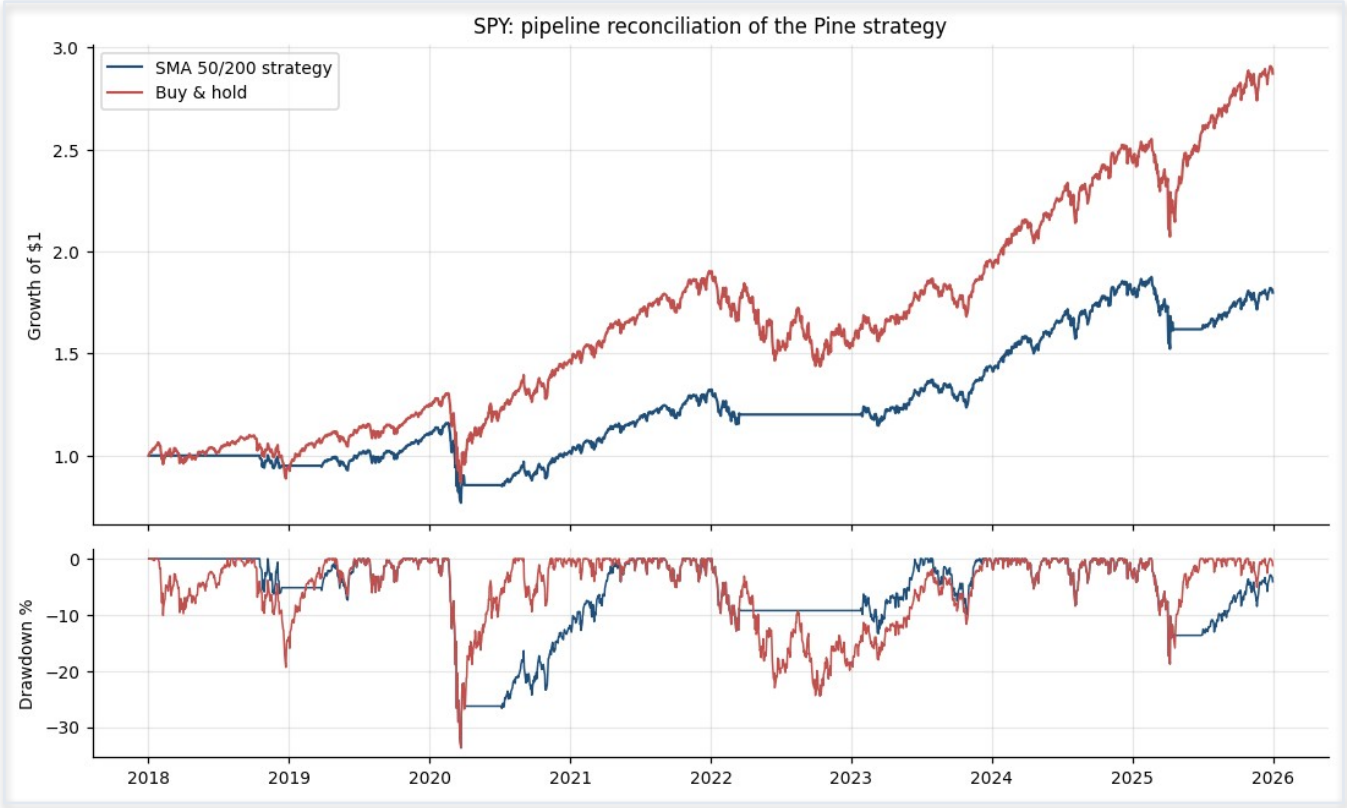
**A Pine MA-crossover strategy with costs, reconciled with the pipeline and paper-traded from the chart.**

Duration
**30 min**


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LAB 06 · OUTPUT

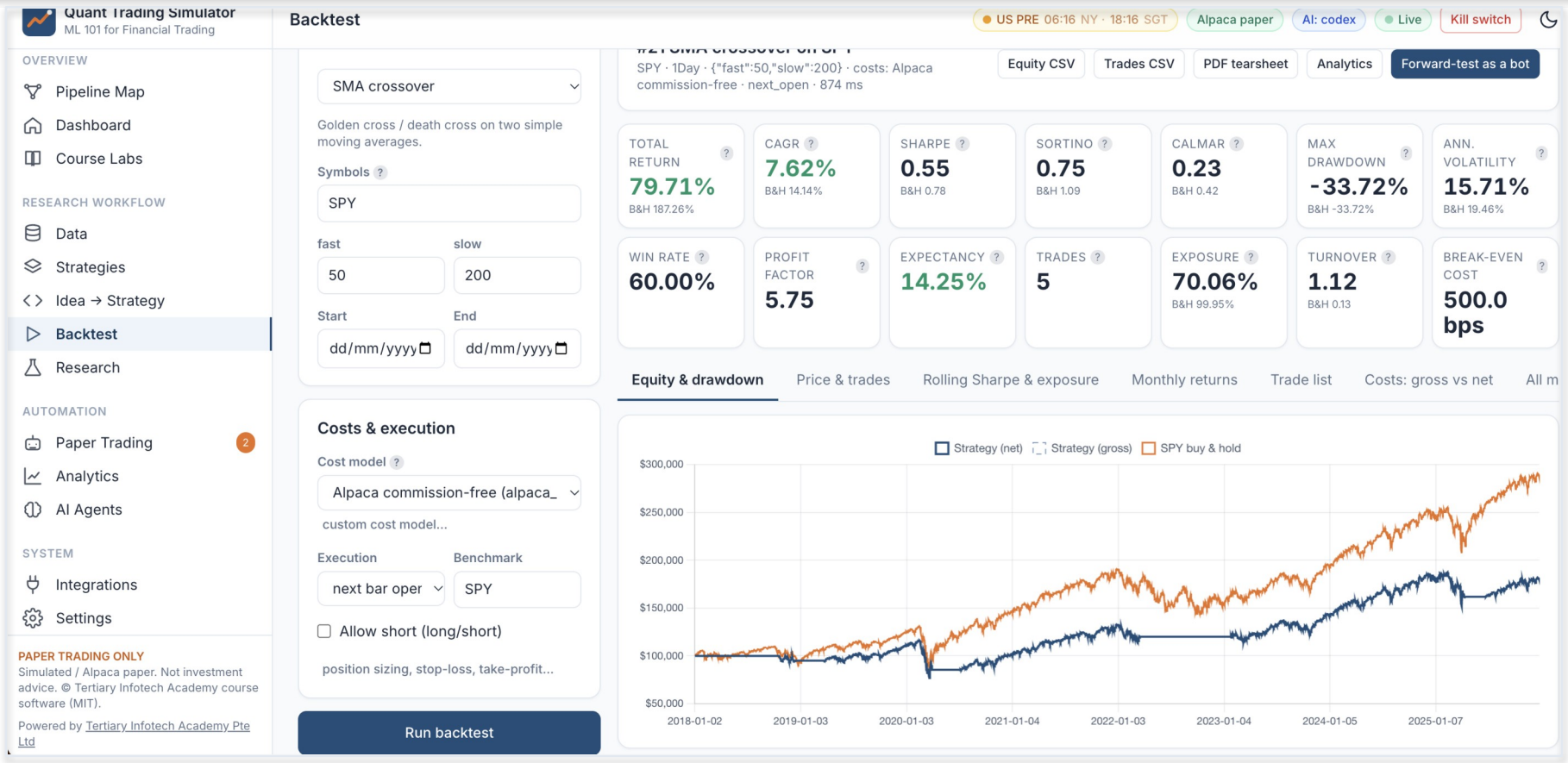
# Lab 06 — What You Should See




Output of Lab06-pine-strategy-tester in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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# Lab 06 — The SMA 50/200 Backtest in the Simulator




Backtest: SMA crossover 50/200 on SPY with Alpaca commission-free costs and next-bar-open fills - return, CAGR, Sharpe, drawdown and trades beside buy-and-hold, ready to reconcile with TradingView's Strategy Tester.

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# Tea Break

**15 minutes · see you at 15:40**


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A2 K3 K7

# Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals

**LAB 07**

Build pipeline/signals.py with trend, mean-reversion, momentum and breakout signals that return target positions with next-bar alignment, a registry, an automated look-ahead test and a universe target file for the execution stage.

## WORKFLOW

1. Agree the signal contract
2. Build the trend signals
3. Build mean-reversion, momentum and breakout signals
4. Add the registry, next-bar alignment and the look-ahead test
5. Build the signals command
6. +3 more — see the Learner Guide

## YOU'LL PRODUCE

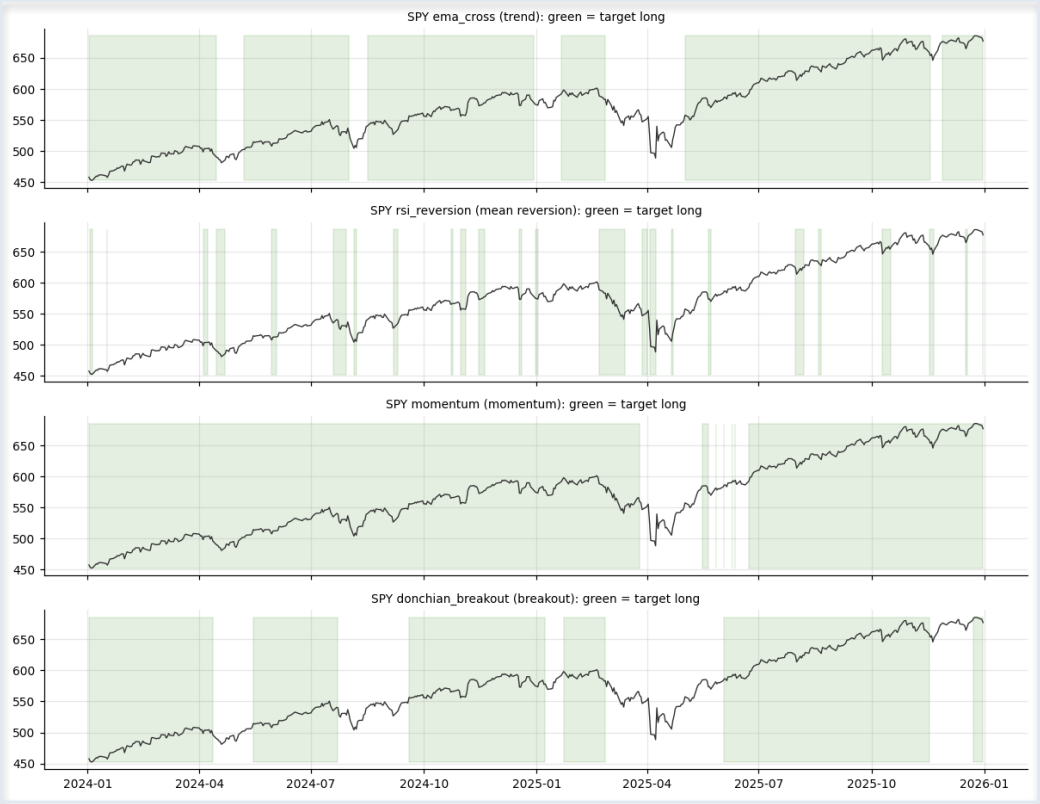
**A signal library (trend, mean reversion, momentum, breakout) with next-bar alignment and look-ahead tests.**

Duration
**25 min**


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LAB 07 · OUTPUT

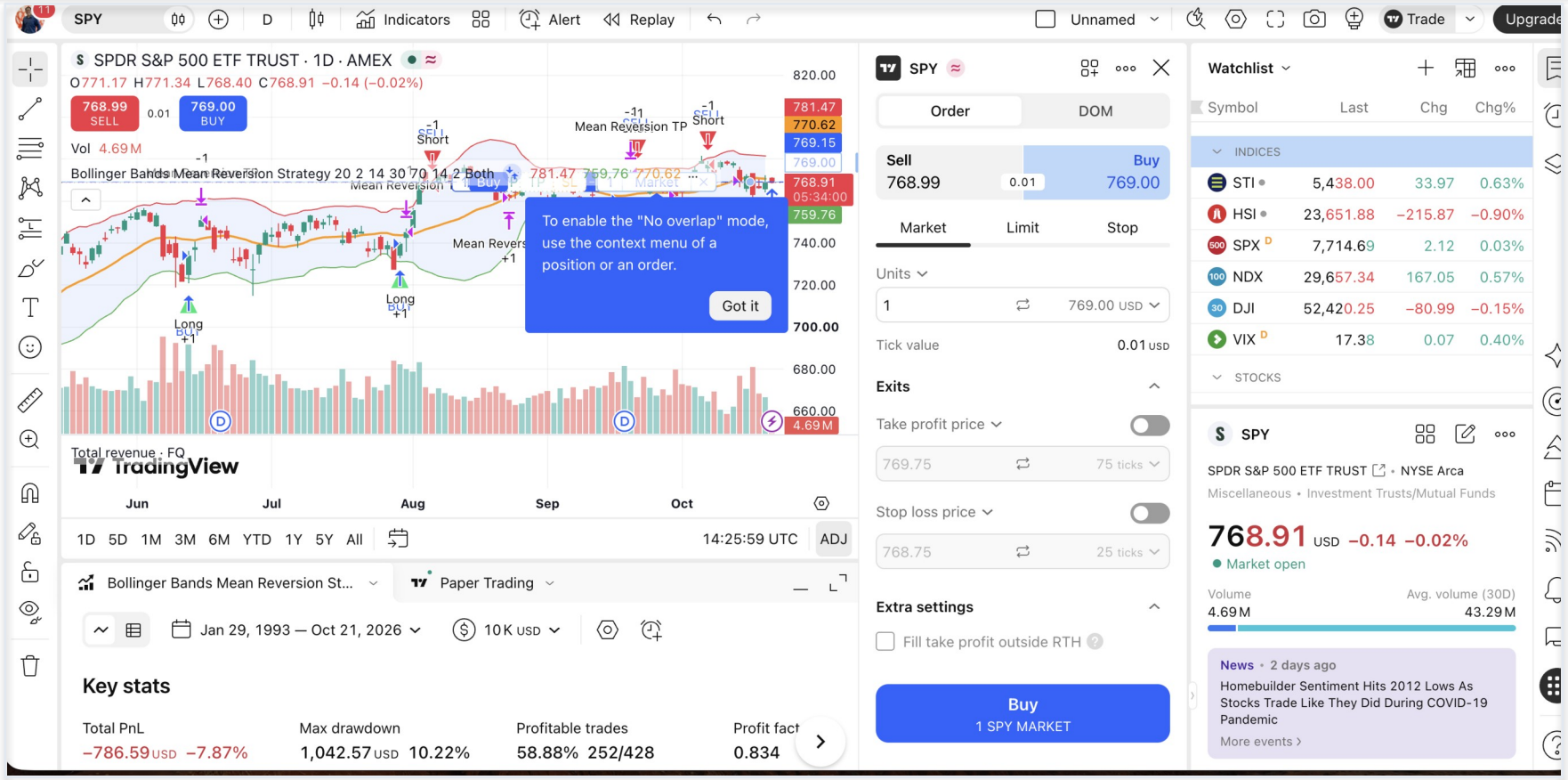
# Lab 07 — What You Should See




Output of Lab07-technical-indicators in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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# Lab 07 — Mean Reversion in TradingView, with Paper Trading




Bollinger Bands mean reversion (20, 2) on SPY with TradingView Paper Trading's order panel open. It lost 7.87% since 1993 though 59% of trades won: buying dips in a trending index hurts, so the lab runs it on KO.


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A3 K5

# Execution Stage: an Order Manager that Trades the Paper Account in Any Session

**LAB 08**

Build pipeline/execution.py: a session-aware, idempotent order manager that sizes orders from target weights, sends market, extended-hours limit, GTC-queued or bracket PAPER orders, polls their lifecycle, cancels protective legs before exits, and falls back safely when an order is rejected.

## WORKFLOW

1. Learn the order rules for each session
2. Build the planning logic: target vs actual
3. Build session-aware, idempotent order submission
4. Add polling, protective-order handling and safe fallbacks
5. Run the order manager in the current session
6. +5 more — see the Learner Guide

## YOU'LL PRODUCE

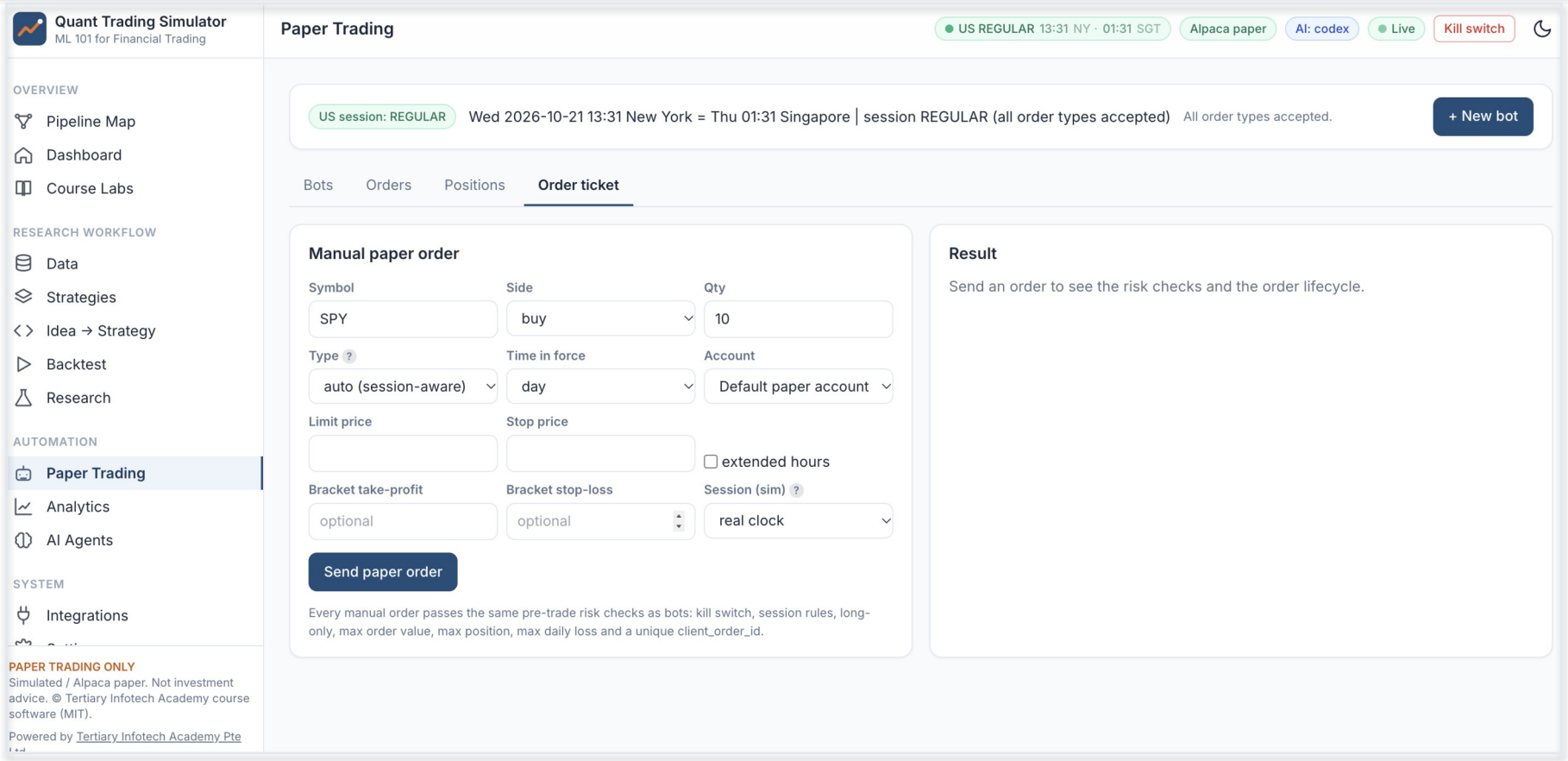
**An order manager that turns target positions into paper orders in regular or extended hours.**

Duration
**30 min**


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LAB 08 · SIMULATOR

# Lab 08 — The Paper Order Ticket

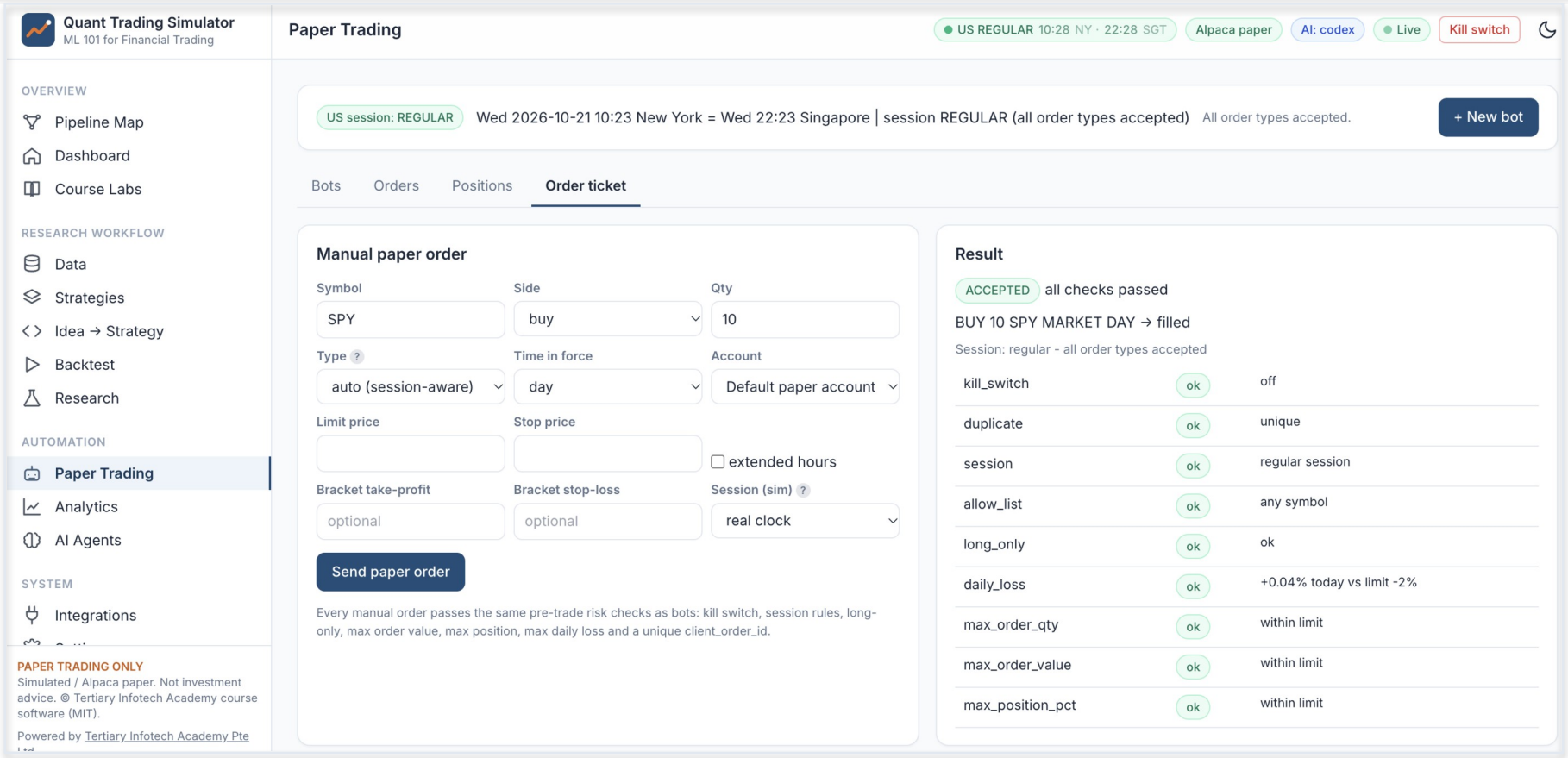



Paper Trading > Order ticket: buy 10 SPY, type auto (session-aware), time in force day, default paper account, Session (sim) = real clock. The banner shows the live session - here REGULAR, so all order types are accepted.


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LAB 08 · SIMULATOR

# Lab 08 — Risk Checks Passed, Order Filled




After Send paper order: ACCEPTED, all checks passed - BUY 10 SPY MARKET DAY -> filled. Every manual order passes the same pre-trade checks as the bots: kill switch, duplicate, session, allow list, long only, daily loss, max order quantity and value, max position.

# Evaluation Stage: Performance and Risk Metrics with a Benchmark Table

**LAB 09**

Build `python -m pipeline evaluate` to compute total return, CAGR, volatility, Sharpe, Sortino, maximum drawdown, Calmar, win rate, profit factor, exposure, turnover, alpha and beta for every signal against a buy-and-hold benchmark, and interpret the ranking.

## WORKFLOW

1. Read the metric formulas
2. Build the evaluation stage
3. Run the evaluation
4. Check one metric by hand
5. Interpret trade and market metrics
6. +2 more — see the Learner Guide

## YOU'LL PRODUCE

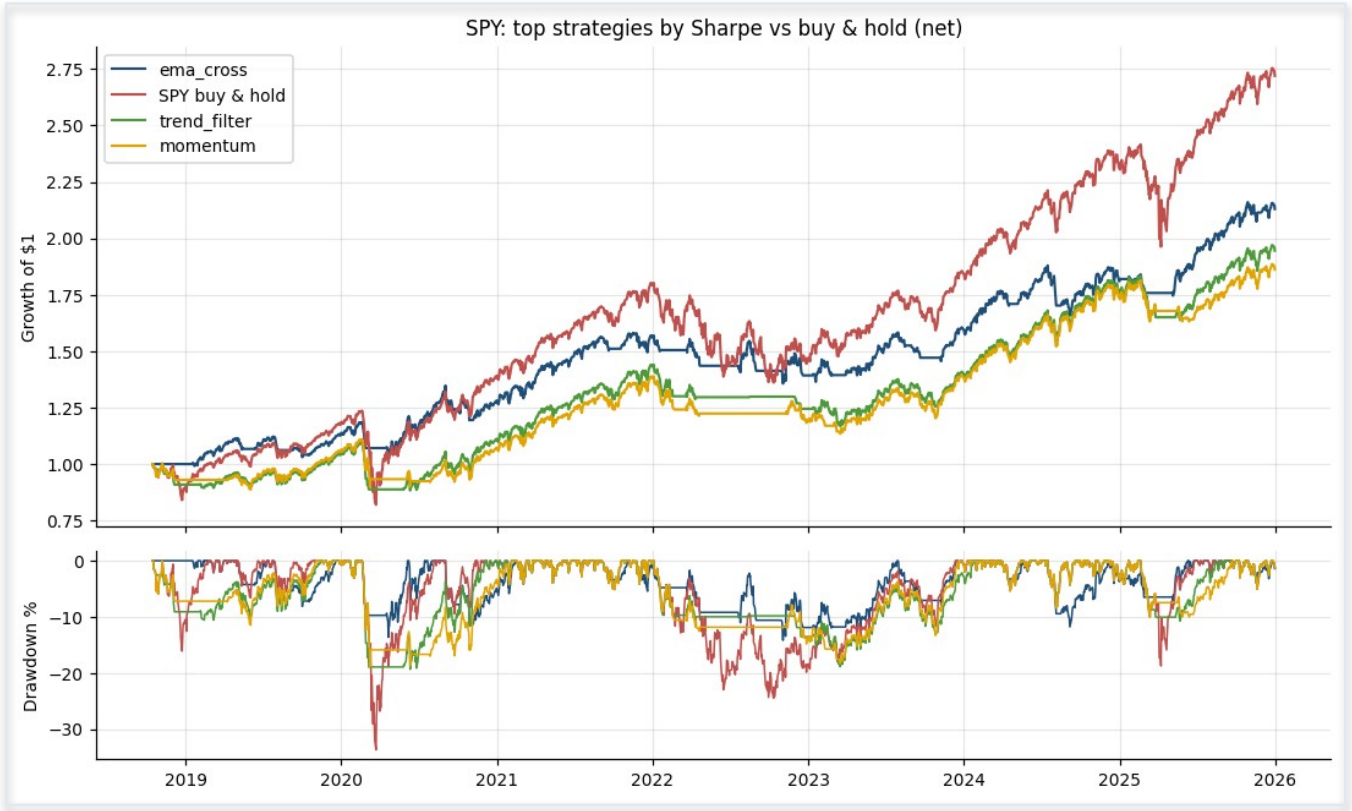
**An evaluation stage producing the metrics and benchmark table for every strategy.**

Duration
**25 min**


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LAB 09 · OUTPUT

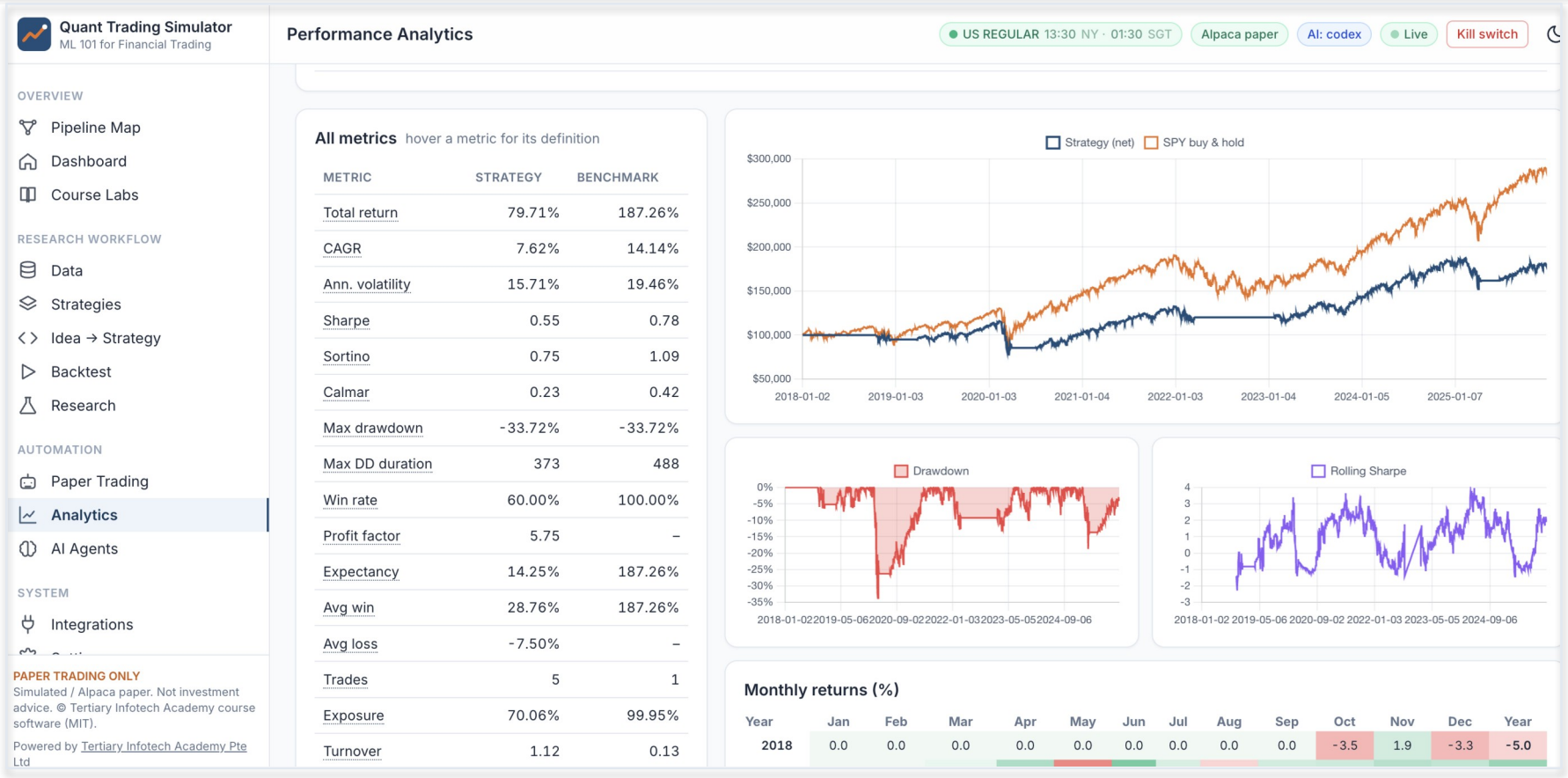
# Lab 09 — What You Should See




Output of Lab09-performance-metrics in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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# Lab 09 — Strategy versus Benchmark in Analytics




Performance Analytics for the SMA 50/200 backtest against SPY buy-and-hold: 79.71% versus 187.26% total return, Sharpe 0.55 versus 0.78 and the same -33.72% maximum drawdown, with drawdown, rolling Sharpe and monthly returns. Hover any metric for its definition.


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A2 A3 K3 K5 K7

# Topic 2 Recap

1. **A2**
   Identify appropriate statistical algorithms and data models to test hypotheses and/or theories

2. **A3**
   Use appropriate analytics platforms, analytical tools, statistical methods and analytical approaches given specific analytics and reporting requirements

3. **K3**
   Range and application of various statistical methods and algorithms

4. **K5**
   Usage of analytics platforms and tools

5. **K7**
   Coding languages for programming of algorithms and signals


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# End of Day 1

See you at 09:00 tomorrow


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TOPIC 3

03

# ML Models and A/B Testing

**LEARNING OUTCOME**

Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.

Abilities A4, A5 · Knowledge K2, K4, K6 · Day 2 · Labs 10–14 · Assessed by PP Task 3 and WA Q2, Q4, Q6


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K2

# Types of Statistical Analyses

1

## Descriptive

What happened — mean return, volatility, skew, correlation.

2

## Inferential

Is it real? — t-test, bootstrap, confidence intervals.

3

## Predictive

What next? — regression, classification.

4

## Prescriptive

What to do? — optimisation, position sizing.


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K2

# Families of Learning Algorithms

1

## Supervised

Features → known target: up/down, next return.

2

## Unsupervised

Structure without a target: regimes, PCA factors.

3

## Reinforcement

Learn a policy from rewards — execution research.

4

## Search & simulation

Grid search, Monte Carlo, bootstrap, walk-forward.


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# Supervised Models for Trading Signals

1

## Linear / logistic regression

Interpretable baselines; P(up) from a weighted sum.

2

## Decision tree & random forest

Threshold splits; averaged trees reduce variance.

3

## XGBoost

Sequential boosted trees — strongest on tabular features.

4

## Neural networks

MLP, LSTM, Transformer — data-hungry, easy to overfit.


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# Unsupervised Models for Regimes

1

## K-means

Minimise within-cluster variance; choose k by elbow or silhouette.

2

## Hierarchical clustering

Dendrogram of similar assets for diversification.

3

## PCA

First component ≈ the market; yield curve = level, slope, curvature.

4

## Fit on the past only

Assign new days to regimes as they arrive.


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# Statistical Modelling Techniques

1

**Returns, not prices**

Closer to stationary; avoids spurious regression.

2

**Time-ordered validation**

TimeSeriesSplit and walk-forward — never shuffle.

3

**Leak-free features**

Lags and rolling windows known at decision time.

4

**Regularise**

L1/L2, depth limits, early stopping, fewer features.


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# A/B Testing Two Strategy Variants

1

## State H0 and H1 first

H0: equal mean returns; H1: B is higher. α = 5%.

2

## Welch's t-test

Compare mean daily returns with unequal variances.

3

## Bootstrap the Sharpe difference

Resample returns; 95% CI excluding 0?

4

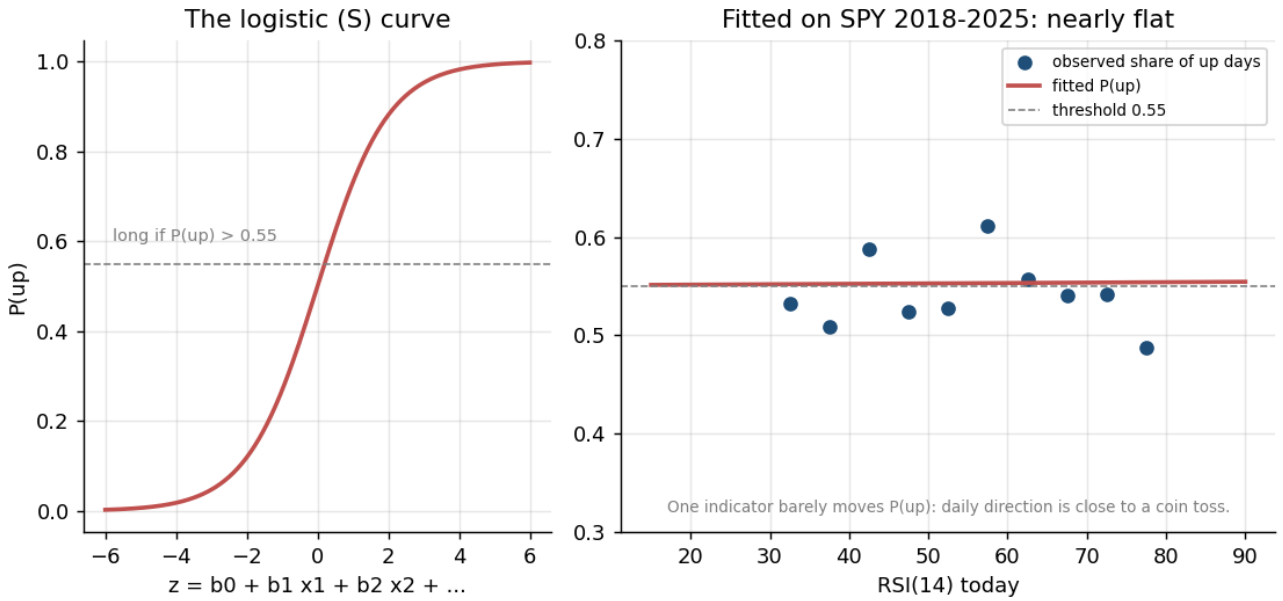
## Mind multiple testing

100 variants → ~5 false 'winners' by chance.


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K4

# Algorithm: Logistic Regression for Direction




**HOW IT WORKS**

Fits P(up tomorrow) = 1 / (1 + e^-(b0 + b1 x1 + ...)) from features known today.

**RULE**

Go long when P(up) exceeds a threshold such as 0.55; otherwise stay flat.

**WORKS BEST WHEN**

A transparent baseline: each coefficient shows how a feature moves the probability.

**WATCH OUT FOR**

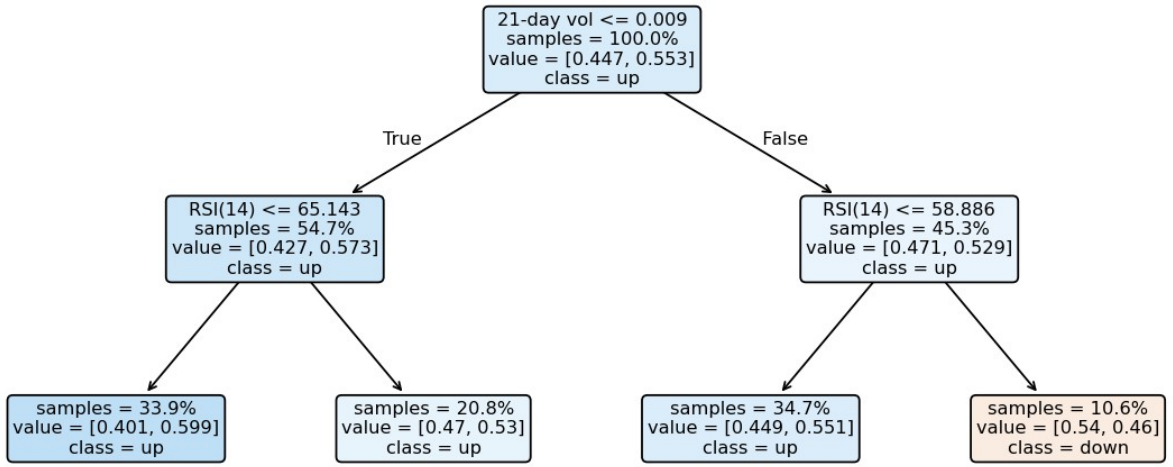
Only linear effects unless you engineer features; daily direction is close to a coin toss.


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K4

# Algorithm: Decision Tree and Random Forest

Decision tree (depth 2) on SPY: split on feature thresholds; a random forest averages many such trees




**HOW IT WORKS**

A tree splits the data on feature thresholds (e.g. RSI < 30) into leaves with an up/down share.

**RULE**

A random forest averages hundreds of trees, each grown on a random sample of rows and features.

**WORKS BEST WHEN**

Non-linear, interacting signals - e.g. oversold AND low volatility.

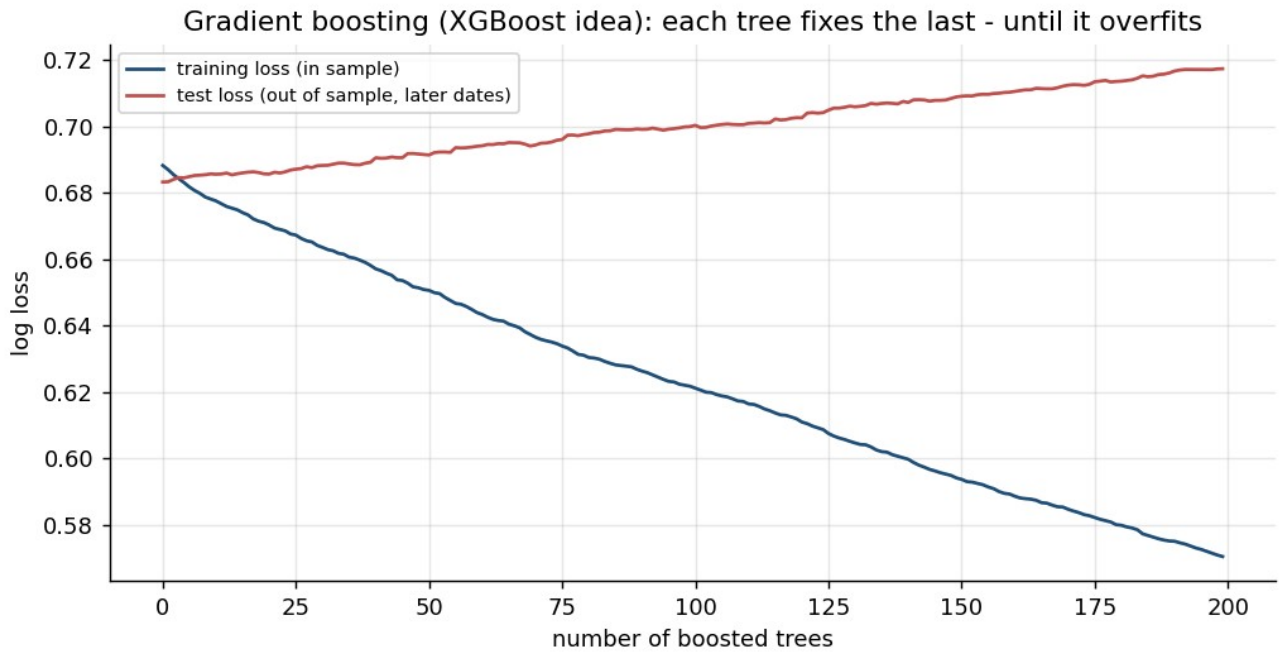
**WATCH OUT FOR**

Deep trees memorise noise; limit depth and leaf size and validate walk-forward.


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K4

# Algorithm: Gradient Boosting (XGBoost)




**HOW IT WORKS**

Builds trees one after another; each new tree corrects the errors of the ensemble so far.

**RULE**

Prediction = sum of many small trees x learning rate; stop early when test loss stops falling.

**WORKS BEST WHEN**

Tabular features such as returns, indicators and volatility - usually the strongest model.

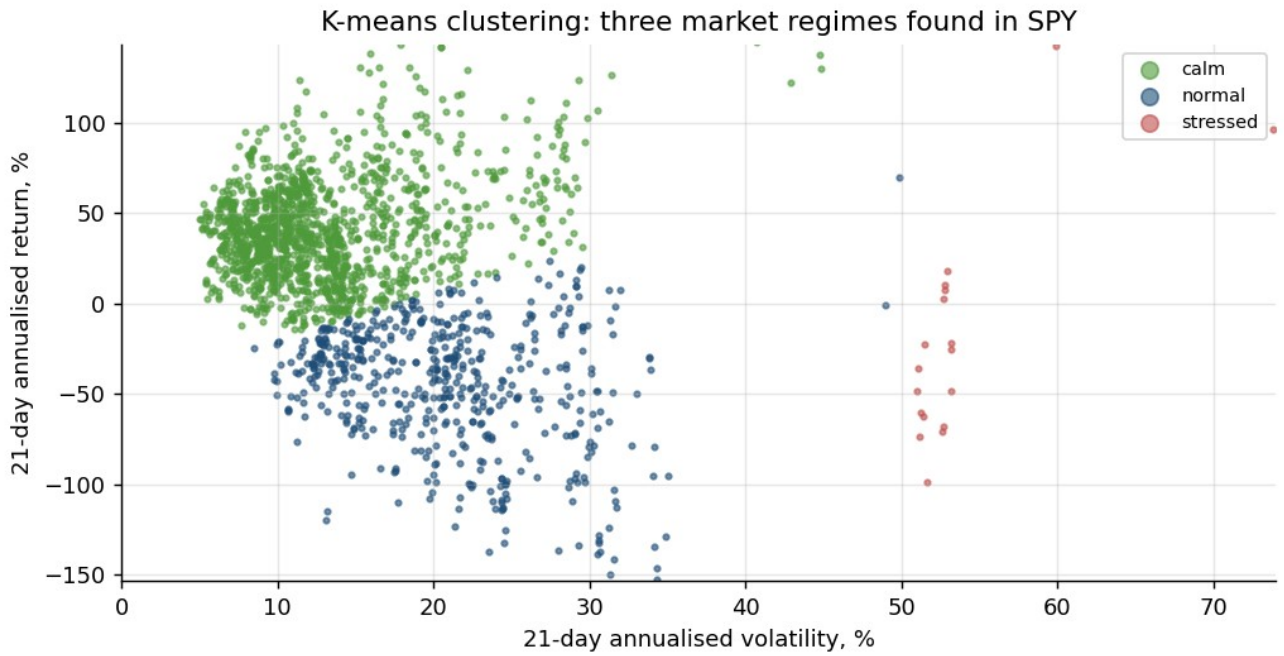
**WATCH OUT FOR**

Training loss keeps falling while test loss rises: classic overfitting (see the chart).


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K4

# Algorithm: K-Means Clustering for Market Regimes




**HOW IT WORKS**

Groups days into k clusters by minimising the distance of each day to its cluster centre.

**RULE**

Features: rolling return and volatility (standardised). Name clusters by their averages.

**WORKS BEST WHEN**

Switching or sizing strategies by regime: trend in calm markets, cut risk when stressed.

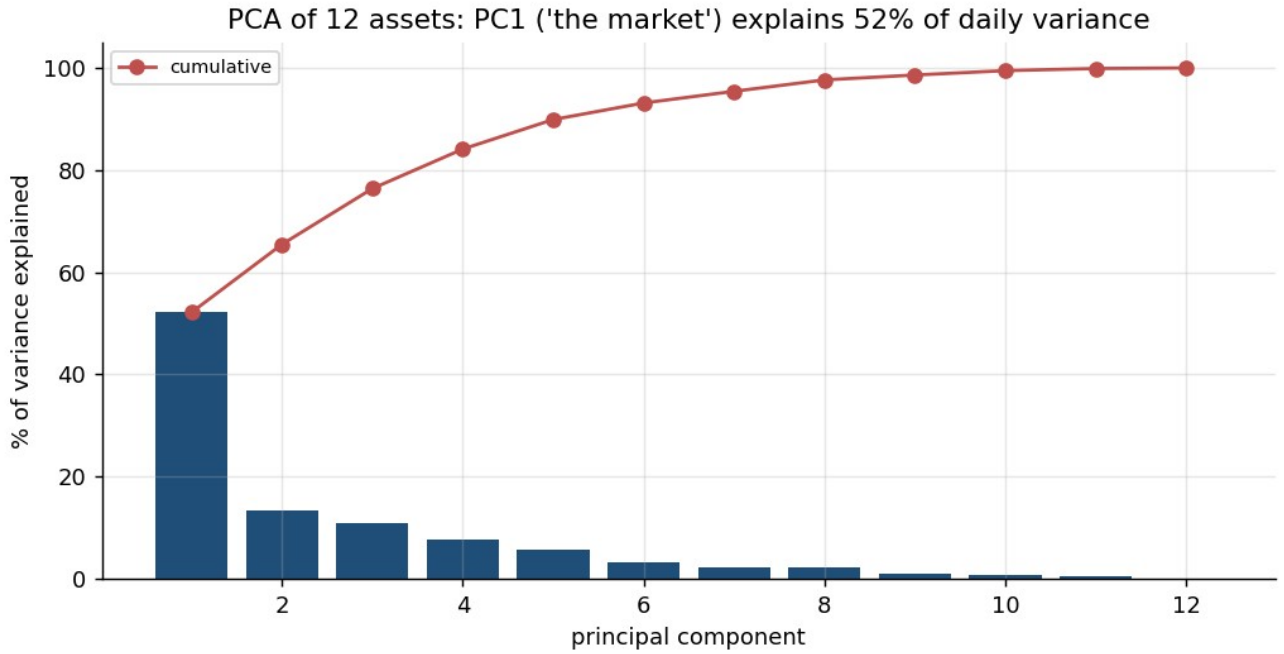
**WATCH OUT FOR**

Fit on past data only; choose k with the elbow or silhouette method; regimes shift.


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K4

# Algorithm: Principal Component Analysis (PCA)




**HOW IT WORKS**

Finds uncorrelated combinations of assets that explain the most variance of returns.

**RULE**

PC1 of a stock universe is 'the market'; later PCs are sector or style factors.

**WORKS BEST WHEN**

Reducing many correlated features; building factor-neutral and pairs portfolios.

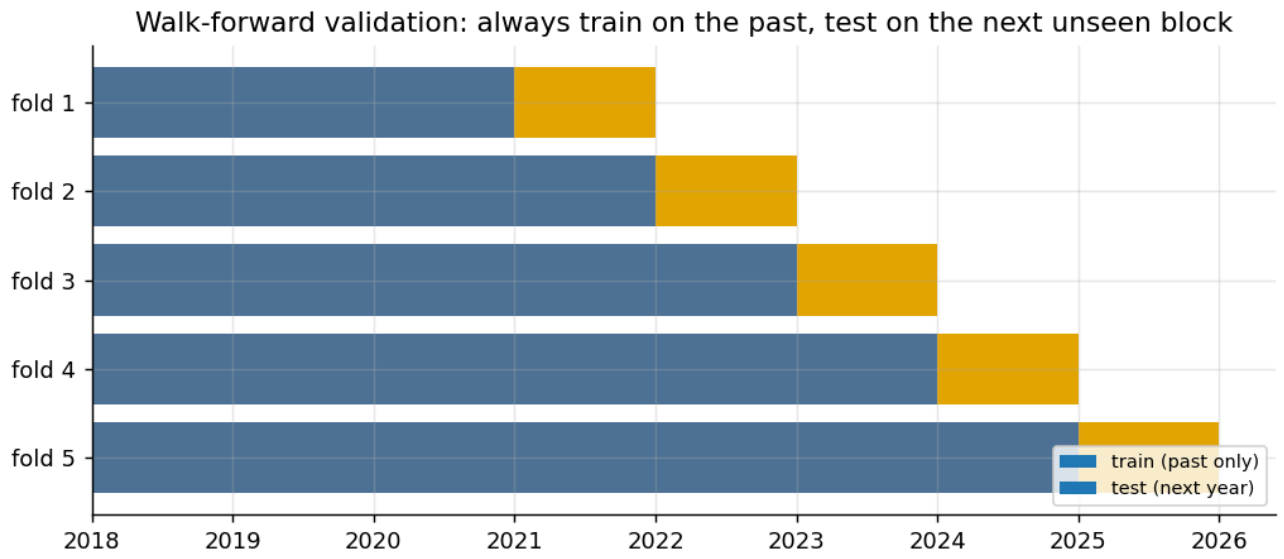
**WATCH OUT FOR**

Standardise first; components change over time; PCs are statistical, not economic, factors.


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K6

# Method: Walk-Forward Validation




**HOW IT WORKS**

Train on a past window, test on the next unseen block, then roll forward and repeat.

**RULE**

Never shuffle time series; each test block is genuinely out of sample.

**WORKS BEST WHEN**

Estimating how a model or parameter set would really have performed live.

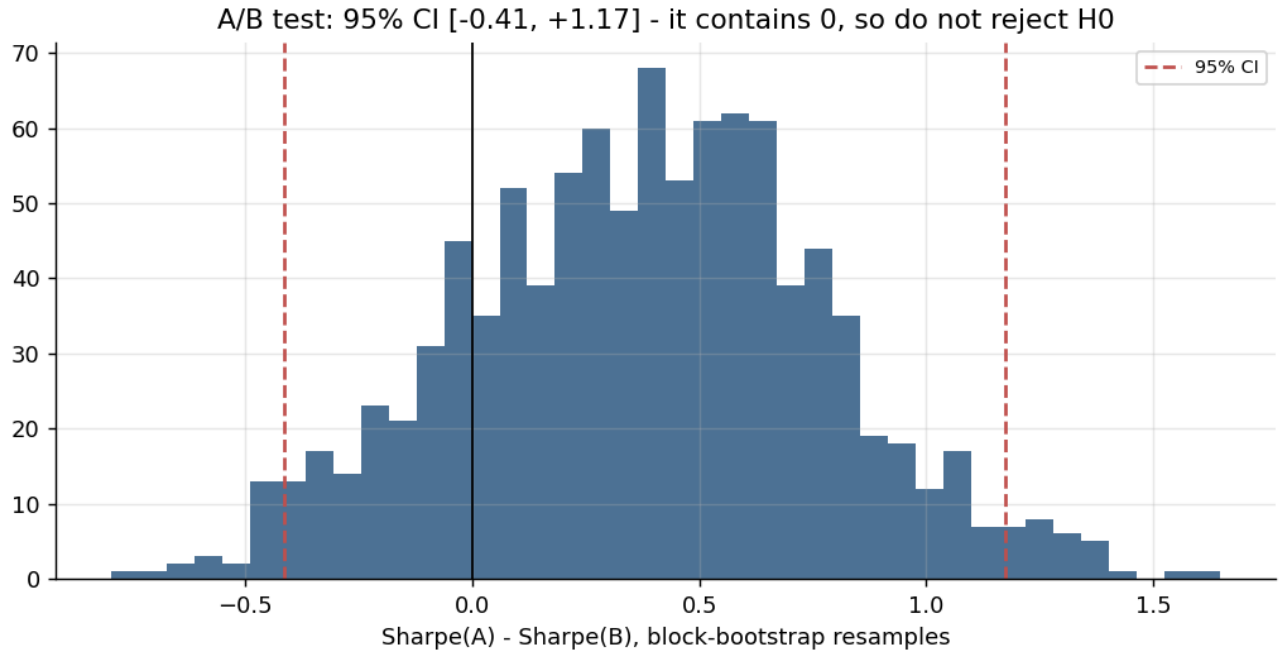
**WATCH OUT FOR**

Overlapping targets leak information - leave a gap between train and test.


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A5

# Method: A/B Testing Two Strategy Variants




**HOW IT WORKS**

Compare variant A and B on the same days: H0 'no difference' vs H1 'B is better'.

**RULE**

Welch / paired t-test on daily returns; bootstrap a 95% CI of the Sharpe difference.

**WORKS BEST WHEN**

Deciding which parameter set or filter to promote to paper trading.

**WATCH OUT FOR**

If the CI contains 0, do not reject H0; testing 100 variants gives ~5 false winners.


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# From Probability to Position

1. Build leak-free features
2. Train on past window
3. Predict P(up) for tomorrow
4. Long if P(up) > 0.55, else flat
5. Backtest net of costs

**Judge the model by the net-of-cost strategy, not by accuracy alone.**


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# Pairs Trading: a Hedging Algorithm

1

## Why it is a hedge

Long the cheap stock, short the rich one in hedge-ratio proportion: market moves largely cancel

2

## Choose the pair

Economically linked stocks (e.g. KO and PEP) that are cointegrated, not merely correlated

3

## Hedge ratio and spread

Regress log price A on log price B: slope = hedge ratio, residual = the spread (Engle-Granger)

4

## Test the spread

ADF test on the residual: a stationary spread tends to revert to its mean

5

## Trade the z-score

Enter when |z| > 2, exit as z returns towards 0; trade both legs on the next bar

6

## Risks

Cointegration breaks, hedge ratio drifts, double costs, short-borrow, multiple testing


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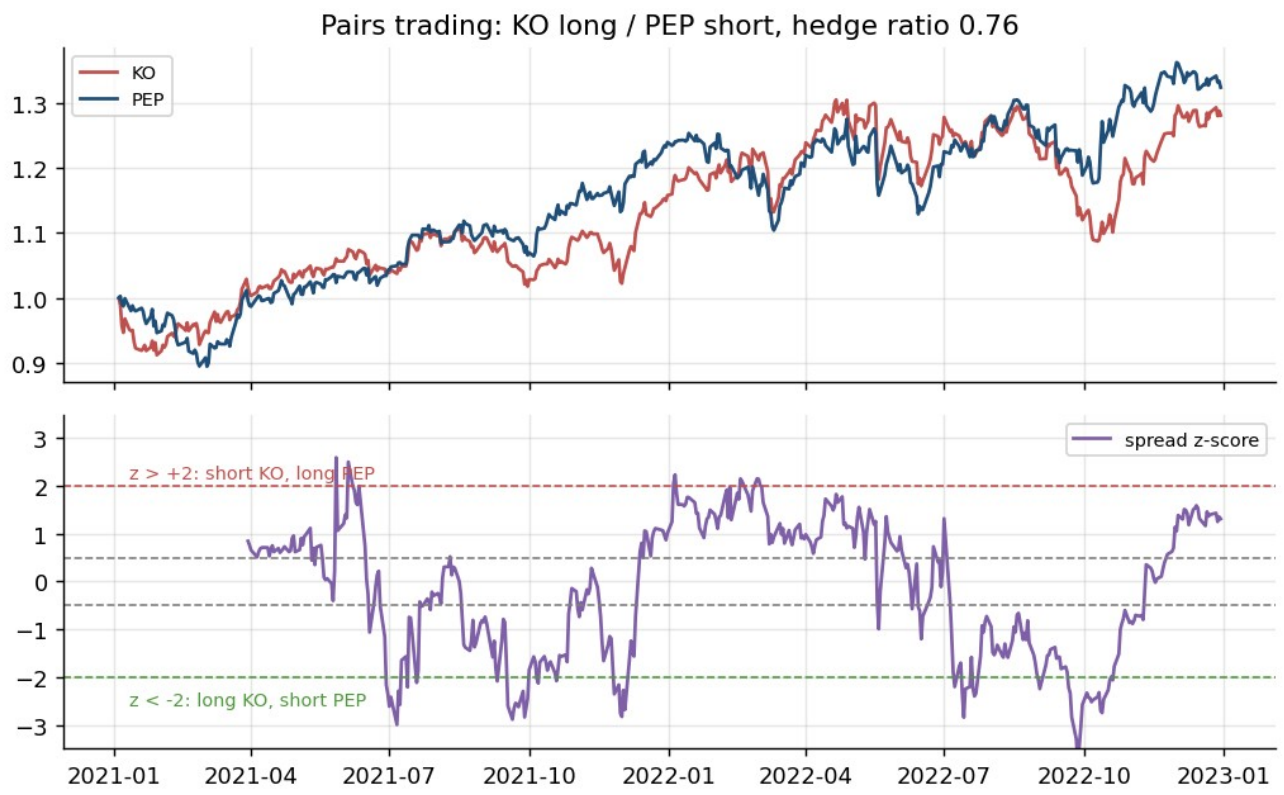
# The Pairs-Trading Algorithm Step by Step

1. Select an economically linked pair
2. Estimate the hedge ratio (OLS)
3. Test cointegration (ADF on residual)
4. Compute the spread z-score
5. Enter / exit both legs
6. Monitor and re-test

**Gatev, Goetzmann and Rouwenhorst (2006): a simple pairs rule on US stocks earned average annualised excess returns of up to 11% (1962-2002); later studies find smaller and less consistent profits.**


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# Algorithm: Pairs Trading (Hedged Statistical Arbitrage)




**HOW IT WORKS**

Long one stock, short a cointegrated partner in hedge-ratio proportion: market moves cancel.

**RULE**

Spread z > +2: short A / long B. z < -2: long A / short B. Exit as z returns to 0.

**WORKS BEST WHEN**

Economically linked stocks (KO/PEP) whose spread has a stable mean.

**WATCH OUT FOR**

Cointegration can break; costs on both legs; short-borrow; screening many pairs.


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A2 A4 K2

# Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay

## LAB 10

Build `python -m pipeline search` to grid-search three signal families, rank parameter sets in sample, measure out-of-sample decay, prefer a robust parameter region, and decide whether anything deserves promotion into config.yaml.

### WORKFLOW

1. Define the search space
2. Build the grid search
3. Add ranking, robustness and the decay report
4. Run the search
5. Read the overfitting evidence
6. +2 more — see the Learner Guide

### YOU'LL PRODUCE

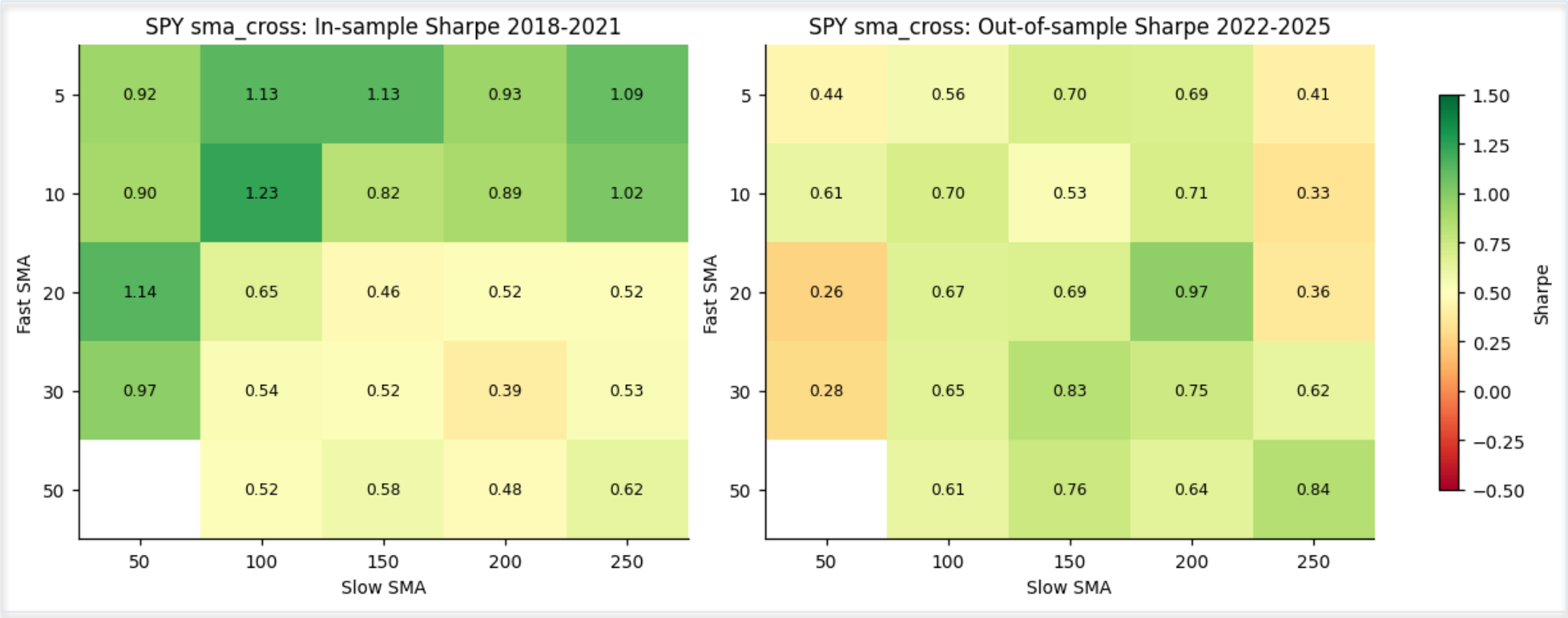
**A strategy-search stage: grid search, out-of-sample ranking and the decay report.**

Duration
**20 min**


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LAB 10 · OUTPUT

# Lab 10 — What You Should See




Output of Lab10-strategy-grid-search in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A4 K4 K6

# ML-Signal Stage: a Walk-Forward Classifier from Probability to Position

## LAB 11

Build pipeline/ml.py: engineer causal features, train logistic regression, random forest and XGBoost with walk-forward TimeSeriesSplit validation, turn P(up) into a target position and backtest it after costs.

### WORKFLOW

1. Build causal features and the label
2. Add walk-forward validation
3. Turn probability into a position and backtest it
4. Run the ML stage
5. Read the strategy result
6. +3 more — see the Learner Guide

### YOU'LL PRODUCE

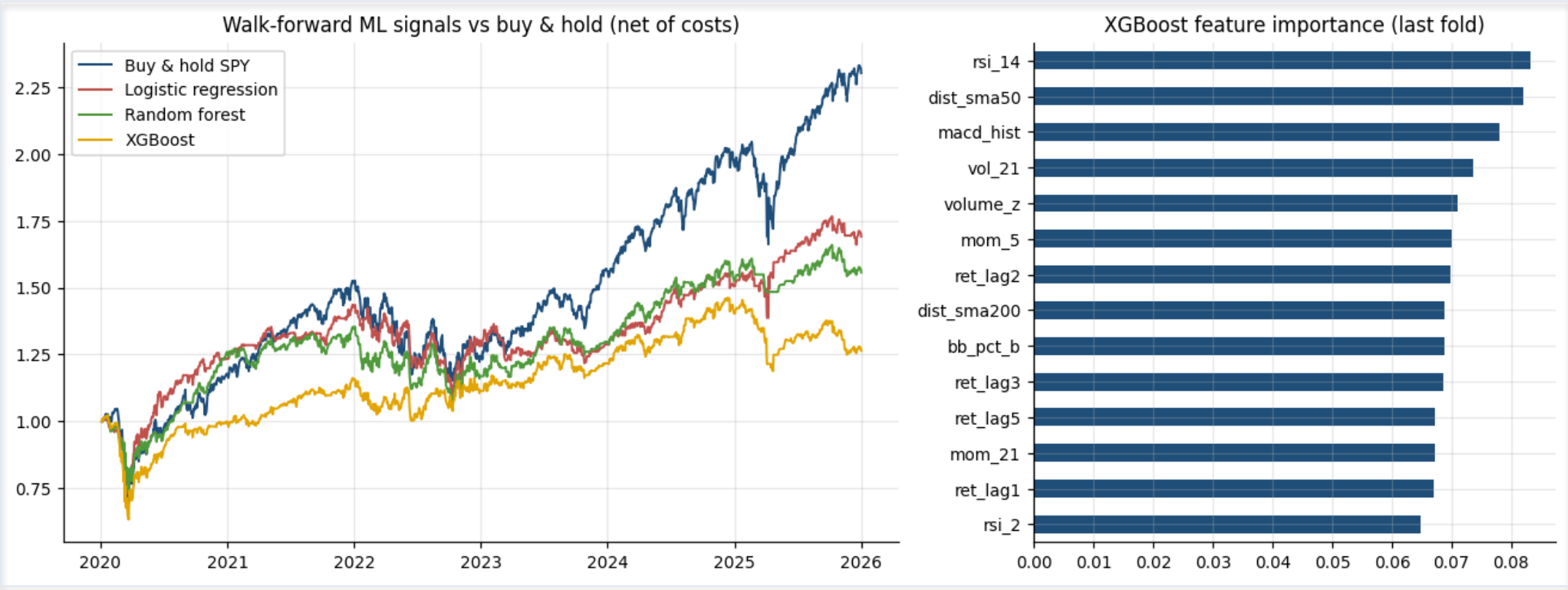
**A walk-forward ML signal stage from predicted probability to position, with a leakage check.**

Duration
**20 min**


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LAB 11 · OUTPUT

# Lab 11 — What You Should See




Output of Lab11-ml-signal-classifier in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A4 K4 K6

# Regime-Filter Stage: PCA and K-Means as a Gate on the Signals

**LAB 12**

Build pipeline/regime.py: PCA on the universe's returns, K-means regimes on backward-looking SPY features, walk-forward regime labels, and a gate that forces signals flat in the stress regime, evaluated after costs.

## WORKFLOW

1. Build PCA and the regime features
2. Add walk-forward labels and the gate
3. Run the regime stage and read the PCA
4. Read the regime profiles
5. Check the walk-forward labels
6. +2 more — see the Learner Guide

## YOU'LL PRODUCE

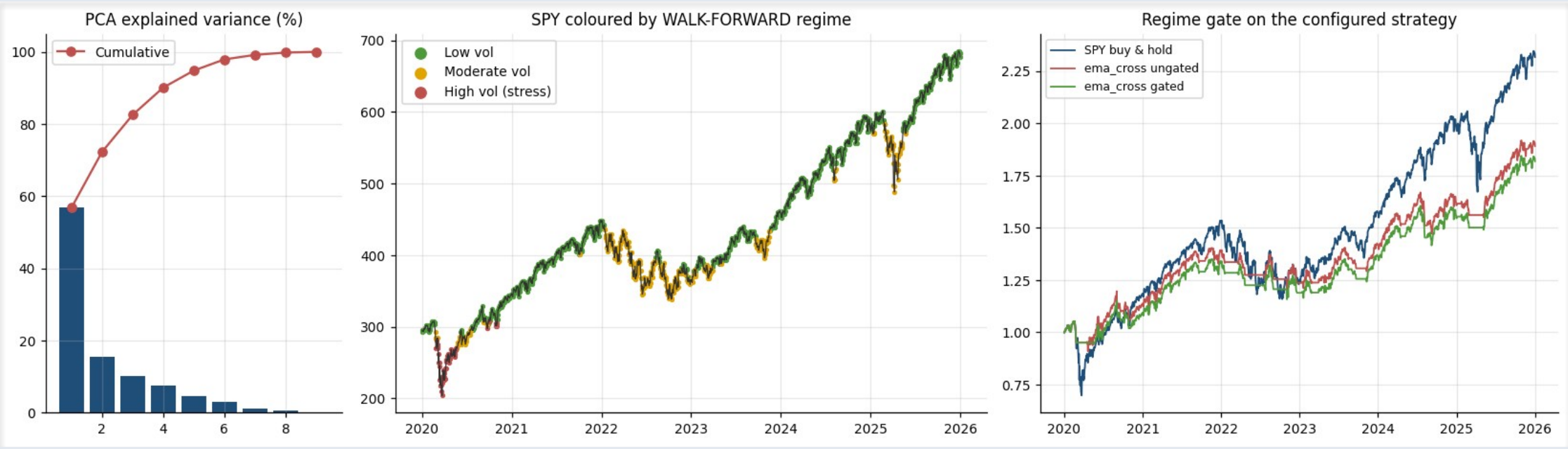
**A PCA / K-means regime gate applied to the signals, with before-and-after performance.**

Duration
**15 min**


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LAB 12 · OUTPUT

# Lab 12 — What You Should See




Output of Lab12-market-regimes in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A5 K2

# A/B-Test Stage: a Statistical Decision before Capital Moves

**LAB 13**

Build `python -m pipeline abtest` to compare any two library signals with a Welch t-test, a paired t-test and a block-bootstrap confidence interval for the Sharpe difference, and to record an accept or reject decision with a multiple-testing caution.

## WORKFLOW

1. State the hypotheses
2. Build the statistics
3. Build the A/B command
4. Run the A/B test
5. Compare the independent and paired tests
6. +2 more — see the Learner Guide

## YOU'LL PRODUCE

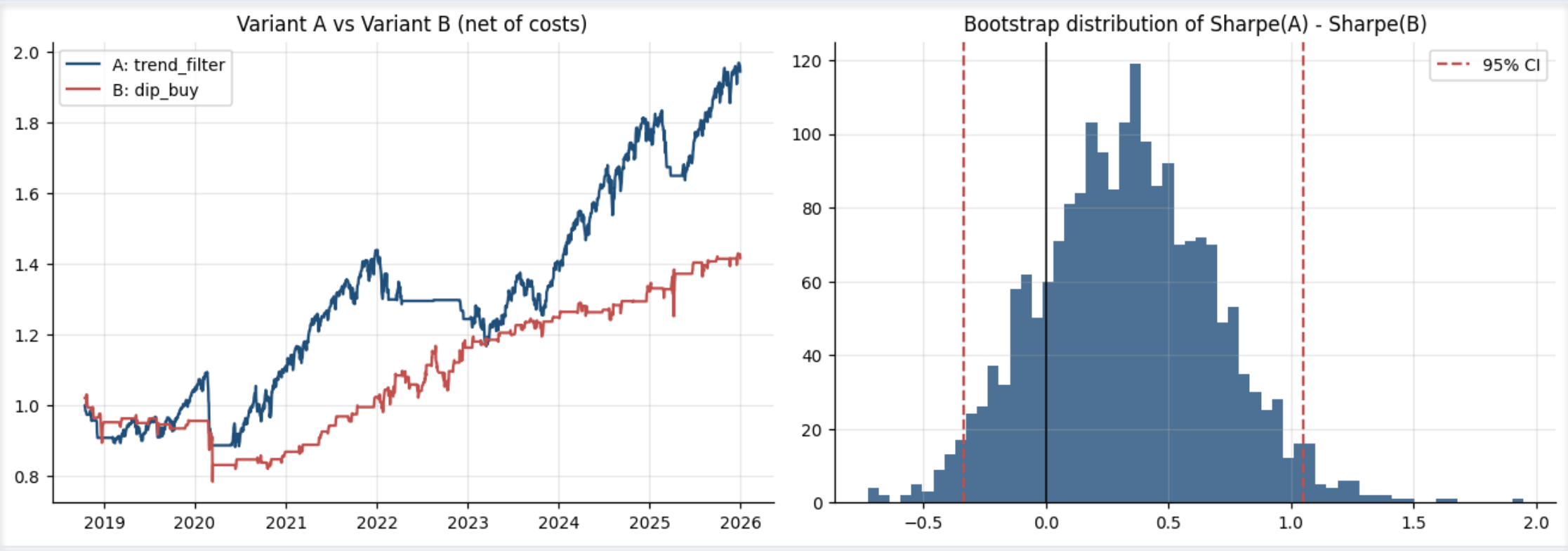
**An A/B-test stage that issues a statistical decision before capital moves.**

Duration
**15 min**


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LAB 13 · OUTPUT

# Lab 13 — What You Should See




Output of Lab13-ab-testing in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A4 K6

# Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve

## LAB 14

Build pipeline/pairs.py: screen pairs with the Engle-Granger cointegration test, estimate the hedge ratio, trade the spread z-score out of sample with next-bar execution and costs on both legs, confirm market neutrality and output leg targets.

### WORKFLOW

1. Read the method
2. Build the cointegration tools
3. Build the spread trading rules and leg targets
4. Run the pair screen
5. Interpret the hedge ratio
6. +3 more — see the Learner Guide

### YOU'LL PRODUCE

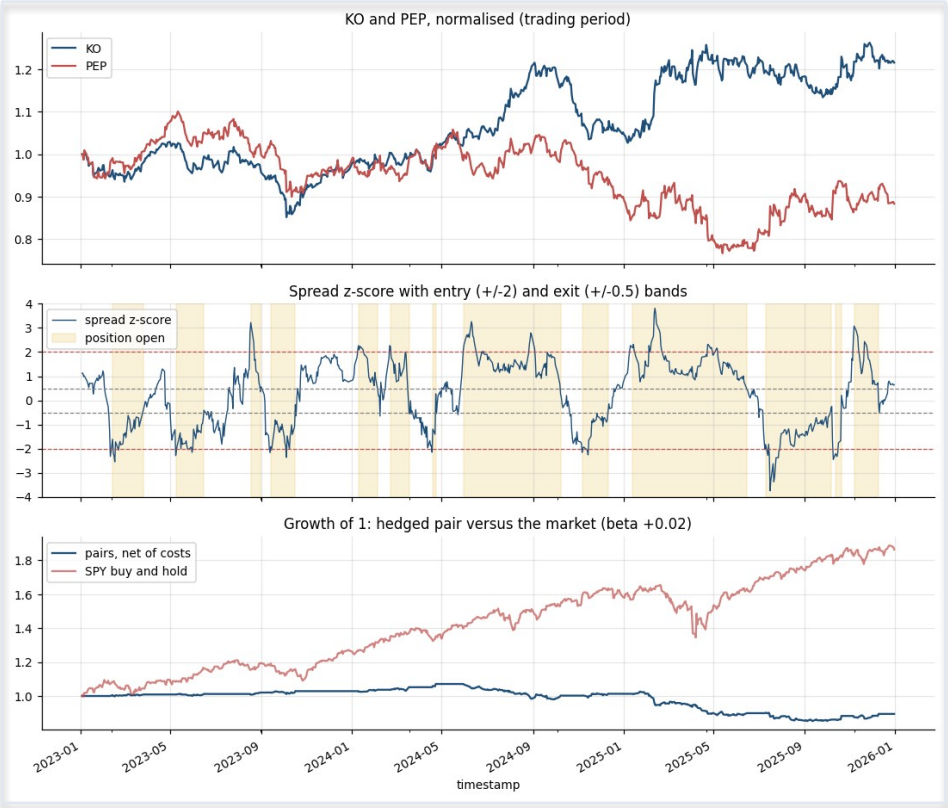
**A hedged pairs sleeve: cointegration screen, hedge ratio and a market-neutral spread strategy.**

Duration
**20 min**


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LAB 14 · OUTPUT

# Lab 14 — What You Should See




Output of Lab14-pairs-trading in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A4 A5 K2 K4 K6

# Topic 3 Recap

1. **A4**
   Conduct statistical modelling of data to derive patterns and solutions

2. **A5**
   Perform A/B testing based on different hypotheses

3. **K2**
   Types of statistical analyses, data models, algorithms and advanced computational methods

4. **K4**
   Range and application of various types of data models

5. **K6**
   Statistical modelling techniques


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TOPIC 4

04

# End-to-End Trading Pipeline

**LEARNING OUTCOME**

Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.

Abilities A7, A8, A9, A10 · Knowledge K7, K8 · Day 2 · Labs 15–20 · Assessed by PP Task 4 and WA Q7, Q8


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# Pre-Trade Risk Controls

1

## Position and order limits

Maximum shares, maximum order value per symbol.

2

## Maximum daily loss

Stop trading for the day when breached.

3

## Market-hours and duplicate checks

Alpaca clock; never resend an open order.

4

## Kill switch

One file or flag halts all trading immediately.


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# Trading Costs You Must Model

1

## Commission

Zero on Alpaca US equities — not on most brokers.

2

## Spread and slippage

Half-spread + slippage on every fill, in bps.

3

## Regulatory fees on sells

SEC Section 31 fee and FINRA TAF.

4

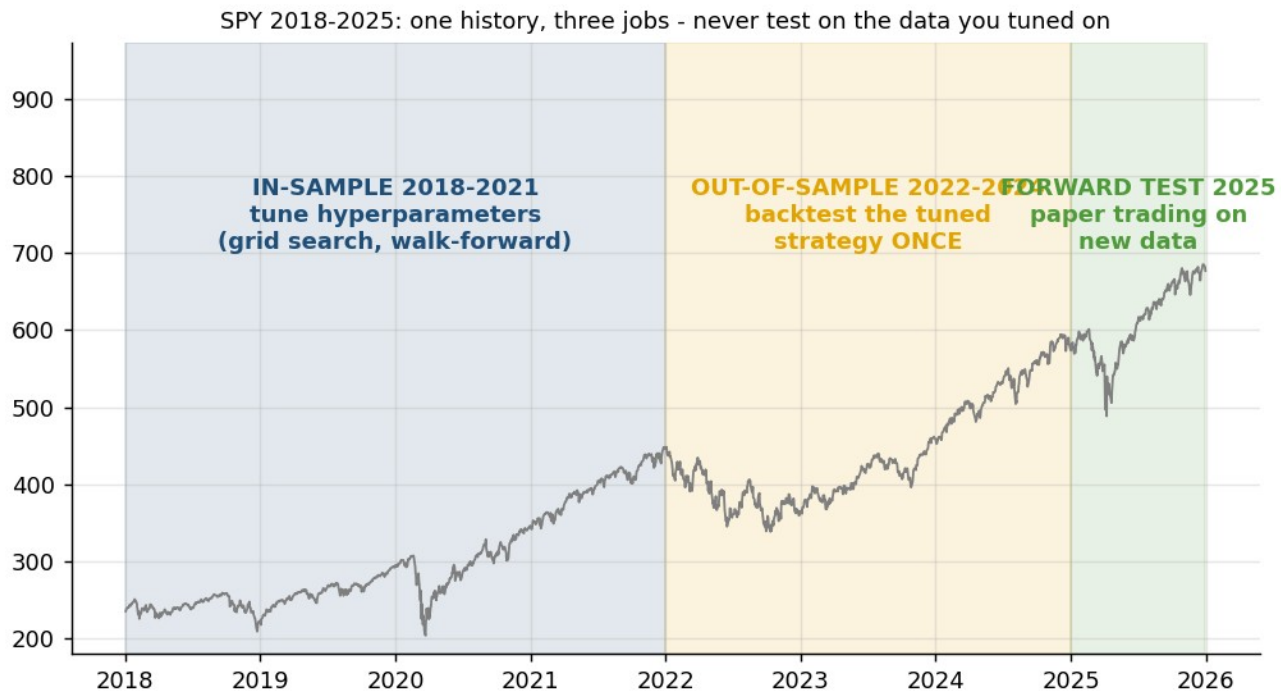
## Borrow and impact

Short-borrow costs; your own market impact.


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A8 K8

# Historical Data: Backtest, Tune, Then Forward Test




**BACKTEST ON HISTORY**

Replay the strategy's rules over past prices - next-bar fills, trading costs - to see how it would have performed.

**TUNE HYPERPARAMETERS (IN-SAMPLE)**

Search the settings (MA lengths, RSI levels, model parameters) on the OLDER history only, by grid search or walk-forward.

**CONFIRM OUT-OF-SAMPLE**

Backtest the tuned settings ONCE on later history the search never saw. A big drop from in-sample = overfitting.

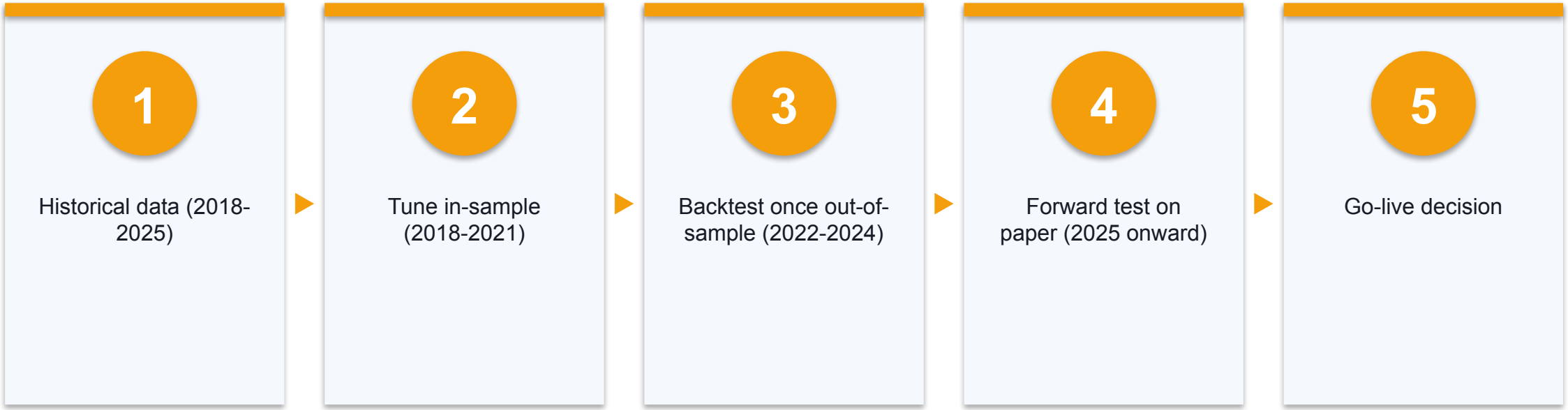
**FORWARD TEST (PAPER)**

Run the tuned strategy on NEW data as it arrives - paper orders, real fills and risk checks - before any capital.


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# From Historical Data to a Forward-Tested Strategy

The same rule moves through four tests; each uses data the previous step never touched.




**Tuning on the test period, or re-tuning after seeing out-of-sample results, turns the test into more tuning - the result will look better than reality (Labs 10, 15, 16).**


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# Potential Reasons for Unintended Outcomes

1

## Overfitting and data snooping

Great in-sample, collapses out of sample.

2

## Look-ahead and survivorship bias

Same-bar fills; only today's survivors.

3

## Cost and execution drag

Live returns trail the backtest by turnover × cost.

4

## Regime change, data and code errors

Drift, bad splits, time zones, duplicate orders.


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# What Goes on a Tearsheet

1

## Equity versus benchmark

Growth of USD 100,000, strategy and buy-and-hold.

2

## Drawdown and rolling Sharpe

Depth, length and stability of the edge.

3

## Monthly returns heatmap

When did it make — and lose — money?

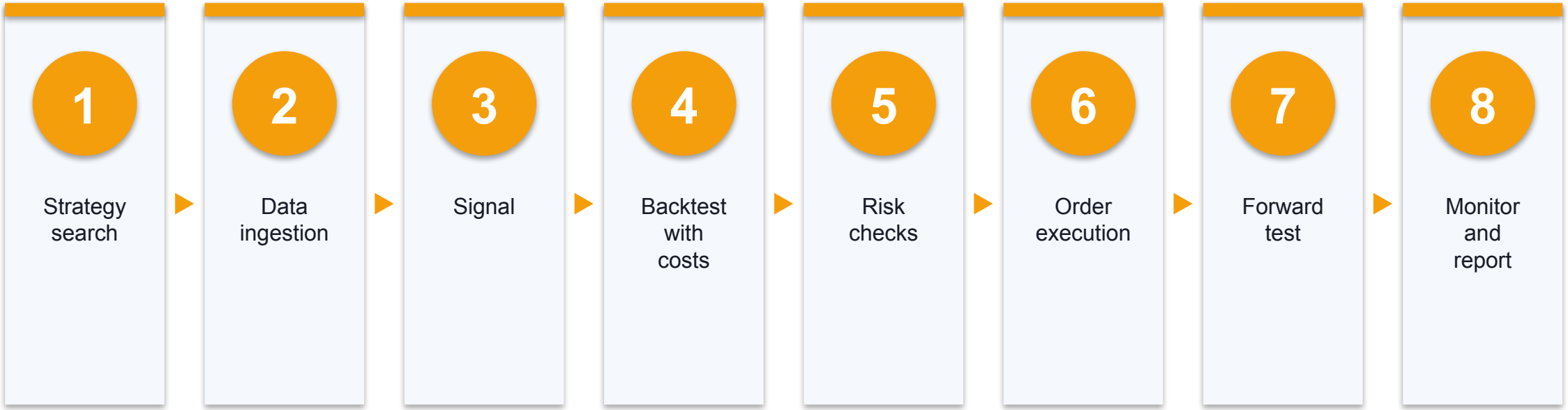
4

## Metrics and cost table

Gross versus net; backtest versus forward.


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# The End-to-End Algorithmic Trading Pipeline

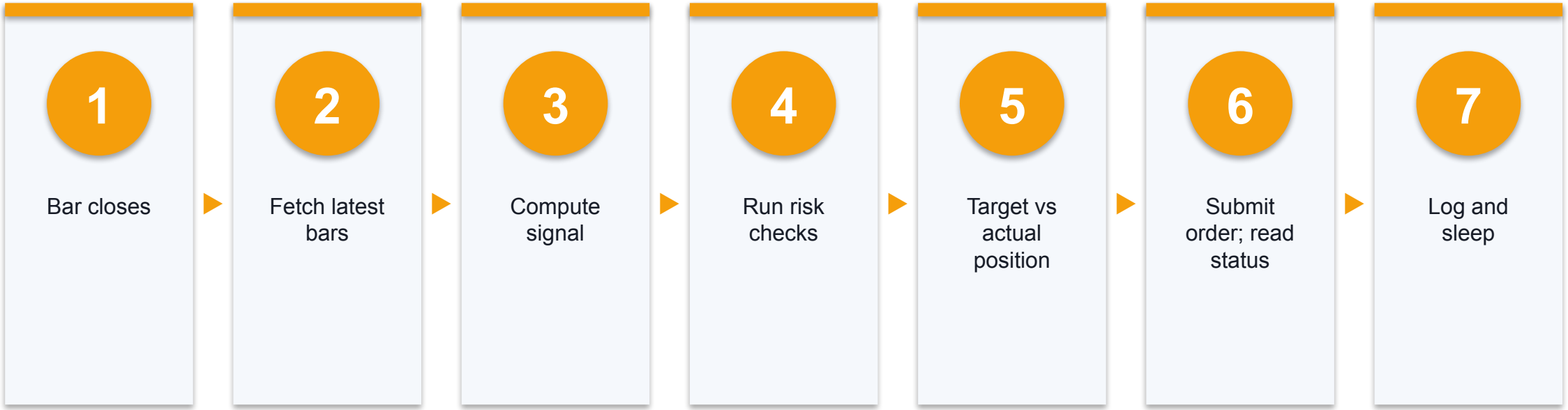



**Configuration — symbols, parameters, limits, keys — lives outside the code.**


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# The Automated Paper-Trading Bot Loop

One cycle per closed bar.

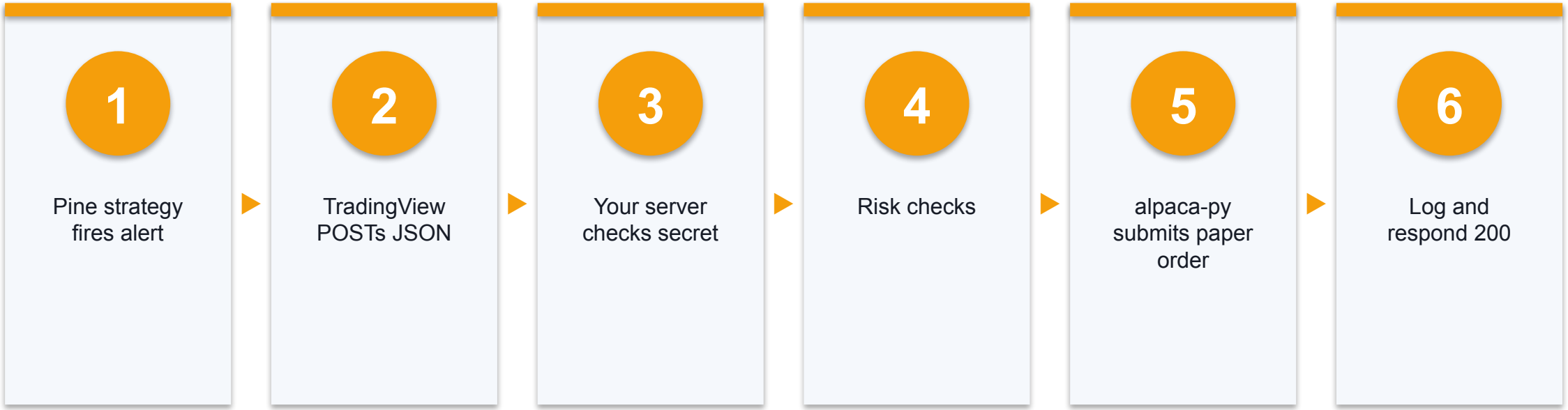



**Every cycle writes to the trade log — the evidence for performance and audit.**


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# TradingView Alert to Alpaca Order

Webhook-driven execution.




**Webhook alerts need a paid TradingView plan and a public URL (e.g. a tunnel).**

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# Diagnosing an Unintended Outcome

1. Reproduce the result
2. Isolate the first divergent step
3. Data · signal · order · fill?
4. Compare with a known-good run
5. Fix and re-test

**Read the first failure, not the last — everything after it is a consequence.**


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# Lunch Break

1 hour · see you at 13:30


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A7 A8 K7

# Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even

## LAB 15

Build pipeline/backtest.py around the common backtester: run any library signal under every cost scenario (and a custom one), break the cost into commission, spread, slippage and regulatory fees, find each strategy's break-even cost and test next-open versus next-close execution.

### WORKFLOW

1. Read the cost model
2. Build the engine helpers
3. Build the scenario report
4. Run the cost scenarios
5. Analyse the cost breakdown
6. +2 more — see the Learner Guide

### YOU'LL PRODUCE

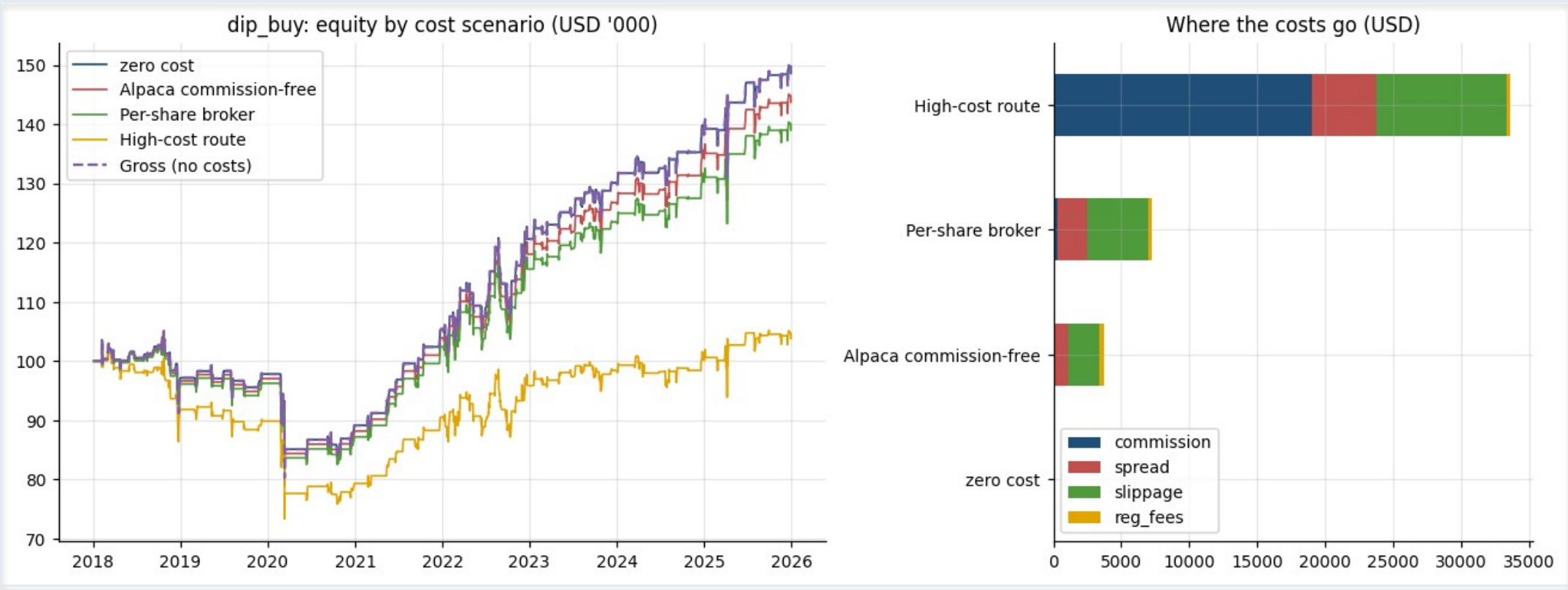
**A backtest engine with three cost models, execution timing and break-even cost.**

Duration
**25 min**


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LAB 15 · OUTPUT

# Lab 15 — What You Should See




Output of Lab15-trading-costs in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A7 A8 K7

# Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading

**LAB 16**

Build pipeline/risk.py and pipeline/runner.py: `python -m pipeline run --once | --loop` with the market clock, risk checks, kill switch and trade log; run a 2025 forward test; run the strategy live on the paper account for a few cycles; and schedule it with cron, launchd or Windows Task Scheduler.

## WORKFLOW

1. Build the risk layer and the kill switch
2. Build one decision cycle
3. Build the forward-test replay and the logs
4. Add the live paper loop and scheduling
5. Run the 2025 forward test
6. +4 more — see the Learner Guide

## YOU'LL PRODUCE

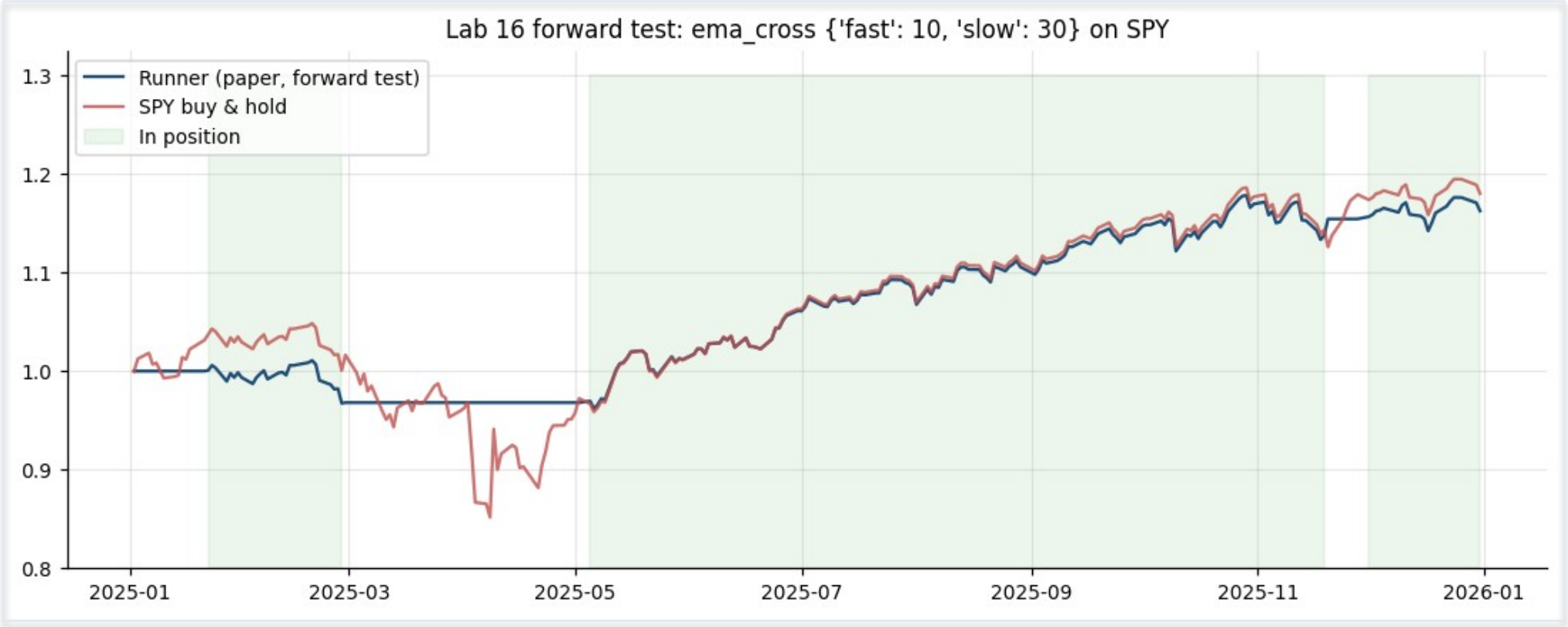
**A scheduled, risk-checked bot trading the paper account live, with a trade log and kill switch.**

Duration
**25 min**


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LAB 16 · OUTPUT

# Lab 16 — What You Should See




Output of Lab16-paper-trading-bot in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A7 K5 K7

# Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution

## LAB 17

Build pipeline/webhook.py: a webhook receiver that turns TradingView alert JSON into a pipeline decision (validation -> signal -> risk -> execution), test it with a local alert simulator and a self-test, and configure a real TradingView strategy alert.

### WORKFLOW

1. Set the shared secret
2. Build alert validation
3. Route valid alerts through signal, risk and execution
4. Build the server, the simulator and the self-test
5. Run the self-test
6. +4 more — see the Learner Guide

### YOU'LL PRODUCE

**Event-driven automation: TradingView alert to webhook to signal, risk and paper execution.**

Duration
**20 min**


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A9 K8

# Diagnostics Stage: Automated Bias Checks before Go-Live

## LAB 18

Build `python -m pipeline diagnose` to run automated look-ahead (features and execution), cost-drag, overfitting and survivorship checks on any strategy and on the junior quant's exhibit, and write a PASS / WARN / FAIL report with a go-live verdict.

### WORKFLOW

1. Run the suspicious backtest
2. Find the problems
3. Build the look-ahead checks
4. Build the cost, overfitting and survivorship checks
5. Run the diagnostics
6. +3 more — see the Learner Guide

### YOU'LL PRODUCE

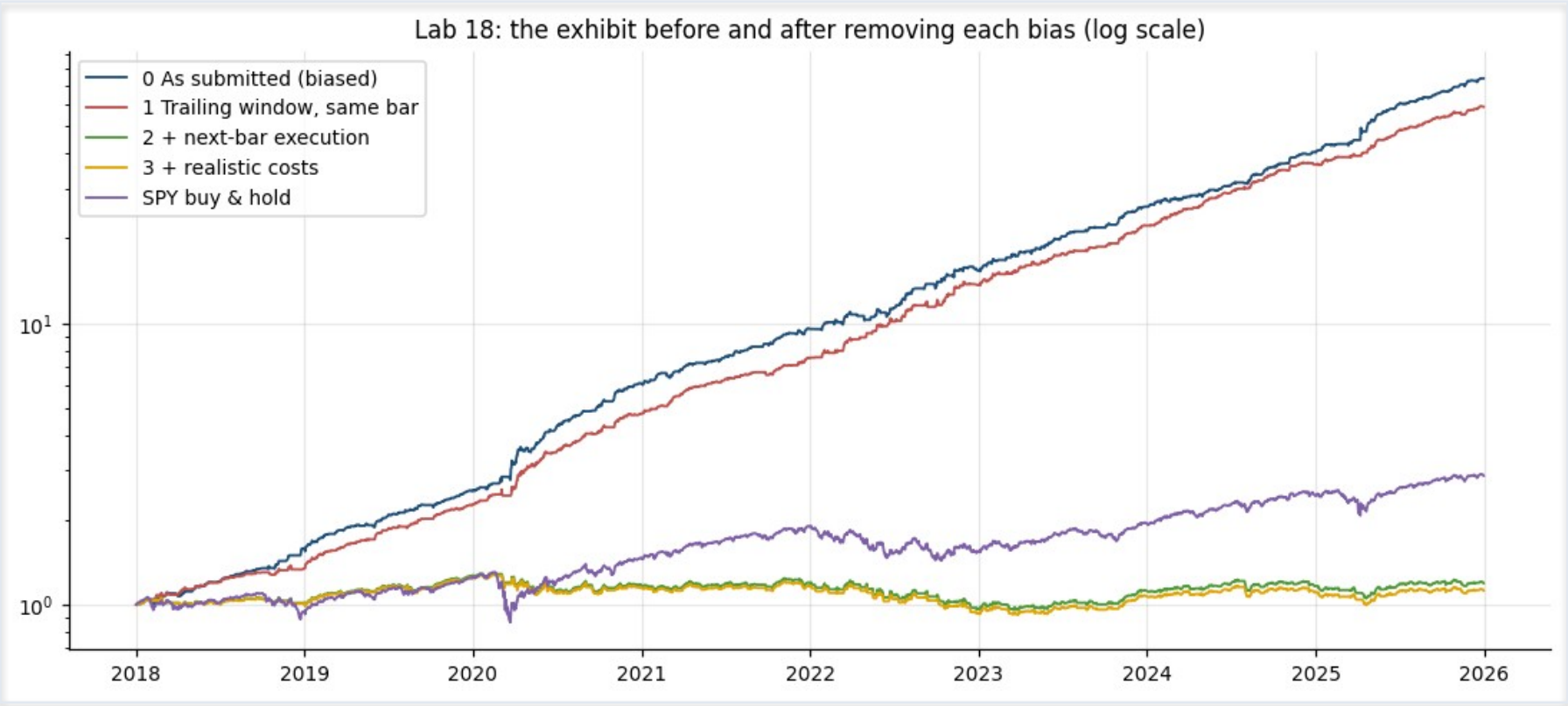
**Automated bias diagnostics (look-ahead, costs, overfitting, survivorship) before go-live.**

Duration
**20 min**


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LAB 18 · OUTPUT

# Lab 18 — What You Should See




Output of Lab18-diagnose-bias in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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# Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report

**LAB 19**

Build `python -m pipeline report` to compare the research backtest with the runner's forward-test logs, run monitoring checks with thresholds, and generate a tearsheet and a markdown report with data-driven insights and a recommendation.

## WORKFLOW

1. Make sure the forward-test logs exist
2. Build the backtest-vs-forward comparison
3. Add monitoring checks, the tearsheet and the report
4. Generate the report
5. Read the monitoring checks
6. +3 more — see the Learner Guide

## YOU'LL PRODUCE

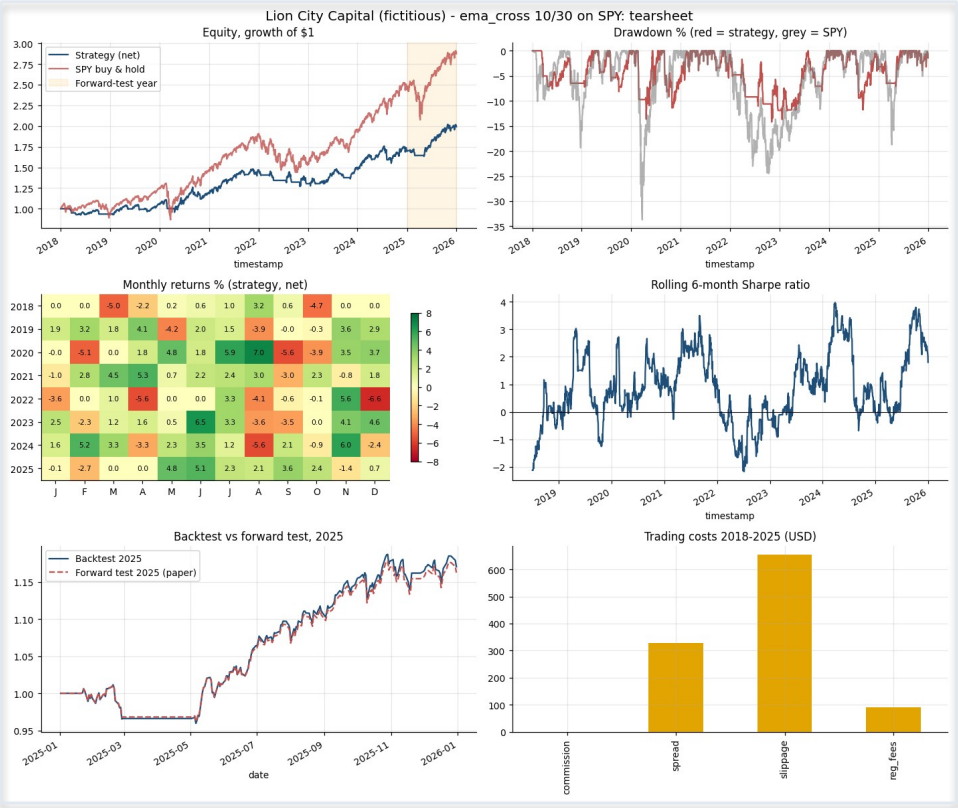
**An automated tearsheet and backtest-versus-forward-test report with monitoring checks.**

Duration
**25 min**


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LAB 19 · OUTPUT

# Lab 19 — What You Should See




Output of Lab19-performance-report in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A7 A10 K1

# Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval

**LAB 20**

Build pipeline/agents.py (analyst, researcher, trader and risk agents; rule-based by default, Claude optional) and pipeline/e2e.py (`python -m pipeline e2e`) that runs every stage in order with gates, then evaluate the pipeline and the role of human oversight.

## WORKFLOW

1. Build the agent layer
2. Build the end-to-end orchestrator
3. Run the whole pipeline
4. Trace the run
5. Read the agents' decisions
6. +5 more — see the Learner Guide

## YOU'LL PRODUCE

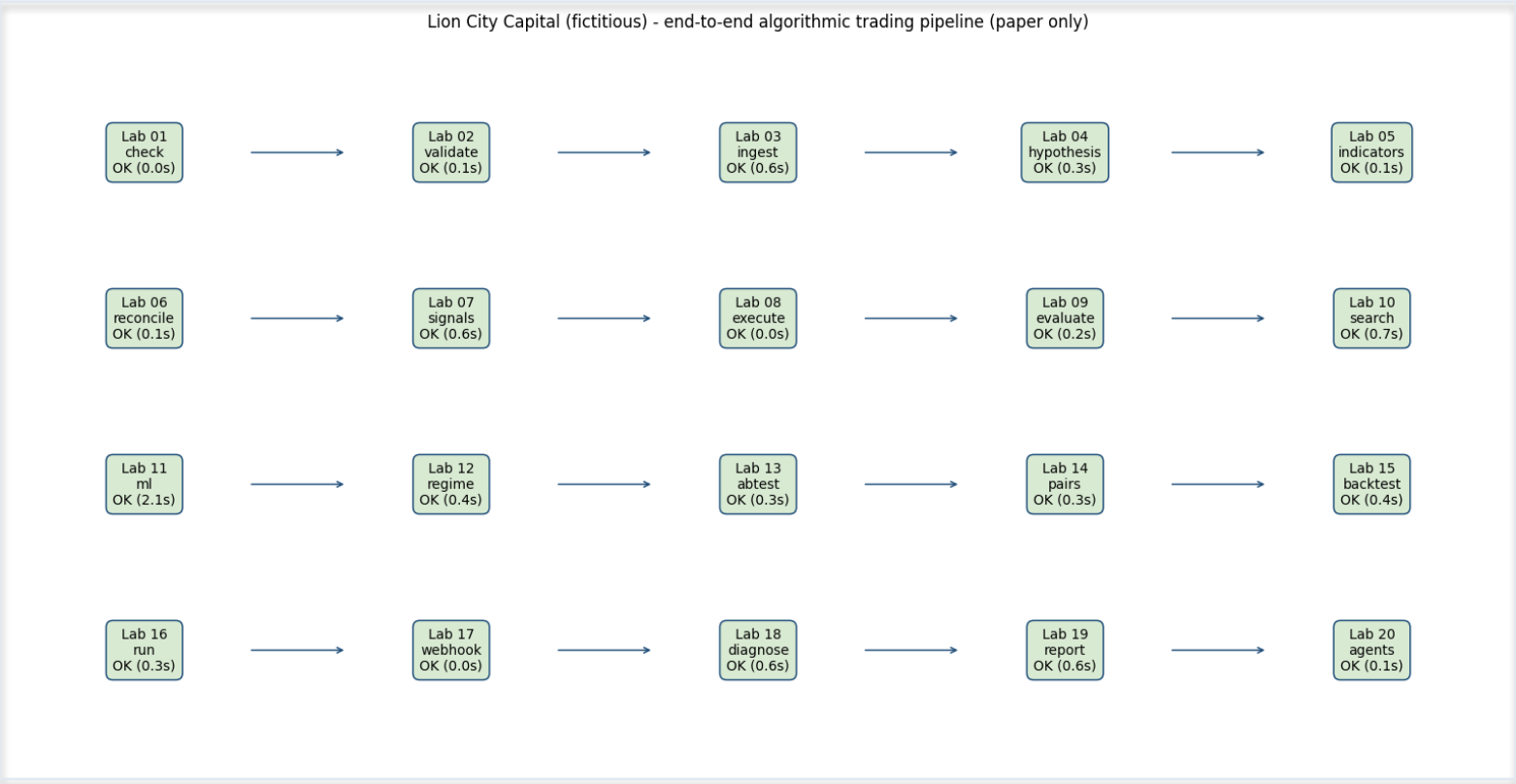
**One command running the whole pipeline end to end, with a human-approved agentic layer.**

Duration
**20 min**


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LAB 20 · OUTPUT

# Lab 20 — What You Should See




Output of Lab20-agentic-capstone in offline mode (bundled data). With Alpaca keys set, the same script uses live paper-trading data.


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A7 A8 A9 A10 K7 K8

# Topic 4 Recap

1. **A7**
   Perform coding and configuration of software agents or programs based on selected models or algorithms
2. **A8**
   Conduct tests on the actions taken and outcomes to assess effectiveness of the models
3. **A9**
   Diagnose unintended outcomes produced by statistical models
4. **A10**
   Draw relevant trends and insights from data analysis to support decisions and report accordingly
5. **K7**
   Coding languages for programming of algorithms and signals
6. **K8**
   Potential reasons for unintended outcomes


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SUMMARY

# What You Achieved

1. **Framed testable hypotheses**
   Instrument, signal, holding period, effect, benchmark.

2. **Built indicators and strategies**
   Pine Script on TradingView; Python on Alpaca data.

3. **Modelled and A/B-tested**
   ML classifiers, regimes, Welch's t-test and bootstrap.

4. **Backtested net of costs**
   Commission, spread, slippage and regulatory fees.

5. **Forward-tested automatically**
   A paper-trading bot with risk checks and a trade log.

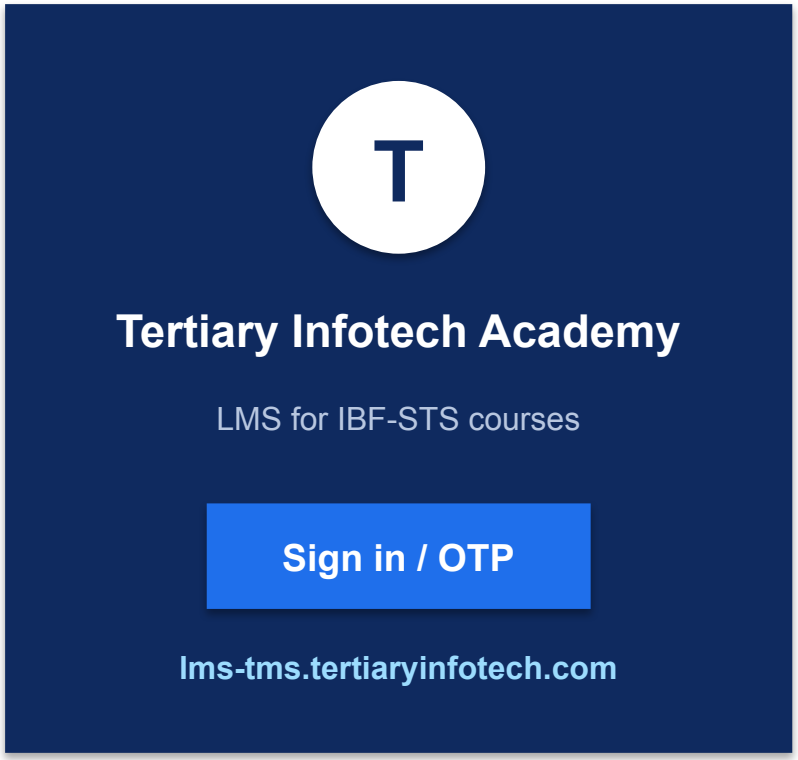
✓ **Diagnosed and reported**
   Found the bias, fixed it, and wrote the recommendation.


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COURSE PORTAL

# Courseware & Assessment on the LMS

Everything for the open-book assessment — slides, Learner Guide and submissions — lives on the LMS.




1. **Sign in**
   Log in with your registered email (OTP or password).
2. **Download the courseware**
   Trainer slides, Learner Guide and lab artefacts.
3. **Mark attendance**
   Sign in for each session — IBF-STS funding needs 75% attendance.
4. **Submit the assessment**
   Upload your WA answers and PP evidence here.


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ASSESSMENT

# Assessment — Reminder

## Written Assessment

45 minutes · open book · 9 short-answer questions covering K1 - K9.

## Practical Performance

75 minutes · open book · 4 tasks (Python + Alpaca paper trading), A1 - A10.


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ON ASSESSMENT DAY

# Assessment Flow

1. Sign the assessment attendance record
2. Assessor briefing (not assessed)
3. Sit the WA (45 min) then the PP (75 min) — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

**Sign the assessment attendance record BEFORE you begin. Submit on the LMS.**


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CLOSING

# Before You Leave

1

## Attendance

Make sure every session is signed — 75% is required for IBF-STS funding.

2

## Course feedback

Complete the feedback form on the LMS.

3

## Keep practising on paper

Your Alpaca paper account, the Trading Simulator and the lab code are yours to keep.

!

## Not investment advice

Scenarios are fictitious; trade real money only with proper advice.


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# Thank You

Machine Learning 101 for Financial Trading · FSE-DAT-3019-1.1

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